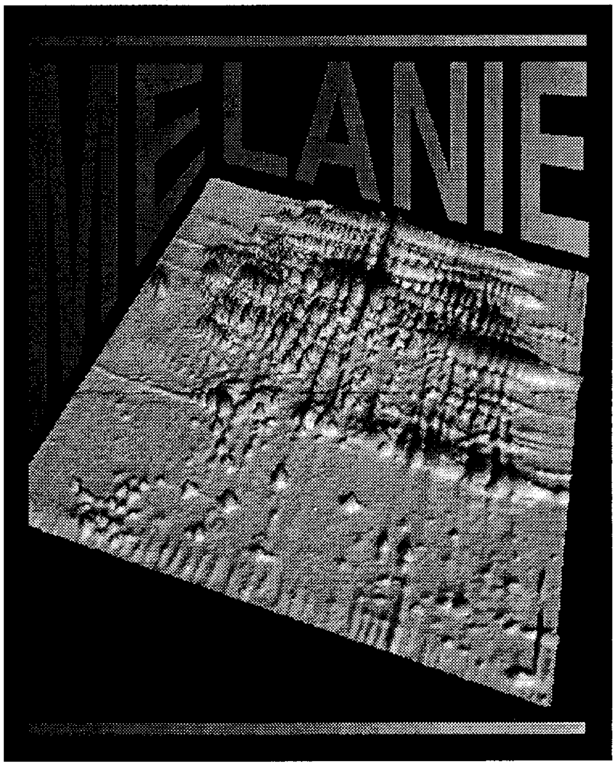


MELANIE II
2-D PAGE

release 2.1
for Unix

User Manual

Catalog Number
170-7566

For Technical Service
Call Your Local Bio-Rad Office or
in the U.S. Call **1-800-4BIORAD**
(1-800-424-6723)

4000052-01 Re vA

Warranty

Bio-Rad Laboratories warrants that the Melanie II software shall substantially conform, in all operational features, to Bio-Rad's current specifications as published in Bio-Rad's user and installation guides and that, when properly installed, it will be free of material defects which affect system performance.The Purchaser must notify Bio-Rad in writing, within 30 days of delivery of the software (not including delivery of any subsequent modifications to the software), of any defect. If the software is found to be defective by Bio-Rad, Bio-Rad's sole obligation under this warranty is to remedy the defect in a manner consistent with Bio-Rad's regular business practices. For a defect which adversely affects the performance of the software, Bio-Rad shall use its best efforts to cure such defect as soon as reasonably practicable after receipt of Purchaser's notice. For minor defects, Bio-Rad shall use its best efforts to correct such minor defects in the next release of its software. If, however, Bio-Rad is unable to cure a major defect within 90 days of receipt of Purchaser's notice, Purchaser shall have the option to cancel this agreement, whereupon Bio-Rad shall refund only the software fees paid.The warranties set forth in this agreement are in lieu of all other representations and warranties, expressed or implied, including warranties of merchantability and fitness for a particular purpose and any other statutory or common-law warranty. Bio-Rad on its own behalf expressly disclaims and excludes any and all such other representations and warranties. Liability of Bio-Rad to Purchaser, if any, for breach of warranty, or any other claim relating to this agreement, shall be limited to the total amount of software fees paid by purchaser to Bio-Rad. In no event shall Bio-Rad be liable for incidental or consequential damages, loss of business or profits, special or indirect damages of any nature whatsoever. No amendment, waiver, or other alteration of the warranties in this agreement may be made except by mutual agreement in writing.

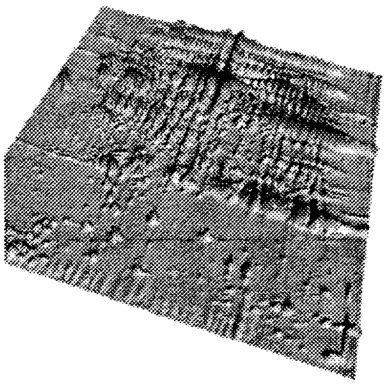
Purchaser agrees that Bio-Rad's liability arising out of contract, negligence, strict liability in tort or warranty shall not exceed the amount of software license fees paid by Purchaser. Under the copyright laws, this manual and the software program contained herein may not be copied, in whole or in part, without the prior written consent of Bio-Rad, except in the normal use of the software or to make a backup copy. This exception does not allow copies to be made for others, whether or not sold, but all of the materials purchased (with all backup copies) may be sold, given or loaned to another person. Under the law, copying includes translating into another language or format. A multi-use license may be purchased to allow the software to be used on more than one computer owned by the purchaser, including a shared disk system.

Copyright

Trademarks

Production notes

This document was created electronically using FrameMaker® 5.

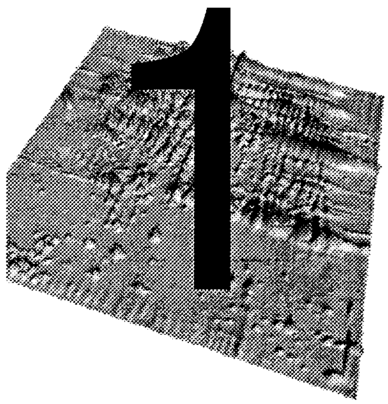


CONTENTS

PART III: ADVANCED FEATURES

PART I

BASICS



OVERVIEW

Using this manual

Welcome to **Melanie II 2-D PAGE analysis software**. Melanie II 2-D PAGE is a complete state-of-the-art software to view, process and analyze images of two-dimensional electrophoresis (2-DE).

This manual is divided into six parts. The first three parts are devoted to **MelView**, Melanie's main program. Part I gives an overview of the Melanie II 2-D PAGE software, as well as the basic procedures for analyzing 2-DE images. Part II details the necessary steps for comparing images. Part III presents advanced features, such as editing, image creation and working with remote databases. Part IV is dedicated to **MelBatch**, a companion program that provides 2-DE analysis in batch (non interactive) mode. The two tutorials in part V describe typical sessions with the MelView software, using 2-DE gels provided on the distribution tape. We strongly suggest you read through this section before working with Melanie II 2-D PAGE. Part VI provides technical support information. Finally, an index is supplied at the end of the manual.

Appendix A details the algorithms and methods used in Melanie II 2-D PAGE.

Important points in the manual, as well as useful hints are marked by a small hand in the margin in order to draw your attention.

The manual is also available on-line in the MelView program (See *On-line help* on page 3-1).

Hardware and software requirements

Environment

Melanie II 2-D PAGE release 2.1 for Unix requires the following minimal hardware and software environment:

- Sun Microsystems® workstations (SPARCstation® series), 32 MB of memory, SunOS® 4.1.x (Solaris 1) or Solaris 2, and either X-Windows X11R4, X11R5 or OpenWindows3®. We recommend at least 100 MB of swap space to process several gels simultaneously.

Programs and files distributed with the Melanie II 2-D PAGE package

Melanie II 2-D PAGE programs

- `MelView` - Melanie II main program
- `MelBatch` - program to perform actions in batch mode
- `MelLicense` - license manager
- `MelProgress` - progress report program
- `elmadmin` - Melanie II license administrator
- `elmd` - Melanie II license demon

Public domain programs

- `Melxwd`
- `Melpgmtolpr`
- `Melpgmtops`
- `Melpnmscale`
- `Melppmtogif`
- `Melppmtopgm`
- `Melxwdtoppm`

Master protein maps

- `CSF.mel` - Human cerebrospinal fluid
- `ECOLI.mel` - Escherichia coli
- `ELC.mel` - Human erythroleukemia cell
- `HEPG2.mel` - HepG2
- `HEPG2SP.mel` - HepG2 secreted proteins

- `LIVER.mel` - Human liver
- `LYMPHOMA.mel` - Human lymphoma
- `PLASMA.mel` - Human plasma
- `PLATELET.mel` - Human platelet
- `RBC.mel` - Human red blood cell
- `U937.mel` - Macrophage like cell line
- `YEAST.mel` - Yeast (Saccharomyces cerevisiae)

Example images

- Three images in directory `Gels`
- Same images after complete analysis in directory `Gels-analyzed`

Other files

- Manual and tutorial files in directory man.

Netscape Navigator®

In order to use the on-line manual, as well as to access remote databases over Internet, you must install the Netscape Navigator® World-Wide Web browser from Netscape Communications Corp., Mountain View, California, USA. You can purchase this product from Netscape Communications Corp., or download a free copy for non commercial use by anonymous FTP from the following FTP sites that are available at Netscape Communications Corp.:

FTP sites:

- ftp.netscape.com
- ftp1.netscape.com
- ftp2.netscape.com
- ftp3.netscape.com
- ftp4.netscape.com
- ftp5.netscape.com
- ftp6.netscape.com
- ftp7.netscape.com

Directory

/pub/netscape/unix/

Mirror sites exist in other geographical areas and you may download Netscape from them. To obtain an updated list of FTP sites, get the following document from Netscape Communications Corp.'s Web site:

http://home.netscape.com/comprod/netscape_products.html

or Netscape Communications Corp.'s main Web page:

http://home.netscape.com/

To download this software, type the following command:

```
> ftp FTP-site
```

where `FTP-site` is one of the FTP sites above. Enter `anonymous` as user name and give your full electronic mail address as password. Then enter the following sequence of commands:

```
> cd /pub/netscape/unix/

> bin

> dir
```

Find the correct version corresponding to your computer and operating system (SunOs or Solaris), then type:

```
> get Right-Version

> quit
```

where `Right-Version` is the version you want to download. Uncompress the downloaded file and follow the installation instructions.

Overview of Melanie II 2-D PAGE

Melanie II 2-D PAGE is a complete state-of-the-art software application designed to analyze images of two-dimensional electrophoresis (2-DE). You may perform qualitative and quantitative image analysis of 2-DE gels. **MelView** is the main program and provides interactive as well as automatic gel analysis.

Manipulating images

2-DE images may be manipulated in various ways. Images may be displayed in tile mode (one gel next to the other) or in stack mode (two gels on top of each other). They may be zoomed. Various color modes are provided in order to best visualize spots. Selected 2-DE images or the whole MelView screen may be printed. Images may be imported from or exported to various file formats such as TIFF, GIF and PPM.

Detecting and quantifying features

Melanie II automatically detects **features** (spots) on several gels at once. Various reports on feature values such as optical density (or instrument counts for non optical images), area and volume are computed and may be displayed, printed or saved to files.

Landmarks and labels

You may add **landmarks** to gels in order to mark specific points on the image. Landmarks may just hold a name, or they can be associated to *pI* and *Mw* values. They may also be used as starting points for the gel matching program.

Labels may be defined and associated to detected features. They are most often used to mark identified proteins. They hold the accession numbers (identification numbers) and names of the proteins.

Aligning and matching gels

Gels may be **aligned**, that is distorted gels will be stretched pixel wise, in order to make a set of images completely superimposable. Aligned or non aligned gels may be automatically **matched**, that is corresponding features will be paired across gels.

Grouping features

A set of gels may be matched to one reference gel. All features paired with the same spot in the reference gel may then be clustered to form **groups**. A group represents the same spot over a set of gels. Reports on groups, as well as histograms may be displayed, printed or saved. Groups may also be exported to ASCII tables in order to be used with external programs such as Excel or other data analysis packages.

Creating new images

Gel images may be duplicated, flipped, scaled or cropped. Filters are provided to improve image quality, such as contrast enhancement, smoothing and background subtraction. **Synthetic gels** may be produced by merging spot information from several gels. Gaussian modeling of features is also supplied.

Querying databases

Master gels may be defined, or those provided on the distribution tape may be used. A master gel contains labels for each identified protein. They may then be automatically linked to remote databases such as SWISS-2DPAGE or SWISS-PROT through the use of Netscape Navigator®.

MelBatch

In the MelView program, actions may be carried out on open gels only. The **MelBatch** program is supplied to perform certain actions on gels without the need of displaying them. In this way, feature detection and quantification followed by gel matching may be performed on a set of gels as a background process.

The Melanie II 2-D PAGE file format

The Melanie II programs work with Melanie II release 2 file formats, as well as with TIFF files. Analyzed gels are always saved into the Melanie II release 2 file format, unless otherwise specified by the user.

Gel images may be imported from or exported to other formats such as TIFF, GIF and PPM.

Running MelView

To run the MelView program, type the following command:

```
> MelView &
```

The `&` sign indicates that the MelView program has to run in background. You have to make sure that the directory in which the MelView program resides is part of your path. Type

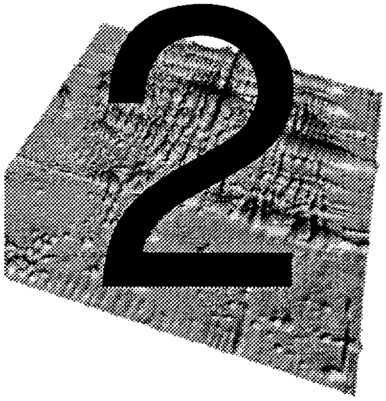
```
> echo $path
```

to verify this. If it is not in the path, please refer to the Unix manual to include it.

You may also run the MelView program and automatically open gel images. Type the MelView command followed by the gel names. For example, to open gels *GelA* and *GelB*, type:

```
> MelView GelA GelB &
```

Upon loading the program the Melanie II 2-D PAGE logo appears. Click on the logo to get rid of it.



THE MELVIEW COMPONENTS

The MelView window is divided into three parts: the horizontal **menu bar** on the top, the vertical **tool bar** on the left side of the window and the rectangular **display area** (Fig. 2-1). You may use the tools to select objects in the display area,

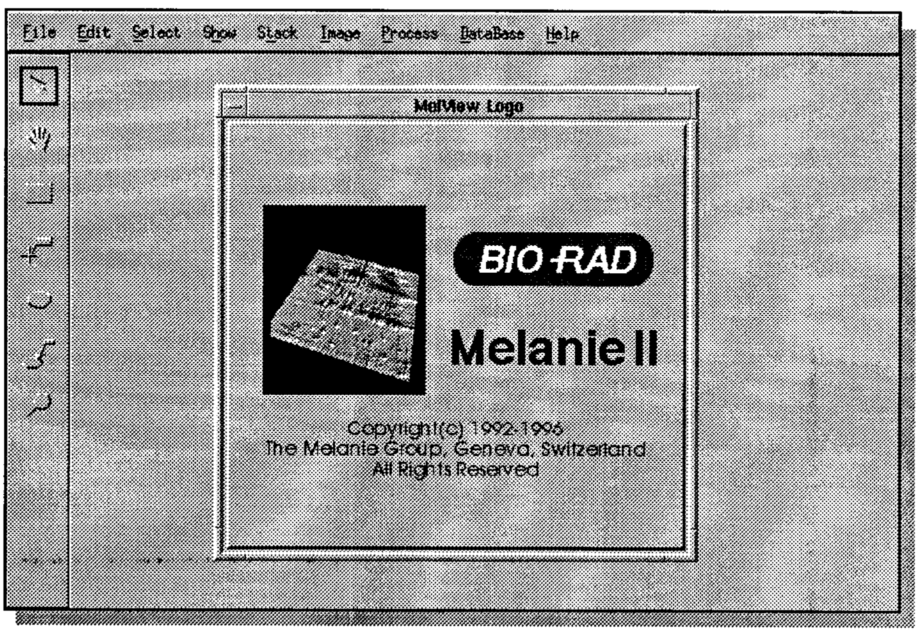


Figure 2-1. The MelView window

then apply actions from the various menu options to the selected objects. Objects are gel images, landmarks associated to specific points on the image, features (detected spots), labels associated to features, feature pairs between two gels, groups of features across several gels, or comments on gels.

This chapter describes the functionality of the various MelView components. Their practical use will be discussed in details in the following chapters.

Objects

A Melanie II object is an entity that may be displayed and manipulated. The objects are gels, as well as gel components such as regions, landmarks, features, labels, pairs, groups or comments.

Gels

A **gel** is a 2-DE image that has been digitized and stored on disk in the Melanie II 2-D PAGE release 2 format. Images in other formats (TIFF, GIF, PPM) may also be opened as gels, but they have to be saved into the Melanie II format in order to keep objects such as detected feature shapes, feature values, pairs, etc. Other file formats may only contain plain unstructured images. If a gel is saved with its objects into a foreign format, then the objects will simply be part of the saved image and will have lost any structure or value.

The gel image is the raw input data to the MelView program, from which features may be detected and quantified. Gels may also be duplicated, erased, cropped, filtered, flipped or scaled. Synthetic gels may be created by merging spots from other gels.

Regions

A **region** is a rectangular subpart of a gel that can be defined using the Region tool. When a region is defined, certain actions may be limited to it (Fig. 2-2).

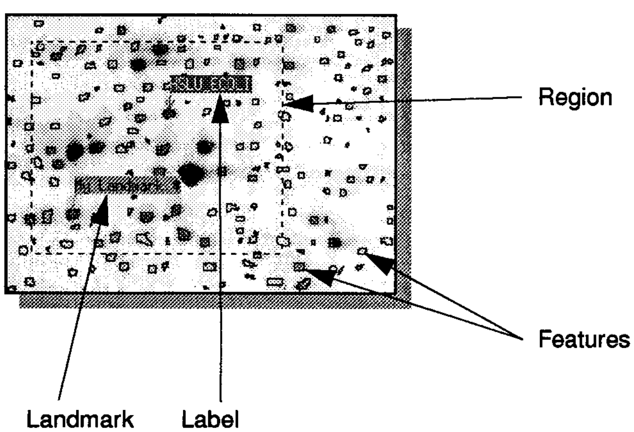


Figure 2-2. Feature shapes, a landmark and a label in a region

Landmarks

Individual points (pixels) in a gel may be marked by **landmarks**, using the Landmark tool. A landmark is defined by its position (row and column number) and its name. It may be assigned *pI* and *MW* values. *pI* and *MW* values of landmarks are used to compute an approximated pI and Mw value for any point in a gel. Landmarks may also be used as reference points for operations such as gel alignment and gel matching (Fig. 2-2).

Features

The elementary component of a gel is the **feature**. Features are spot shapes automatically detected by Melanie II's feature detection algorithm, or manually adjusted by the user (Fig. 2-2). Gels may be displayed with or without their feature shapes. Each feature in a gel has an associated **feature ID** (a unique sequential number) automatically given when it is created. Features may be quantified, that is their optical density (or instrument count for non optical images), area and volume are computed. Alternatively, features may be modeled by Melanie's Gaussian modeling algorithm. In this case their quantification values will be the bi-Gaussians peak values and volume, as well as the area of the Gaussian's half-height section.

Features represent the proteins from the original image. Features may automatically be paired between two gels, or groups of features may be formed across a series of gels. Quantification values of spot groups may then be displayed or printed as histograms, or exported as tables for use with external data analysis programs.

Labels

The **label** is a small tag associated to a feature and identifying the corresponding protein. The label holds the protein's accession number (AC) and name taken from a user selected database, or it may just contain any textual data. The label is the link to MelView's remote database query engine (Fig. 2-2).

Pairs and groups

Gels may be compared using Melanie's powerful automatic gel matching program. When two gels are matched the corresponding features are **paired**. Selected feature pairs may be displayed when gels are in tile mode. Alternatively, all pairs may be shown as small vectors when two gels are displayed in stack mode. Gels can be matched pair wise, or a set of gel images can be matched to a user chosen reference gel. A reference gel may be any 2-DE gel image, a master gel image or even a synthetic gel image. The pair is the link between corresponding features, that is the same protein in several gels.

When several gels have been matched to a given reference gel, the latter provides a unique feature numbering scheme across all gels. Each paired feature in a gel image may then be associated to the corresponding feature ID in the reference gel. The features in a set of gels that are paired to one given feature in the reference gel form a feature **group**. The group is the basic element for exporting data as tables, and for producing histograms of feature values across a series of gel images.

Comments

Textual data may be associated to individual gels in the form of comments. Comments may be displayed, printed or saved to file.

Selecting and showing objects

Objects may be either **shown**, **hidden** or **selected**. When an object is shown, it is visible on screen. Objects may also be selected using one of the tools from the left hand side vertical tool bar or using certain menu options (for example *Select → Features → All Features*). Once selected, an object is marked in green. The green color is always used to highlight selected objects. If an object is shown, that is, if it is visible, this does not mean it has been selected. As a general rule, actions from the menus may only be performed on selected objects. Objects may also be hidden, so that they are not visible on screen.

Mouse buttons

The mouse may be used in various manners, each having a specific behavior depending on the active tool (See *Tools* on page 2-5).

Mouse click

The most common way to use the mouse is to move it so as to position the cursor over a given object, then press the left mouse button, then quickly release it. In given situations, you may also click the middle or the right mouse button. Generally, the right mouse button is used to manipulate corresponding objects on several visible gels. The left mouse button being the most used one, we shall sometimes in this manual refer to it by just saying *the mouse button* or *click on the mouse*.

Dragging

You may also push down the left mouse button, and then, without releasing it, move the mouse in order to drag objects from one place to the other. In given situations, this is also possible using the middle or the right mouse buttons.

Tools

Objects on screen may be displayed, manipulated or processed using the various options from the MelView menus. Prior to choosing an option from the menu bar, individual objects have to be selected using the **tools** provided in the left hand side tool bar (Fig. 2-3).

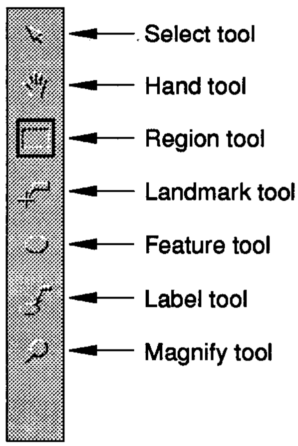


Figure 2-3. The MelView tool bar

The Select tool

The Select tool is used to select individual gels. To select one gel, pick the Select tool in the tool bar. The Select tool will be surrounded by a small black rectangle showing that it is active. Put the cursor in a gel and click on the left mouse button. The gel will be surrounded by a green line indicating that is has been selected. To select more than one gel, select the first one as described above, then hold the shift key on the keyboard and select additional gels by clicking on them.

This rule holds for any kind of objects in MelView: to select several objects of the same kind, pick the corresponding tool from the tool bar, select the first object by clicking on it with the left mouse button, then hold the shift key on the keyboard and select additional objects by clicking on them.

The Hand tool

The MelView window is often not large enough to display the whole gel images, and only a small part of the gels will be visible at one given time. The Hand tool moves gels in order to show other parts. Pick the Hand tool in the tool bar, put the cursor in a gel, hold down the left mouse button and drag the mouse. The gel image will follow your movement.

If instead you hold down the middle mouse button, then all selected gels will move together.

By holding down the right mouse button, you will position all gels according to the absolute cursor position in the pointed to gel. That is, the top left corner of each visible gel area will correspond to the same pixel (in absolute row and column position).

The Region tool

The Region tool is useful to select a rectangular subregion of a gel. Position the cursor at the top left position of the desired region, hold down the left mouse button, then drag the cursor to the bottom right position. A rectangle will be drawn around the region while selecting it. Release the mouse button at the desired point. You may move the region by dragging the rectangle's top left corner. You may also change its size by dragging its bottom right corner.

If gels have been aligned (see *Aligning gels* on page 7-1) then you may use the right mouse button to select the same region in all aligned gels simultaneously.

The Landmark tool

The Landmark tool is used to create or select landmarks at specific points on the gel image. To create a landmark, pick the Landmark tool, then double click the left mouse button at the position you want the landmark to be. To select an existing landmark, just click on it. To select more than one landmark, select the first one by clicking on it with the left mouse button, then hold the shift key on the keyboard and select additional landmarks by clicking on them. To select all landmarks in a given region, position the cursor at the top left position of the desired region, hold down the left mouse button, then drag the cursor to the bottom right position. All landmarks in the selected region will be selected and highlighted in green.

If gels have been aligned (see *Aligning gels* on page 7-1) then you may use the right mouse button to select landmarks in all aligned gels simultaneously. See detailed discussion on landmarks in *Landmarks* on page 4-1.

The Feature Tool

The Feature tool lets you select individual features. To select a feature, pick the Feature tool and click on the chosen feature. To select more than one feature, select the first one by clicking on it with the left mouse button, then hold the shift key on the keyboard and select additional features by clicking on them. To select all features in a given region, position the cursor at the top left position of the desired region, hold down the left mouse button, then drag the cursor to the bottom right position. All features in the selected region will be selected and highlighted in green. The selected region delimits an area of the gel in which given actions will be limited to.

If gels have been aligned (see *Aligning gels* on page 7-1) then you may use the right mouse button to select features in all aligned gels simultaneously. See detailed discussion on features in *Features (spots)* on page 5-1.

The Label tool

With the Label tool you may select labels. To select a label, pick the Label tool and click on the chosen label. To select more than one label, select the first one by clicking on it with the left mouse button, then hold the shift key on the keyboard and select additional labels by clicking on them. To select all labels in a given region, position the cursor at the top left position of the desired region, hold down the left mouse button, then drag the cursor to the bottom right position. All labels in the selected region will be selected and highlighted in green.

If gels have been aligned (see *Aligning gels* on page 7-1) then you may use the right mouse button to select labels in all aligned gels simultaneously. See detailed discussion on labels in *Labels* on page 6-1.

The Magnify tool

The Magnify tool lets you zoom in on the region of a gel around the cursor position. Just pick the Magnify tool, then hold down the left mouse button. Similar to a magnifying glass, you may then move the mouse to change the zoomed region.

If gels have been aligned (see *Aligning gels* on page 7-1) then you may use the right mouse button to magnify the same region in all aligned gels simultaneously.

Menus

Once objects have been selected you may choose actions from the menus to be performed on them. Move the cursor to the desired menu on the menu bar located at the top of the MelView window, hold down the left mouse button, then choose a menu option. Certain menu options have sub-options you may choose from by dragging the cursor over the small arrow on the right hand side of the option (see Fig. 2-4).

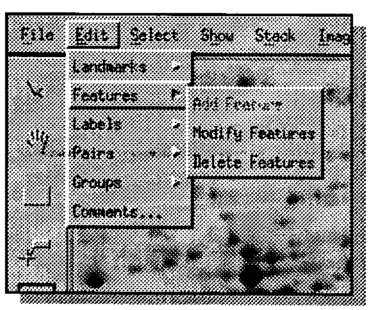


Figure 2-4. Choosing a sub-option from the menu

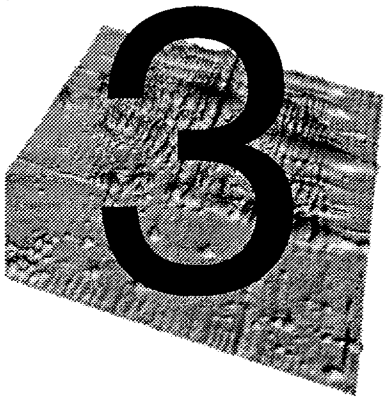
The following menus are provided in the MelView program (from left to right in the menu bar):

1 **File** - to handle whole gels and data on disk (opening and closing gel images, saving data, printing images, exporting data).
2 **Edit** - to edit (add, modify, delete) selected objects.
3 **Select** - to select specific objects.
4 **Show** - to show or hide specific objects, as well as displaying reports.
5 **Stack** - to handle stacked gels (two gels on top of each other).
6 **Image** - to create new gel images from selected ones (duplicating, cropping, filtering gels, creating synthetic or Gaussian gels, flipping and scaling gels, as well as erasing gels).
7 **Process** - to detect and quantify features, and to align and match gels.
8 **Database** - to handle master gels and to access remote 2-DE databases.
9 **Help** - to obtain on-line help.

At given points in time, depending of the status of the MelView window, certain menu options will not be available. In this case, they will appear faded in the menus.

Shortcuts

Several menu options may be activated by keyboard shortcuts. These are indicated at the right hand side of the corresponding menu option. For example, you may activate the *Open Gels* action by holding down the Control key and the O key simultaneously (lower or upper case). This is indicated next to the *Open* menu option by the `^O` sign. Similarly, you may exit MelView by hitting `^Q`.



BASIC PROCEDURES

This chapter presents basic MelView procedures such as opening, saving and closing gels, manipulating gel images (such as changing colors and zoom modes) and printing them.

On-line help

The manual is also available on-line in the MelView program. To consult it on-line, choose *Preferences* in the *File* menu. Set the *Melanie Home* directory to the directory in which the *man* directory resides, usually the same as the Melanie II programs. Then, in the *Help* menu, choose *Tutorial*. On-line user manual and index are available too. To access the on-line tutorial and user manual, you must have the Netscape Navigator® World-Wide Web browser installed.

You may choose the table of content document from which you can read any of the on-line manual chapters, or you can request the on-line index, then click on the desired concept to obtain the related on-line chapter. You may also directly refer to the on-line tutorial. Go to the *Help* menu and choose one of the available options.

If Netscape was running before MelView, and if the colors in Netscape do not look correct, quit Netscape and choose the *Help* option again.

Setting preferences

A few preferences may bet set and preserved between MelView sessions. In menu *File*, choose *Preferences*. You may set the following preferences:

1 **The Netscape Navigator® full pathname**. To use the on-line manual as well as the database options (see *Databases* on page 13-1) you must give the exact location of the Netscape program (for example: `/usr/local/bin/netscape`).

2 **The database server**. To use the database options (see *Databases* on page 13-1) you must specify the hostname of the World-Wide Web server on which the database resides (for example: **expasy.hcuge.ch**).

3 **The database name**. To use the database options (see *Databases* on page 13-1) you must specify the remote database (for example: **SWISS-2DPAGE**).

4 **The printer**. You have to give the printer name (such as for example: **lp**).

5 **The Melanie Home directory.** This is the directory in which the on-line manual files, the *man* directory resides, usually the same as the Melanie II programs.

Databases on page 13-1)*Databases* on page 13-1)In the Preferences window, click on the OK button. If preferences were already found on disk, MelView will ask you to confirm the new preferences.

Opening gels

Opening Melanie 2-DE images

Click in the *File* menu and select the *Open* option. The *Open* window will be displayed (Fig. 3-1). Hold down the Control key and, in the *Files* column, click

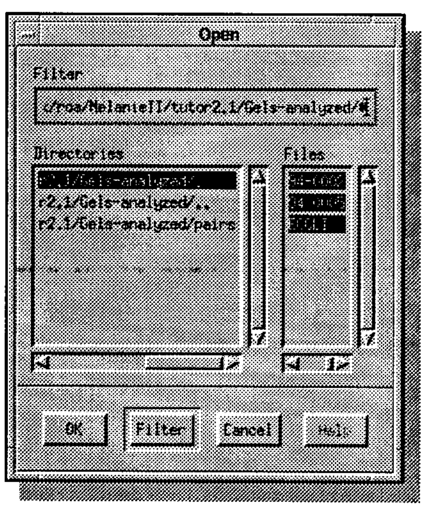


Figure 3-1. The Open window allows the selection of the gels that will be opened and loaded into MelView

on each of the gel file names you want to open. Click on the *OK* button. The gels will be opened.

The *Open* menu option allows you to open gel images stored on disk in the Melanie II release 2 or Melanie 1 format. Melanie 1 images will be converted to Melanie II release 2 file format. MelView will ask you for confirmation. You cannot open Melanie 1 images without converting them to the new format. If you want to keep the image in the old Melanie 1 format, use the *Open As* menu option (see next section).

Opening foreign image formats

MelView lets you import images stored in foreign image file formats, such as Elsie, Melanie 1, TIFF (8 or 16 bits), GIF or PPM. Choose *File → Open As*. In the *Open As* window that pops up, choose one of the available formats. Click *OK*. In the next pop up window, select the images you want to open following the above described procedure. Click on the *OK* button. Opening a Melanie II image in this way lets you open a copy of the gel image.

Selecting gels

To perform the actions described in the remaining part of this chapter, you must first select one or more gels. This may be achieved using either the Select tool or one of the *Gels* options in menu *Select*.

To select one gel, pick the Select tool in the tool bar. The Select tool will be surrounded by a small black rectangle showing that it is active. Put the cursor in a gel and click on the left mouse button. The gel will be surrounded by a green line indicating that is has been selected. To select more than one gel, select the first one as described above, then hold the shift key on the keyboard and select additional gels by clicking on them.

Alternatively you may select all visible gels in the MelView window by choosing *Select → Gels → Visible Gels*, or only the aligned gels by taking *Select → Gels → Aligned Gels* (See *Aligning gels* on page 7-1) or just the stacked gels by taking *Select → Gels → Stacked Gels* (See *Stacking gels* on page 3-11).

Showing gels / reporting on gels

Once you have opened a given number of gel images, you can decide which ones you want to display and which ones you prefer to hide at a given point in time. The *Gels* option in menu *Show* provides the following choices to this purpose:

- *Selected Gels* - displays all selected gels and hides the other ones.

- *Aligned Gels* - displays all aligned gels (See *Aligning gels* on page 7-1) and hides the other ones.
- *Stacked Gels* - displays all stacked gels (See *Stacking gels* on page 3-11) and hides the other ones.
- *All Gels* - displays all opened gels, including the currently hidden ones.
- *Show Comments* - displays the comments for the selected gel (See *Editing / showing comments* on page 11-4).
- *Report On Gels* - Displays a report summarizing information about the selected gels, such as image size in pixels, minimum and maximum grey levels, number of detected features (Fig. 3-2).

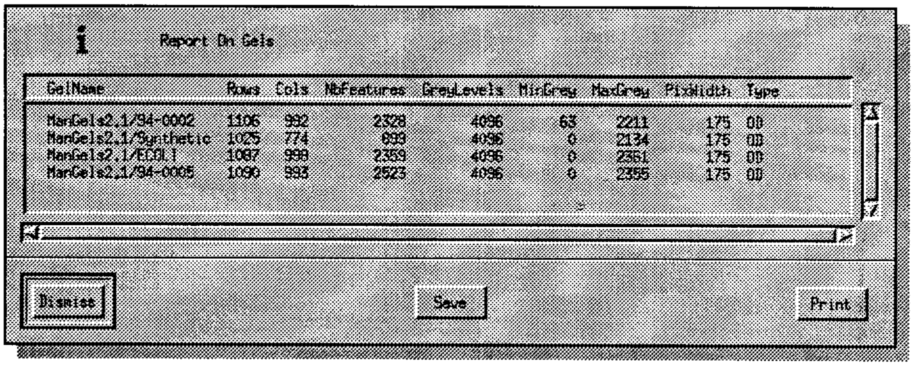


Figure 3-2. Report on gels

- *Report On Matches* - Displays information on matching between the selected gels and all open gels, that is the number of paired features between each pair of gels (See *Matching gels* on page 8-1).
- *Report All Matches* - Displays information on matching between all gels in a given directory (See *Matching gels* on page 8-1). A pop up window requests the directory name (default: current directory).

Moving gels

The size of the MelView window usually does not allow to display whole gels. At any point in time you may determine to move gel images in order to show other image parts. You may also change the way in which gels are arranged on screen.

Showing corners

The *Show Corner* option lets you choose to display the top left, top right, bottom left or bottom right part of the selected gels. Just take *Show* —> *Show Corner*, then choose corresponding sub-option.

Using the Hand tool

The Hand tool allows you to interactively move one or several gels. Pick the Hand tool in the tool bar, put the cursor in a gel, hold down the left mouse button and move the mouse. The gel image will follow your movement.

If instead you hold down the middle mouse button, then all selected gels will move together.

By holding down the right mouse button, you will position all gels according to the absolute cursor position in the pointed to gel. That is, the top left corner of each visible gel area will correspond to the same pixel (in absolute raw and column position).

Using the Select tool

You may change the order in which gel images are arranged on screen. Pick the Select tool, put the cursor in the gel you want to shift, then hold down the left mouse button and move the cursor over another gel. Release the mouse button. The two gels will be swapped.

Zooming

Another method to decide what part of a gel should be displayed is to change the zoom mode. You may zoom in to enlarge the image (you will see less of the image at once) or zoom out to see more of the gel image.

Selecting zoom mode

Select the gels you want to zoom, then take *Select → Select Zoom* and choose the zoom mode: 1 is the default mode and displays images in true size; 2 zooms in by a factor of 2; 1/2 and 1/4 zoom out by a factor of respectively 2 and 4.

Using the magnifying glass

An alternative way to enlarge parts of gels is to use the magnifying glass. Pick the Magnify tool, then hold down the left mouse button. Similar to a magnifying glass, you may then move the mouse to change the zoomed region (Fig. 3-3).

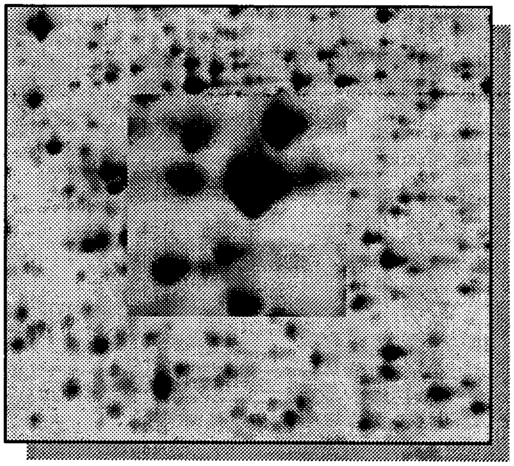


Figure 3-3. The magnifying glass

If gels have been aligned (see *Aligning gels* on page 7-1) then you may use the right mouse button to magnify the same region in all aligned gels simultaneously.

Changing colors

A 2-DE gel image is traditionally displayed as a grey level image, where grey levels represent the gel's optical density (respectively instrument counts for non optical images). You may opt to display images using pseudo colors. You may also change the way grey levels are displayed.

Adjusting colors

Modern scanners are usually able to scan 2-DE images with 12 or even 16 bits per pixel, that is with 4096 or 65536 grey levels, respectively. Because common screens are only able to display 256 colors or grey levels, mapping has to be determined between the 4096 or 65536 grey levels, respectively and the 256 screen grey levels. By default, MelView uses a linear mapping function, where the lightest point in the image is mapped to 0 (white) and the darkest point is mapped to 255 (black) (Fig. 3-4). You may though choose other mapping func-

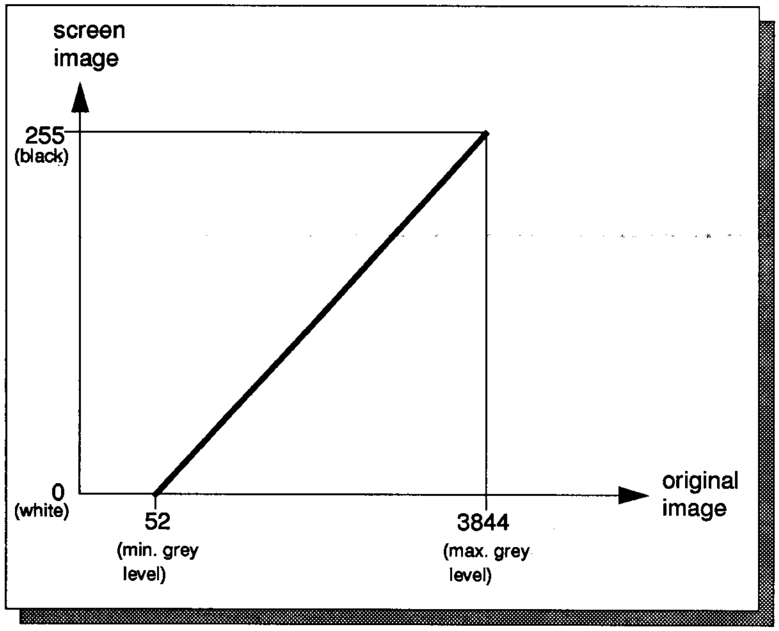


Figure 3-4. default linear mapping: minimum image grey level is mapped to white and maximum grey level is mapped to black.

tions, in order to highlight small light spots for example.

MelView lets you change the grey levels mapping in two different manners. You can define the minimum (0) and maximum grey levels (255), that is you may look for example at only the light or the dark regions of the images (Fig. 3-5). Or you can change the default linear mapping function to an exponential or logarithmic one (Fig. 3-6 and 3-7).

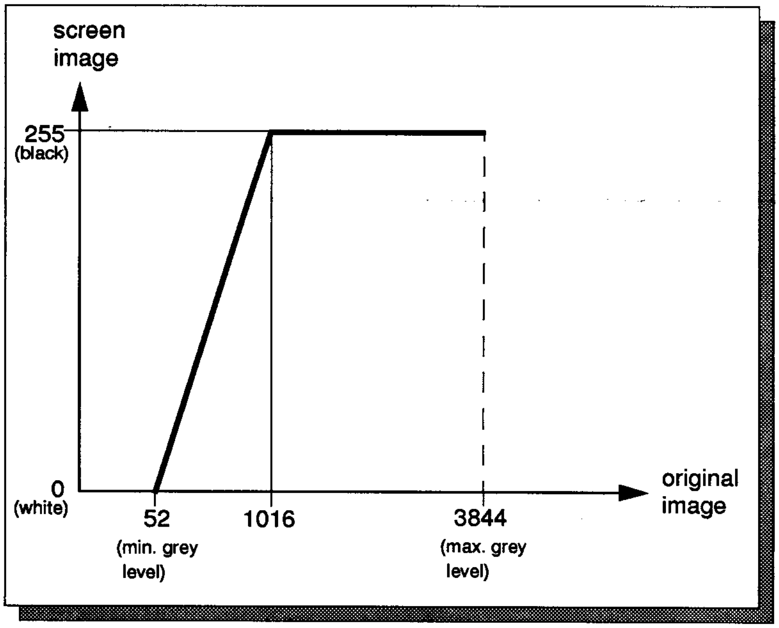


Figure 3-5. Mapping the light image pixels; grey levels 1016 to 3844 in the original image will appear black.

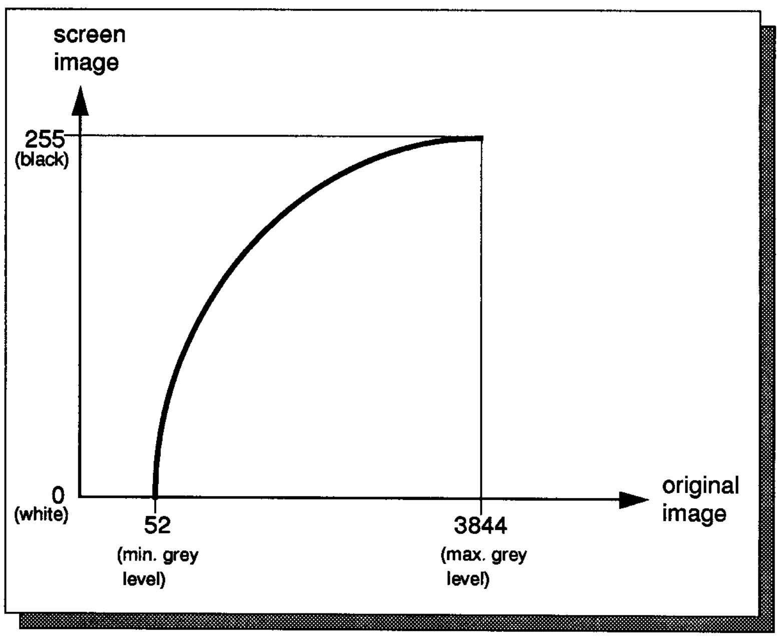


Figure 3-6. Using a logarithmic mapping function, light spots appear darker.

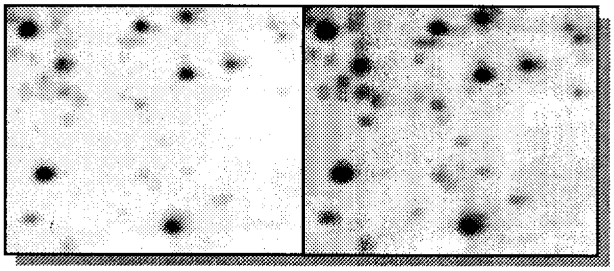


Figure 3-7. Region of a gel showing the color adjustment from Figure 3-6: the left image shows the original image; the right image shows the same gel area after mapping with a logarithmic function.

To change the mapping function, select some gels, then choose *Select → Select Mode → Adjust Colors*. Then, in the Adjust Colors window, change the minimum and maximum grey levels, as well as the mapping function. You may increase or decrease the curvature of the exponential or logarithmic function by adjusting the respective sliders. Hold down the left mouse button and move the slider, or click once on the slider and adjust it using the arrow keys. Click *OK*. Colors will be adjusted in all selected gels.

If you save gels after having performed color adjustments, the latter will be saved with the gel images (see *Saving gels* on page 3-12).

Adjusting colors in real-time

Each change of the mapping function in the Adjust Colors window may immediately be reflected in a given region, if such a region has been defined prior to adjusting colors. Select the Region tool and draw a small rectangle in one of the selected gels by putting the cursor at the top left point of the region, holding down the left mouse button and dragging the cursor to the bottom right point. Release the mouse button. The selected region will be delimited by a rectangle. In the *Select* menu, choose *Select Mode → Adjust Colors.* Each time you change one of the mapping parameters (the function or a slider), the change will immediately be reflected in the selected region (Fig. 3-8).

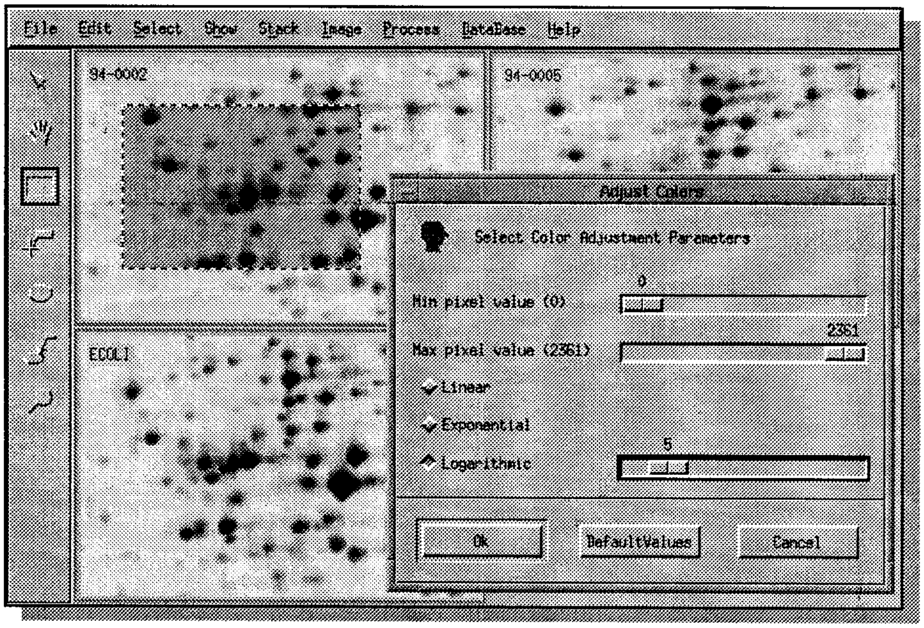


Figure 3-8. The *Adjust Colors* option. The modifications are directly reflected in the selected region.

Click on the *OK* button. The grey levels (or colors if you have set pseudo colors - see next section) will be adjusted in all selected gels.

Selecting pseudo colors

Choose *Select* → *Select Mode* → *Pseudo Colors*. Choose one of the color lookup tables, for example FiveRamps. Click OK. The lookup table will be changed accordingly (Fig. 3-9). To go back to grey levels, choose *Select → Select Mode → Grey Levels.*

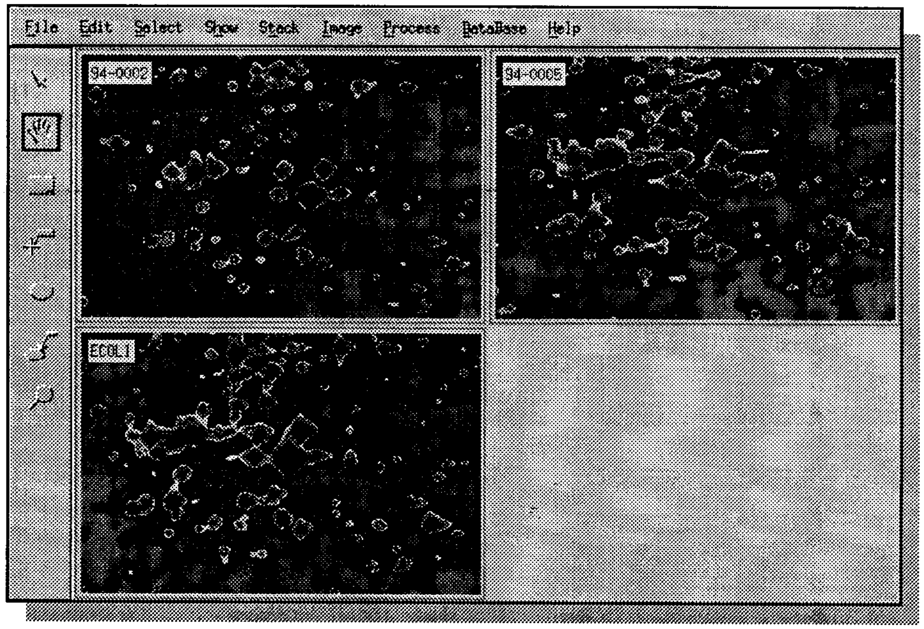


Figure 3-9. Gels displayed using the *FiveRamps* color lookup table.

Selecting grey levels

This is the default mode. To select this mode, choose *Select → Select Mode → Grey Levels.*

Stacking gels

In given situations it is helpful to stack two gels, that is to display one gel on top of the other. By switching from one gel to the other you may then better compare given gel areas. Select two gels, then *Stack → Stack Selected Gels.* The two gels will be displayed on top of each other. To switch from one gel to the other, choose *Stack → Front To Back*, or just Control-F. By pressing Control-F quickly several times, the differences between the two gels can be observed visually. To unstack the gels, choose *Stack → Unstack Selected Gels.*

Moving stacked gels

MelView provides the means to superimpose corresponding features of stacked gels in a simple way. Select the Hand tool. When two gels are stacked, you may use the left mouse button to move the front gel. Holding down the middle mouse button, both gels will be moved at the same time.

To superimpose two features, put the cursor on the given feature in the front gel, then hold down the left mouse button. Press Control-F to display the other gel. Move the cursor to the corresponding feature and release the mouse button. The two features will be superimposed. Verify this by pressing Control-F.

Printing gels

Two print modes are provided in MelView. You may print selected gels, one gel image per page, or print the whole MelView window. You first have to set the printer. Select *Files → Preferences*. Make sure to define an existing printer (example: lp). You may also set the printer from the *File → Print → Printer Setup* option.

Printing the MelView window

Select *File → Print → Print MelView Window.* The image of the whole MelView window is sent to the Postscript® printer.

Printing selected gels

Using the Select tool, select some gels. Then choose *Files → Print → Print Selected Gels.* The selected gels will be printed in full size.

Saving gels

Any changes carried out on gels have to be saved in order to be stored in the gel file. Otherwise the changes will be lost when exiting the MelView program or when closing the gels. Choose *File → Save → Save Changes* to save all changes performed on the selected gels, or any of the individual saving options. For example, *File → Save → Save Features* will only save the detected feature shapes, but not other changes.

If you save gels after having performed color adjustments, the latter will be saved with the gel images.

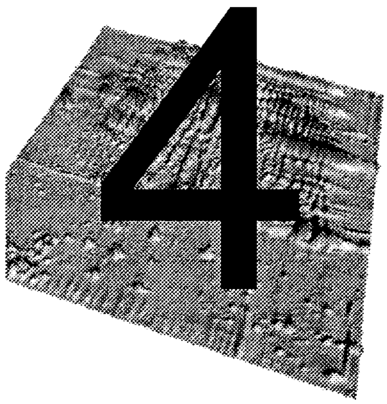
After performing changes on gels, you may revert to the saved status on disk by choosing one of the *Revert* options in menu *File.* Current changes will be lost. If objects have never been saved to disk for the selected gels, then the revert option is not active.

Alternatively, you may save any gel images into one of the supported foreign file formats. In this case, you can choose to save the gel image alone, or the gel image with any of its associated components such as features, labels or landmarks. But the saved file will contain a flat image, without any structure. For example, when saving a gel with its features to a TIFF file, this file will contain a flat image of the gel and its features. Choose *File* —> *Save As*, then select the objects you want to add to the image and click OK. The File Extension window will pop up. You can then enter the names of the new files manually, or use the *Replace Extension* or the *Add Extension* options. *Add Extension* lets you specify an extension to add to each new file name, while *Replace Extension* lets you replace the file name extensions of the selected gels with any other extension.

Closing gels and exiting MelView

Use *File* —> *Close* to close all selected gels. If any of the gels have not been saved since changes were performed, MelView lets you choose to save them before closing them.

To exit MelView, do *File* —> *Quit*. MelView lets you choose to save any unsaved gel images before exiting.



4 LANDMARKS

Individual points (pixels) in a gel image may be marked by a **landmark**. A landmark is defined by its position (row and column number) and its name. It may be assigned *pI* and *Mw* values. *pI* and *Mw* values of landmarks are used to compute an approximated *pI* and *Mw* value for any point in a gel. Landmarks may also be used as reference points for operations such as gel alignment and gel matching (see *Aligning gels* on page 7-1 and *Matching gels* on page 8-1).

Creating landmarks

To create a landmark, pick the Landmark tool and double click at the position in a gel where you want the landmark to be located. The Landmark window pops up. Enter a name. You may also enter the *pI* and/or *Mw* of the gel at the given position. A value of `-1` means that no value is set. The pop up windows also indicates the selected landmark position in row and column numbers. You can adjust these by clicking on one of the row or column sliders and moving them (hold down the left mouse button and drag the cursor). To assure fine position adjustment, click on one of the sliders, then adjust the position using the arrow keys. Click on *OK*. The landmark is created and its name and position are displayed on the gel.

If the landmark contains a *pI* or *Mw* value, then it's tag will contain an asterisk ('*') (Fig. 4-1).

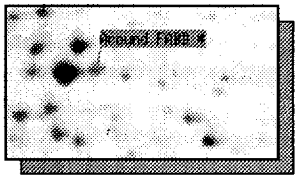


Figure 4-1. A landmark with pI/Mw (shown by the asterisk)

Alternatively, landmarks may be created with the *Edit → Landmarks → Add Landmarks* option. In this case the Landmark window pops up without prior landmark position. You then have to specify the landmark position using the sliders or the arrow keys.

For aligned gels (see *Aligning gels* on page 7-1) double click using the right mouse button, in order to create the same landmark at the same position in all selected aligned gels.

Selecting landmarks

Landmarks may be selected wit the Landmark tool. Click on any landmark to select it. The selected landmark will be highlighted in green. To select more than one landmark, select the first one by clicking on it with the left mouse button, then hold the shift key on the keyboard and select additional landmarks by clicking on them. To select all landmarks in a given region, position the cursor at the top left position of the desired region, hold down the left mouse button, then drag the cursor to the bottom right position. All landmarks in the selected region will be selected and highlighted in green.

Landmarks may also be selected from the *Select → Landmarks* sub-menu. Choose *Select → Landmarks → By Name* and enter a landmark name to select the corresponding landmark. You may use wildcards ('*') to select a group of landmarks. For example, "L*" designates all landmarks whose names start with the letter 'L'.

Select a region on a gel using the Region tool, then *Select → Landmarks → In Region* to select all landmarks in the selected region, or *Select → Landmarks → All Landmarks* to select all landmarks in all selected gels.

Finally, if some landmarks have already been selected, you have the possibility to do *Select → Landmarks → Inverse Selection.* This will unselect the selected landmarks and select all unselected ones.

Showing landmarks

Landmarks may be displayed with the *Show* → *Landmarks* → *Show Landmarks* menu option, and hidden with the *Show* → *Landmarks* → *Hide Landmarks* option. You may show only selected landmarks with *Show* → *Landmarks* → *Selected Landmarks*.

Modifying and deleting landmarks

In the *Edit* → *Landmarks* sub-menu, two options allow you to modify and delete existing landmarks. *Edit* → *Landmarks* → *Modify Landmarks* is available when one landmark is selected. This pops up the landmark window. You may then change the various landmark parameters. *Edit* → *Landmarks* → *Delete Landmarks* deletes all selected landmarks.

You can also manually change a landmark position. Select a landmark with the Landmark tool. Then move the cursor to the cross representing the landmark position. Hold down the mouse button and drag the landmark. Release the mouse button

To interactively change a landmark's tag position, select a landmark with the Landmark tool. Then move the cursor onto the landmark's tag. Hold down the mouse button and drag the tag. Release the mouse button (Fig. 4-2).

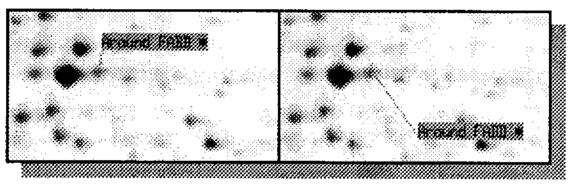


Figure 4-2. Default landmark tag position (left) and modified tag position (right).

Reports on landmarks

Select landmarks using the Landmark tool or the *Select → Landmarks* options. Choose *Show → Landmarks → Report On Landmarks*. A report will be displayed giving specific information on the selected landmarks, such as landmark names, positions and *pi/Mw*. The report may be saved or printed (Fig. 4-3).

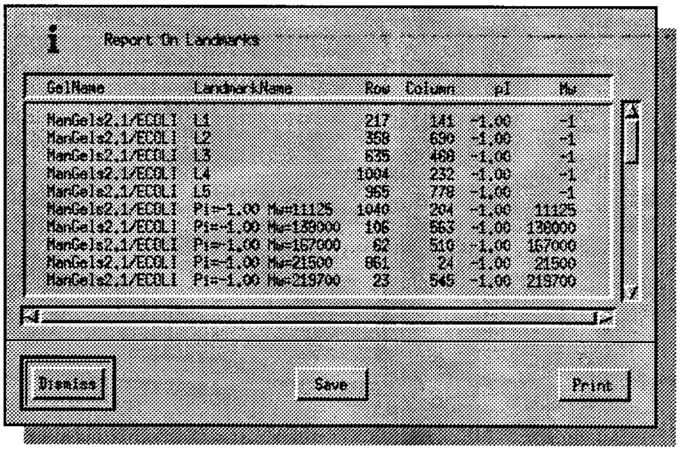


Figure 4-3. Report on landmarks

Showing *pI* and *Mw*

If you have defined a few landmarks in a gel and given *pI* and *Mw* values, you can display the interpolated *pI* and *Mw* values at any point in the gel. Take *Show → ID & pI/Mw*. This will display a small window holding the *pI* and *Mw* at cursor position. Moving the cursor will change the window content accordingly (Fig. 4-4).

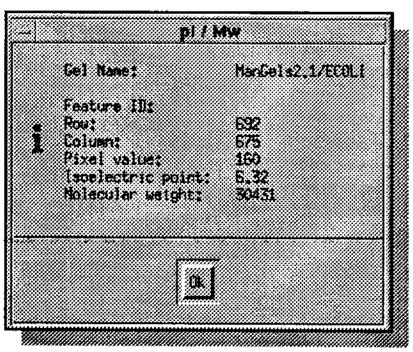
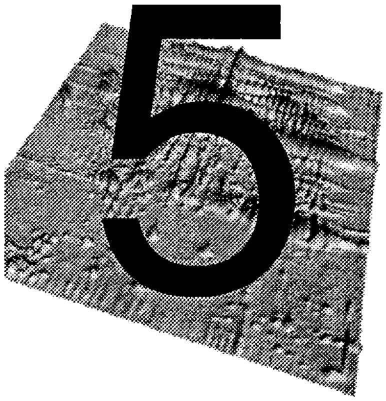


Figure 4-4. The ID & pI/Mw window

Additionally, this option also displays the feature ID, when the cursor is positioned over a detected feature (see *Features (spots)* on page 5-1).



FEATURES (SPOTS)

The elementary component of a gel is the **feature**. Features are spot shapes automatically detected by Melanie II's feature detection algorithm, or manually adjusted by the user. Each feature in a gel has an associated **feature ID** (a unique sequential number) automatically given when it is created. Features may be quantified, that is their optical density (or instrument count for non optical images), area and volume are computed. Alternatively features may be modeled by Melanie's Gaussian modeling algorithm. In this case their quantification values will be the bi-Gaussians peak values and volume, as well the area of the Gaussian's half-height section (see *Creating Gaussian gels* on page 12-5).

Detecting features

Select gels to detect features on, then take *Process → Detect Features*. The Detect Features window pops up. Various detection parameters are available and may be adjusted. The default parameters have been optimized for silver stained 2-D PAGE images run according to the SWISS-2DPAGE protocol [see reference 6]. You may tune the parameters to adapt the detection program to your images. The following parameters may be set (Fig. 5-1):

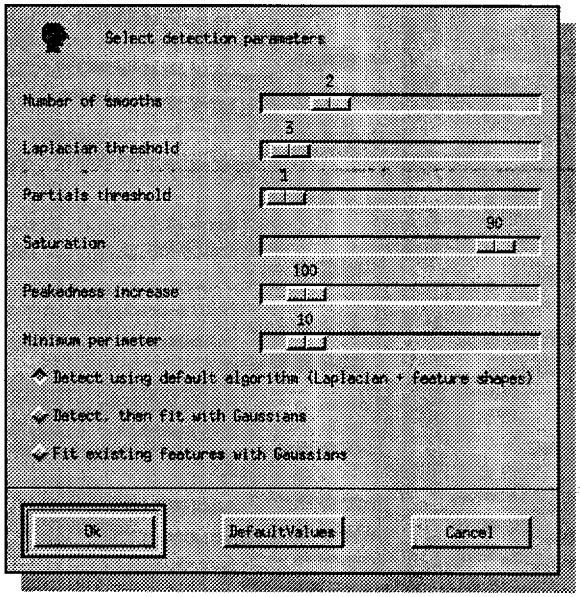


Figure 5-1. The feature detection parameters

- **Number of smooths on the images** - Before detecting features, MelView will smooth the image using a smooth by diffusion algorithm, in order to eliminate parasite noise. Select the number of times the smoothing operation should occur. Note that other smoothing algorithms are available in the *Image* menu (see *Filtering gels* on page 12-2). To avoid smoothing, set parameter to zero.
- **Threshold for Laplacian** - The detection algorithm looks for spot peaks by calculating the Laplacian for each pixel in the image. To detect as many features as possible, set a small value for this parameter. If you set high values, only very curved features will be detected and flat spots will be ignored.
- **Threshold for Partial derivatives** - Melanie also checks the spot curvature in both the X and Y axes, in order to separate features with flat sections, or to eliminate streaks. Augmenting this value increases the separation power of the detection algorithm.
- **OD (resp. counts) above which image is saturated (in percentage)** - Some very dark spots may be saturated (especially when digitized with older scanners) and may therefore be flat on the top. To insure that these features be detected, Melanie II artificially increases the peakedness (curvature) of all pixels whose OD (respectively counts) are above a given threshold. The threshold is expressed as a percentage of the maximum grey level in the image. A value of 90% is usually considered correct.
- **Increase in peakedness for saturated pixels** - This is an internal value expressing how much Melanie increases the curvature of saturated pixels. 100 is the default value and should only be changed with care.

- **Minimum feature perimeter** - With this parameter you may decide to ignore features that are too small. Melanie II will ignore all features whose perimeters (in pixels) are smaller than the given threshold.

  An additional criterion affects the way features are represented. You have to choose between the following three options:

1. **Detect using default algorithm** - This is the default. Features are detected using the Laplacian, and feature shapes are represented that match the original spot shapes.
2. **Detect, then fit with Gaussians** - Features are detected with the Laplacian, then Gaussian modeling is performed on all detected features. Features are represented by bi-Gaussian functions.
3. **Fit existing features with Gaussians** - Same as 2, but modeling is performed on already detected features.

Click *OK*. Features (spots) will be detected according to the parameter values you have set. After detection the feature shapes will be displayed over the gels (Fig. 5-2 and 5-3). Try first to detect features using the default parameter values, then tune them for your data.

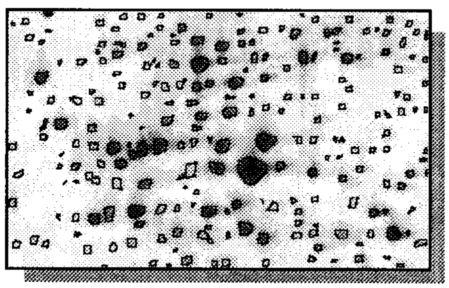


Figure 5-2. Detected feature shapes

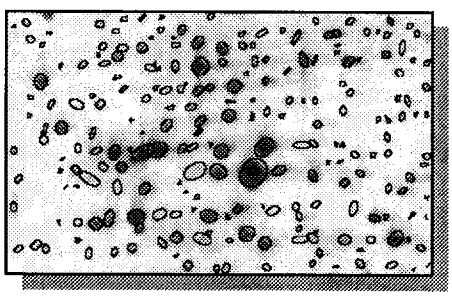


Figure 5-3. Features modeled as Gaussians

Adjusting detection parameters in real-time

Each change of a parameter value may immediately be reflected in a given region, if such a region has been defined. Select the Region tool and draw a small rectangle in one of the selected gels, using the mouse and the left button. In the *Process* menu, choose *Detect Features.* Each time you change one of the parameters, the change will immediately be reflected in the selected region (Fig. 5-4).

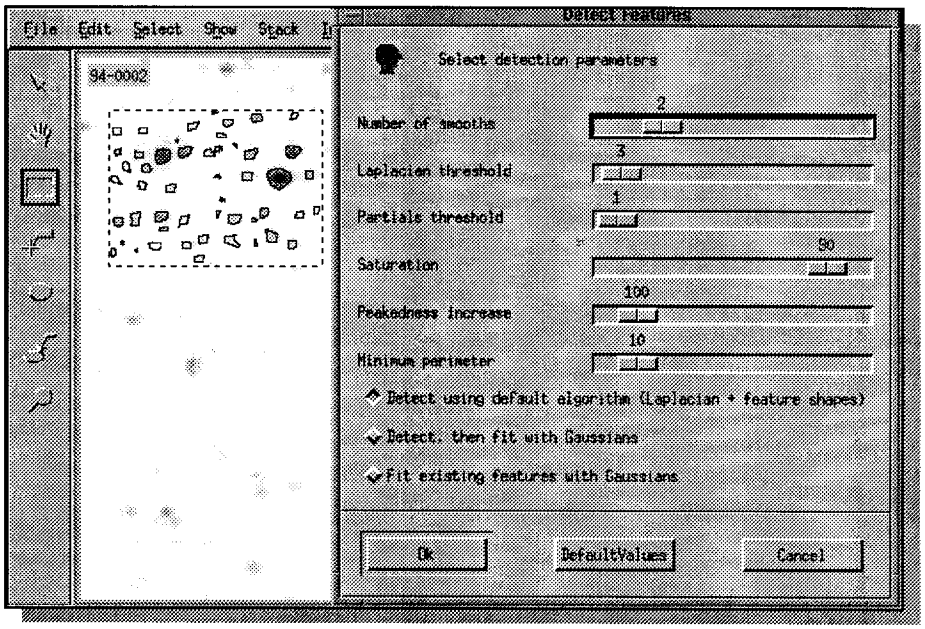


Figure 5-4. Adjusting detection parameters in real time.

Click on the *OK* button. The features will be detected in all selected gels using the parameter values you have set.

Quantifying features

Choose *Process* —> *Quantify Features* to compute the feature values. Depending on the scanner that was used to digitize the images, the following quantification values are calculated for each detected feature:

Images digitized by transparence

The pixel value is the optical density:

- **OD** - the highest optical density in the feature.

- **AREA** - the feature's area.
- **VOL** - the feature's volume, that is the integration of OD over the feature's area.
- **%OD** - the relative OD, that is the OD divided by the total OD over the whole image.
- **%VOL** - the relative VOL, that is the VOL divided by the total VOL over the whole image.

Images digitized by transmittance

The pixel value is transmittance. A conversion table is used during quantification in order to convert transmittance values to optical density:

- **OD** - the highest optical density in the feature.
- **AREA** - the feature's area.
- **VOL** - the feature's volume, that is the integration of OD over the feature's area.
- **%OD** - the relative OD, that is the OD divided by the total OD over the whole image.
- **%VOL** - the relative VOL, that is the VOL divided by the total VOL over the whole image.

Non optical images

The pixel value is instrument counts:

- **Count** - the highest instrument count in the feature.
- **AREA** - the feature's area.
- **VOL** - the feature's volume, that is the integration of Count over the feature's area.
- **%Count** - the relative Count, that is the Count divided by the total Count over the whole image.
- **%VOL** - the relative VOL, that is the VOL divided by the total VOL over the whole image.

Gaussian features

If features have been modeled as Gaussians, the following quantification values are computed:

- **OD/Count** - the optical density (respectively count) of the Gaussian's peak.
- **AREA** - the area of a section of the Gaussian, taken at the peak's half height.
- **VOL** - the Gaussian's volume.
- **%OD** - the relative OD, that is the OD divided by the total OD over the whole image.
- **%VOL** - the relative VOL, that is the VOL divided by the total VOL over the whole image.

Selecting features

Features may be selected with the Features tool. Click on any feature to select it. The selected feature will be highlighted in green. To select more than one feature, select the first one by clicking on it with the left mouse button, then hold the shift key on the keyboard and select additional features by clicking on them. To select all features in a given region, position the cursor at the top left position of the desired region, hold down the left mouse button, then drag the cursor to the bottom right position. All features in the selected region will be selected and highlighted in green.

Features may also be selected from the *Select → Features* sub-menu. Choose *Select → Features → By ID* and enter a feature ID to select the corresponding feature. Choose *Select → Features → By Master ID* and enter a feature ID from the master gel to select the corresponding features (see *Master gels* on page 13-2). Choose *Select → Features → By Group ID* and enter a group ID to select the corresponding features (see *Groups* on page 9-1). Choose *Select → Features → Only Gaussians* to only select features that have been modeled by a Gaussian function.

Select a region on a gel using the Region tool, then *Select → Features → In Region* to select all features in the selected region, or *Select → Features → All Features* to select all features in all selected gels. If certain labels are selected you may also choose *Select → Features → For Labels*. This will select all features associated to currently selected labels (see *Labels* on page 6-1).

Finally, if some features have already been selected, you have the possibility to do *Select → Features → Inverse Selection*. This will unselect the selected features and select all unselected ones.

Showing features

Features may be displayed with the *Show → Features → Show Features Shapes* menu option, and hidden with the *Show → Features → Hide Features Shapes* option. You may show only selected features with *Show → Features → Selected Features*.

You may also show (respectively hide) the ID of selected features. Select some features, then *Show → Features → Show Features ID* (respectively *Show → Features → Hide Features ID).*

Hiding features also has the side effect of displaying images faster.

Reports on features

Select features using the Feature tool or the *Select → Features* options. Choose *Show → Features → Report On Features.* A report will be displayed giving specific information on the selected features, such as features ID, quantification values, *pI* and *MW*, as well as the following data from the master gel, provided the selected gels have been matched to a master gel (see *Matching gels* on page 8-1 and *Master gels* on page 13-2) : serial number, *pI*, *MW* and accession number (M:S/N, M:pI, M:Mw and M:AC, respectively). The report may be saved or printed (Fig. 5-5).

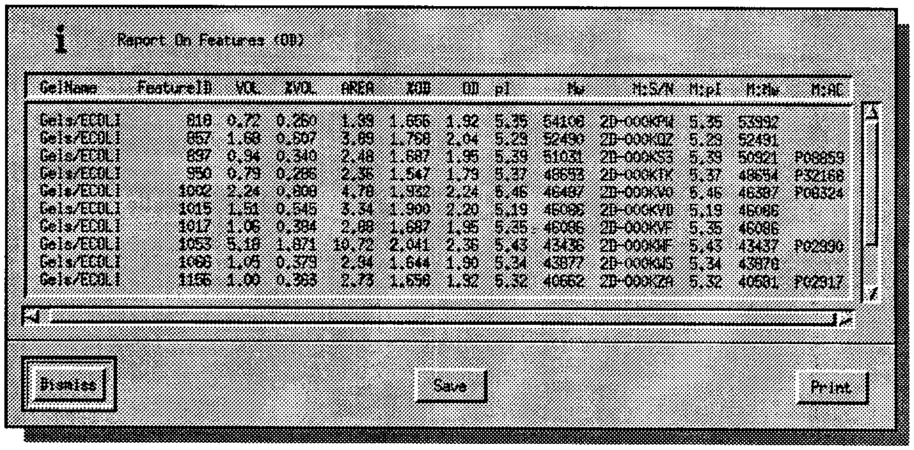


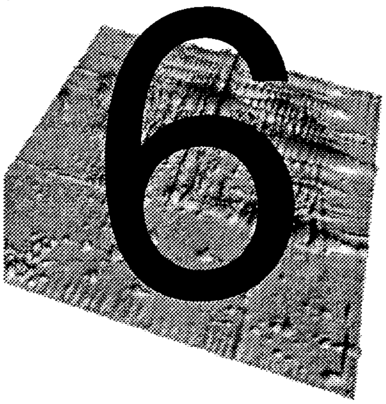
Figure 5-5. Report on features

Selecting / unselecting all

The *Select → Select All* menu option lets you select all defined objects from selected gels (including landmarks, features and labels). The *Select → Unselect All* option lets you unselect all objects from selected gels.

Showing / hiding all

The *Show → Show All* menu option lets you display all defined objects from selected gels (including landmarks, features and labels). The *Show → Hide All* option lets you hide all objects from selected gels. Hiding objects also has the side effect of displaying images faster.



LABELS

The **label** is a small tag associated to a feature and identifying the corresponding protein. The label holds the protein's accession number (AC) and name taken from a user selected database, or it may just contain any textual data. The label is the link to MelView's remote database query engine.

Note that the AC is the label's identification. Two labels with the same AC are considered identical, while the label names may differ.

Also, as labels are associated to features, you may define labels only after having detected features (See *Features (spots)* on page 5-1).

Creating labels

To create a label, pick the Label tool and double click on a feature you wish to associate a label to. The Label window pops up. Fill in the AC (accession number) and name fields. Click on *OK*. The label is created and its AC or name is displayed on the gel, depending on the *Show→ Labels* options previously chosen (Fig. 6-1).

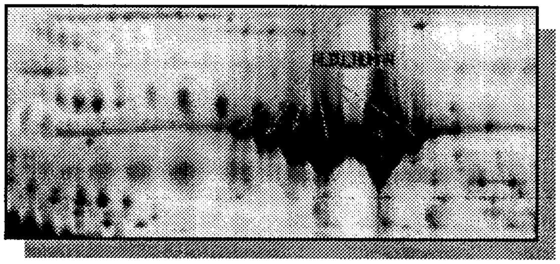


Figure 6-1. A label for Human *albumin*

Alternatively, labels may be created with the *Edit → Labels → Add Labels* option. Select one or more features using the Feature tool, then choose *Edit → Labels → Add Labels*. The label will be added to all selected features.

Selecting labels

Labels may be selected wit the Label tool. Click on any label to select it. The selected label will be highlighted in green. To select more than one label, select the first one by clicking on it with the left mouse button, then hold the shift key on the keyboard and select additional labels by clicking on them. To select all labels in a given region, position the cursor at the top left position of the desired region, hold down the left mouse button, then drag the cursor to the bottom right position. All labels in the selected region will be selected and highlighted in green.

Labels may also be selected from the *Select → Labels* sub-menu. Choose *Select → Labels → By AC* and enter a label accession number to select the corresponding labels. You may use wildcards ("*') to select a group of labels. For example, "P12*" designates all labels whose names start with the letters "P12".

Choose *Select → Labels → By Name* and enter a label name to select the corresponding labels. You may also use wildcards to select a group of labels.

Select a region on a gel using the Region tool, then *Select → Labels → In Region* to select all labels in the selected region, or *Select → Labels → All Labels* to select all labels in all selected gels.

If you selected features before selecting labels, you may take *Select → Labels → For Features* and all the labels associated to the selected features will be selected.

Finally, if some labels have already been selected, you have the possibility to do *Select → Labels → Inverse Selection*. This will unselect the selected labels and select all unselected ones.

Showing labels

Labels may be displayed with the *Show → Labels → Show Labels* menu option, and hidden with the *Show → Labels → Hide Labels* option. If several features hold the same label AC (accession number), then, by default, each feature will be displayed with its own label. You may display only one label tag for all identical labels by choosing *Show → Labels → Pack Labels. Show → Labels → Unpack Labels* separates the label tags again.

You can either display the label names or the label ACs. The option *Show → Labels → Show AC* and *Show → Labels → Show Name* may be used to toggle the AC/name display.

You may show only selected labels with *Show → Labels → Selected Labels.*

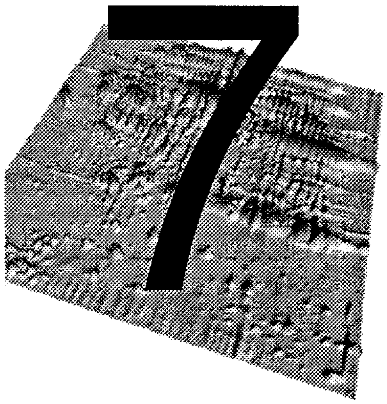
Modifying and deleting labels

In the *Edit → Labels* sub-menu, two options allow you to modify and delete existing labels. *Edit → Labels → Modify Labels* is available when one label is selected. This pops up the label window. You may then change the AC or name. *Edit → Labels → Delete Labels* deletes all selected labels.

You can manually change a label's tag position. Select a label with the Label tool. Then move the cursor onto the label's tag. Hold down the mouse button and drag the tag. Release the mouse button.

PART II

Aligning and Matching



7 ALIGNING GELS

Introduction

The MelView program has been especially designed to work with raw 2-DE images, that is without the need for image preprocessing prior to 2-DE image analysis. Nevertheless several image enhancement facilities are provided that may be useful when working with low quality 2-DE images (see *Filtering gels* on page 12-2).

Gel alignment is a very useful technique for comparing 2-DE images that present too many variations in protein migration. In general, while all Melanie II programs may work with unaligned gel images, alignment might prove helpful when comparing stacked gels, or for matching distorted 2-DE gel images.

You may align two or more gels at once. You must select a reference gel, that is an image the other gels will be aligned to. At least four landmarks (for the default alignment algorithm) must have been defined and must be well distributed over the gel image. The other gels must contain the same landmarks, as the alignment algorithm will try to deform the images in order to best superimpose the landmarks.

Alternatively, if you align gels that have already been matched, you may then choose to use existing pairs as reference points instead of landmarks (see *Matching gels* on page 8-1).

Creating corresponding landmarks

The first step consists in verifying that all gels you are going to align contain at least four landmarks, and that they represent the same points in all gel images. The easiest way to define these reference landmarks is probably to set landmarks in the center of a few spots corresponding to known proteins, and to name the landmarks accordingly.

Only landmarks with identical names in all selected gels will be used for the alignment. This means you do not need to delete existing landmarks, but you can just add new ones. If you already have corresponding landmarks in your gels, you may go to the next section. Otherwise, follow on with the current section.

Decide which gel will serve as the reference gel. Choose four to seven points in this gel and set landmarks on it (see *Creating landmarks* on page 4-1). Select the reference gel and one of the other gels. Stack them (*Stack* → *Stack Selected Gels* or Control-T). Put the reference gel in front (Control-F). Pick the Hand tool. Put the cursor on top of one landmark. Hold down the left mouse button, press Control-F and move the cursor to the corresponding point in the other gel. Release the mouse button. The corresponding point in the second gel should now lie exactly under the landmark. Press Control-F twice to verify this. If it is the case, select the Landmark tool and double click on the point in the second gel. This will create a new landmark. In the Landmark window, enter the same landmark name as in the reference gel. Click *OK*. You do not need to specify *pI* or *Mw* values. Repeat this step for all other landmarks.

Specify landmarks in the other gels following the steps described above. Save the landmarks with *File* → *Save* → *Save Landmarks*.

Aligning gels

You may now align the gels. Select all gels that have to be aligned. Take *Process* → *Align Gels* → *Align Gels*. The selected gels will be aligned according to the reference gel, so that corresponding landmarks are best superimposed. The reference gel is considered to be aligned with itself.

Operations on aligned gels

Most operations work on aligned gels in the same way they do on unaligned gel images. Two types of functions in the MelView program have a different behavior with aligned gels:

1 **Matching** - If the alignment has been performed correctly, then the gel matching program works faster and produces better results, because gel image distortion has partially been eliminated (see *Matching gels* on page 8-1).

2 **Tools** - Most tools that perform actions on one gel using the left mouse button may now be used to perform the actions synchronously on all aligned gels using the right mouse button:

- Hand tool - Select the Hand tool. Hold down the right mouse button, and move the cursor. The aligned gels will move synchronously. If you use the middle button, then only the selected gels will move.
- Region tool - Select the Region tool. Hold down the right mouse button, and drag the cursor. The same region will be selected in all aligned gels (Fig. 7-1).

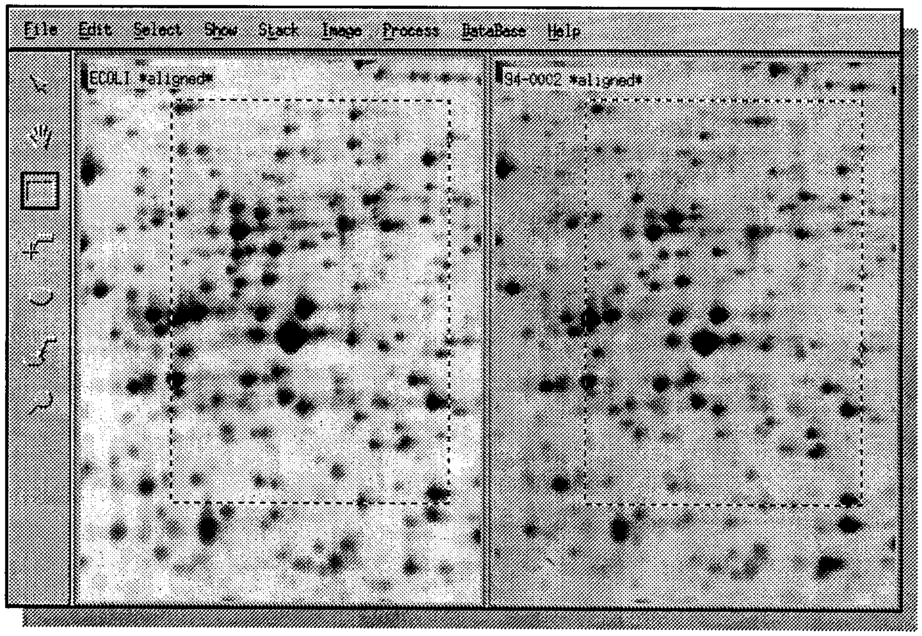


Figure 7-1. Synchronized regions in aligned gels

- Landmark tool - Select the Landmark tool. If you select landmarks using the right mouse button, then landmarks located at the exact same positions in all aligned gels will be selected at once. You may also select landmarks by holding down the right mouse button, and dragging the cursor to select all landmarks in the given region. Landmarks in the corresponding region in all aligned gels will be selected.
- Feature tool - Select the Feature tool. If you select features using the right mouse button, then features located at the exact same positions in all aligned gels will be selected at once. You may also select features by holding down the right mouse button, and dragging the cursor to select all features in the given region. Features in the corresponding region in all aligned gels will be selected.
- Label tool - Select the Label tool. If you select labels using the right mouse button, then labels located at the exact same positions in all aligned gels will be selected at once. You may also select labels by

holding down the right mouse button, and dragging the cursor to select all labels in the given region. Labels in the corresponding region in all aligned gels will be selected.

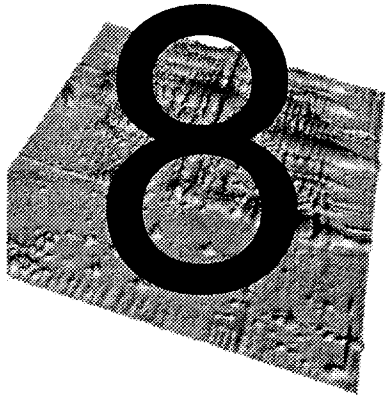
- Magnify tool - Select the Magnify tool. If you magnify an area of a gel using the right mouse button, then the same area in all aligned gels will be magnified. You may then move the magnifying glass simultaneously in all aligned gels.

Unaligning gels

To get back to the original images, select aligned gels, than take *Process* → *Align Gels* → *Unalign Gels*. The original images will be displayed again in place of the aligned ones.

Special alignments

By default gels are aligned according to the X and Y image axes, using a second degree alignment function in both the X and Y axes. You may decide to align images according to only the X or Y axis, or to use another alignment function. Choose *Process* → *Align Gels* → *Other alignments* and select the desired option. First, second and third degree functions are available, that request at least two, three and respectively four landmarks.



MATCHING GELS

Introduction

Gel matching is a key operation in 2-DE gel image analysis. The basic gel matching algorithm consists of comparing two gel images and finding **pairs** of related features, that is features depicting the same protein in both gels. A pair is represented by a doublet *(f_1, f_2)* where f_1 is a feature ID in the first gel and f_2 a feature ID in the other gel. One generally says that gels have been **matched**, and that individual features have been **paired**.

MelView can match two or more gels. Matching two gels means finding all the pairs between features of the two gels. Matching several gels means picking out a **reference gel**, then successively matching each gel to the reference gel. In this way, features in all gels may be compared with features in the reference gel. For consistency, when matching only two gels, MelView will also ask you to specify which one is the reference gel.

All features in selected gels that are paired with the same feature in the reference gel form so-called **groups**. The concept of group is essential to analyze features in a set of gels. Elaborate reports and histograms may be produced from them (see *Groups* on page 9-1).

In MelView you may accept **single pairs** or **multiple pairs**. Matching with single pairs means that one feature in a gel will be paired to exactly one feature in the other gel. Matching with multiple pairs means allowing that one feature in one gel be paired to several features in the other gel (Fig. 8-1).

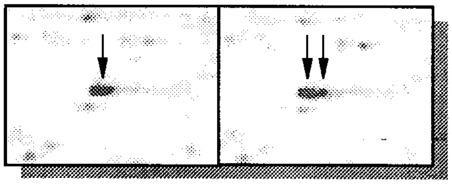


Figure 8-1. One feature paired to two features in the other gel

Matching aligned or unaligned gels

The only prerequisite for matching two gels is that features have already been detected and quantified. Aligning gels prior to matching is optional, but in many cases the matching process will be faster and will produce better and more accurate results, especially with distorted gel images (see *Aligning gels* on page 7-1).

Automatic matching

To match two or more gels, select them using the Select tool or one of the *Select → Gels* menu options. Choose *Process → Match → Match Gels*. MelView will first request that you designate the reference gel among the selected gels. Pick one of the gels, then click *OK*. You will then be asked whether you allow multiple pairs. If you accept multiple pairs, press *OK*. If you want only single pairs, then click *No*. All gels will then be matched to the reference gel. Depending on the computer you are running MelView on, this might take time.

Do not forget to save the pairs (*File → Save → Save Pairs*, or *File → Save → Save Changes*, or just Control-S). Saving pairs creates a pairs file that resides in the same directory as the matched gels. Therefore, only gels that reside in the same directory may be matched.

Creating starting pairs

MelView has been designed to automatically match gels. Nevertheless you may speed and improve the matching process by specifying a few starting pairs. There are three manners to define starting pairs: you may manually add pairs, or you can automatically pair identical landmarks or labels.

You may manually add a pair either in stack mode or in tile mode.

Adding / deleting pairs in tile mode

Select two gels using the Select tool, then select one feature in one gel and one or more features in the other gel using the Feature tool (click on one feature, then hold down the shift key and click on the other ones). Choose *Edit → Pairs → Add Pairs* (or Control-A).

To delete pairs, select paired features, then *Edit → Pairs → Delete Pairs* (or Control-D).

Adding / deleting pairs in stack mode

Select two gels using the Select tool and bring them into stack mode (Control-T). Using the Hand tool, put the cursor on the feature you want to pair in the front gel, then hold down the left mouse button and bring the other gel to the front (Control-F). Move the cursor to the second feature and release the mouse button. The two features should be superimposed. Press Control-A. MelView will pair the two features situated exactly under the cursor. You may also add pairs in the same way as in tile mode, by first selecting them.

To delete pairs, select pairs using one of the *Select → Pairs* options, then press Control-D.

Adding groups

A very convenient way to add pairs between a set of gels and a reference gel is to create groups (see *Groups* on page 9-1 and *Editing groups* on page 11-3). You first have to set a master (See *Master gels* on page 13-2). Choose *Data-Base → Set → Master* and choose your reference gel as the master. Then select one feature in the reference gel and features in the other selected gels. Choose *Edit → Groups → Group Features*. The selected features will all be paired to the selected feature in the reference gel.

Using landmarks as starting pairs

If two or more gels have landmarks with identical names, then you can use these to create starting pairs. Select the gels, then *Process → Match → Match Landmarks*. MelView will ask you to point to the reference gel. It will then, for each feature in the reference gel that has got a landmark defined in it, pair it with features in the other gels that have got the same landmark defined in them. These newly created pairs may then be used as starting pairs for matching.

Note that only landmarks may be used that are located inside of features.

Using labels as starting pairs

If two or more gels have labels with identical accession numbers (AC), then you can use these to create starting pairs. Select the gels, then *Process* —> *Match* —> *Match Labels*. MelView will ask you to point to the reference gel. It will then pair each feature in the reference gel that is associated to a label, to features in the other gels that are coupled to labels with the same AC. These newly created pairs may then be used as starting pairs for matching.

Showing pairs

You can look at the matching results in stack mode. When two selected gels are in stack mode, choose *Show* —> *Pairs* —> *Show Pairs*. The pairs between the two staked gels will be highlighted in the form of blue vectors linking the locations of paired features. The head of the vector points to the feature in the front gel (Fig. 8-2). To hide vectors, choose *Show* —> *Pairs* —> *Hide Pairs*.

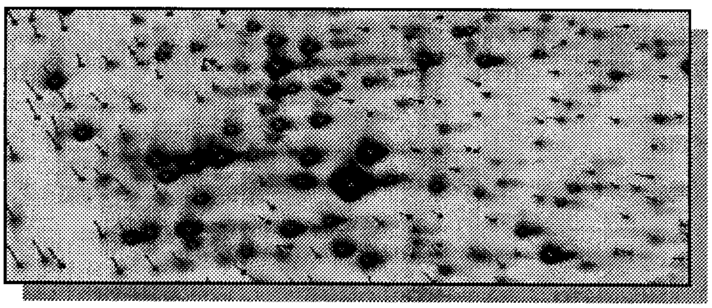


Figure 8-2. Matched gels: blue vectors indicate paired features.

Selecting pairs

To observe paired features when gels are in tile mode, you have to select the pairs. First select matched gels. You then have a number of possibilities. To select all pairs among the selected gels, choose *Select* —> *Pairs* —> *All Pairs*. All paired features will be selected.

You can highlight only pairs for a given set of features. Using the Feature tool, select a number of features in one of the gels. Choose *Select* —> *Pairs* —> *For Features*. Features in selected gels that have been paired to the selected features will be highlighted. If you have selected features in the reference gel, then all pairs for the given features will be selected in the other gels (Fig. 8-3). To

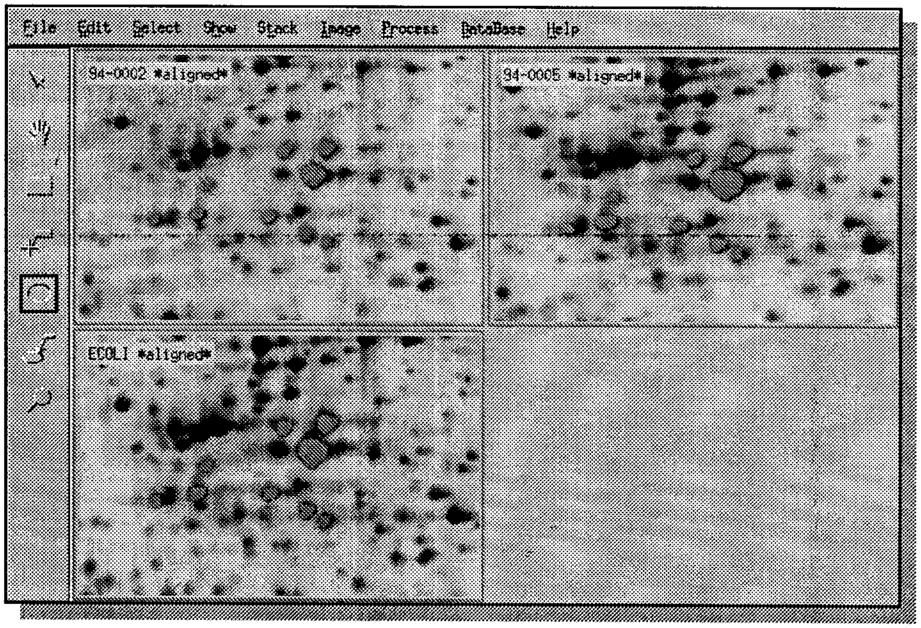


Figure 8-3. Selected paired features across three gels.

observe pairs in a specific region, select a region in one gel using the Region tool, then *Select → Pairs → In Region.* To select only multiple pairs, excluding single pairs, take *Select → Pairs → Multiple Pairs.*

An algorithm to predict the accuracy of pairs has been implemented in MelView. It is based on the length of the vectors between paired features. You may select only good pairs, or only bad pairs. Choose *Select → Pairs → Good Pairs,* respectively *Select → Pairs → Bad Pairs.*

Finally, if you have selected a number of pairs by one of the above described methods, and if your gels are in stack mode, you may display the selected pairs with *Show → Pairs → Selected Pairs.*

Reports on pairs

MelView can display, print or save a report on pairs between two matched gels. Select two matched gels and a number of pairs, using one of the selection methods described above. Then take *Show → Pairs → Report On Pairs.* You will get the list of all paired feature IDs (identifications) between both gels.

Another way to display the matching result is to select two or more matched gels, then *Show → Gels → Report On Matches.* This will display the number of pairs between all the selected gels and any open gels.

Finally, you may choose *Show → Gels → Report All Matches*. This displays information on matching between all gels in a given directory. A pop up window will request the directory name (default: current directory).

PART III

ADVANCED FEATURES

9 GROUPS

Introduction

The **feature group** is an essential concept in the Melanie II software. When several gels have been matched to a given reference gel, the latter provides a unique numbering scheme for features across all gels. Each paired feature in a gel image may then be associated to the corresponding feature ID in the reference gel. The set of features that are paired to one given feature in the reference gel forms a feature group. The **group ID** is the ID of the corresponding feature in the reference gel.

The group is the basic element for exporting data as tables (see *Exporting data* on page 10-1) and for producing reports and histograms of feature values across a series of gel images.

Groups are automatically created when matching a set of gels to a reference gel, but they may also be created and modified manually (see *Editing groups* on page 11-3).

If you have matched your gels with multiple pairs, you may then obtain very large groups, because a given feature may be paired to several features in the reference gel. You should therefore use multiple pairs with care.

Selecting groups

The two main reasons to consider groups are to produce reports or histograms, and to export data to external programs. In both cases you first have to select the groups of interest. To select and display all groups, choose *Select → Groups → All Groups*. To select one specific group, take *Select → Groups → By Group ID* and specify the group ID when requested.

To select the groups one or more features belong to, select features using for example the Feature tool, then *Select → Groups → For Features*. To get all features in a given region, define a region with the Region tool and choose *Select → Groups → In Region*.

Reports on groups

Once you have selected a given number of groups (possibly all), you may display a report on them. Choose *Show → Groups → Report On Groups*. MelView will ask you to pick out the reference gel. If a master gel has already been set, the latter is taken as the reference gel (see *Master gels* on page 13-2). You then have to select one of the feature value types for the report (OD, Count, VOL, AREA, %OD, %VOL, %Count).

The report shows information on each selected group, such as group ID and value of each feature in the group. You may save or print the report.

Histograms on groups

A more visual way to look at groups is to display a histogram. To get histograms on groups, choose *Show → Groups → Histograms On Groups*. Figure 9-1 shows histograms for two selected feature groups. The group IDs of the selected groups are shown on the reference gel.

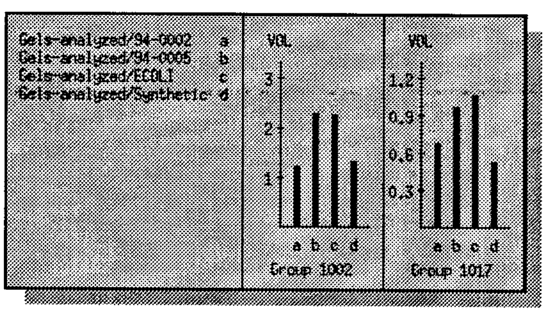
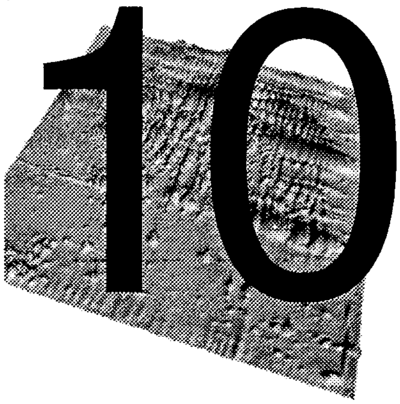


Figure 9-1. Histograms for two feature groups.

Groups with synthetic gels

One of the best ways to study the variation in protein expression among a series of gels is to produce a synthetic gel by merging features from the studied gels, then to match all gels to the synthetic one, and finally to produce reports or histograms on groups (see *Creating synthetic gels* on page 12-4).



10 EXPORTING DATA

Groups of features are automatically created when you match a series of gels to a reference gel. A group clusters features corresponding to the same protein across a set of gels. Groups may therefore be used to study the variations in protein expression among a series of gels. They may be analyzed in reports or histograms (see *Groups* on page 9-1) or they may be exported as tables in order to be used in external programs such as Excel or other statistical packages.

MelView provides three export options. You may produce *Outgroups* tables and standard ASCII tables for groups data, or export ASCII lists with feature data from selected gels.

Exporting to *Outgroups*

Outgroups is the table format used by the Melanie 1 heuristic and statistical classification programs. To export feature values for selected feature groups, follow these steps:

1. Select groups (see *Selecting groups* on page 9-2).
2. Choose *File* → *Export*.
3. Specify *Outgroups*.
4. Pick the reference gel for the groups.
5. Decide if you want to include data from the reference gel, or only from the other gels.
6. Specify the file name.
7. Click *OK*.

Exporting to ASCII tables

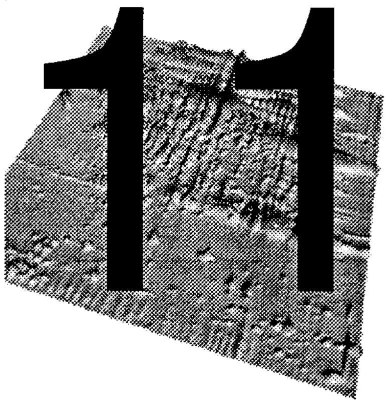
You may export the selected feature groups to ASCII tables. A table contains tabulator separated columns. Each column represents one gel. The lines contain feature data, one group per line. You have to choose one feature value type (OD, Count, VOL, AREA, %OD, %VOL, %Count). Perform the following steps:

1. Select groups (see *Selecting groups* on page 9-2).
2. Choose *File → Export.*
3. Specify *Excel Table.*
4. Pick the reference gel for the groups.
5. Pick one value type.
6. Decide if you want to include data from the reference gel, or only from the other gels.
7. Specify the file name.
8. Click *OK.*

Exporting to ASCII lists

You may export the selected features of any number of gels. A list differs from a table in that it does not contain feature groups data, but feature data from individual gel images. Each list contains the quantification values for selected features, one gel after the other. Perform the following steps:

1. Select gel.
2. Select features in the selected gels.
3. Choose *File → Export.*
4. Specify *Excel List.*
5. Specify the file name.
6. Click *OK.*



EDITING

This chapter details some editing capabilities of the MelView program that have not yet been addressed in the previous chapters.

Editing landmarks

We have already described how to create, modify and delete landmarks (See *Landmarks* on page 4-1). Another convenient option of MelView is the **copy / paste** facility. You may select landmarks in a given gel image, then copy them to another gel. This is a simple means of creating a series of similar landmarks in a set of gels. Perform the following steps:

1. Select landmarks using the Landmark tool or one of the *Select → Landmarks* options.
2. Choose *Edit → Landmarks → Copy Landmarks.* This will copy the selected landmarks into memory.
3. Select one or more gels you want to copy the landmarks to.
4. Choose *Edit → Landmarks → Paste Landmarks.* The copied landmarks will be pasted into each of the selected gels.
5. Adjust the landmarks using the Landmark tool or the *Edit → Landmarks → Modify Landmarks* option.

Editing features

The usual way to create features is to detect them using Melanie's powerful feature detection algorithm (see *Detecting features* on page 5-1)*Detecting features* on page 5-1). Nevertheless, you may manually create, modify or delete selected features.

Adding features

You may add a feature as an ellipse that is inscribed in a pre-selected region:

1 Define a small region using the Region tool.

2 Choose *Edit —> Features —> Add Feature*. This creates an elliptic feature within the selected region and displays a pop up window with an enlarged representation of the new feature (Fig. 11-1).

3 Modify the new feature if necessary. To this purpose, use the left mouse button to add pixels to the feature, the right mouse button to remove pixels, and the middle mouse button to fill in holes.

4 Click *OK*. The new feature will be added to the gel.

Modifying features

You may modify a feature's shape:

1 Select a feature with the Feature tool.

2 Choose *Edit —> Features —> Modify Features*. This displays a pop up window with an enlarged representation of the selected feature (Fig. 11-1).

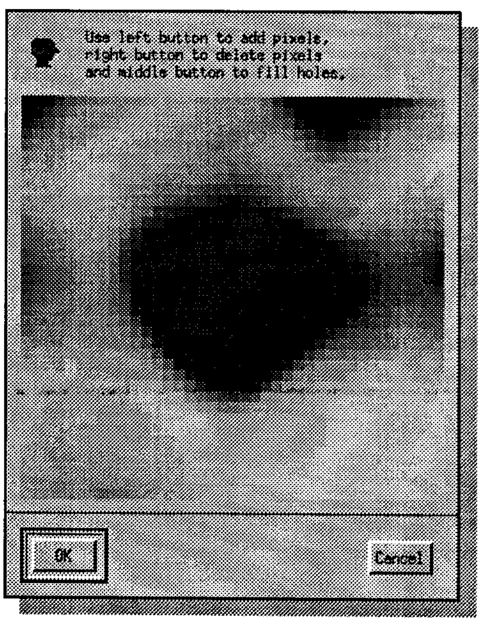


Figure 11-1. The feature modification window

3 Modify the feature. Use the left mouse button to add pixels to the feature, the right mouse button to remove pixels, and the middle mouse button to fill in holes.
4 Click *OK*. The feature will be modified accordingly.

Deleting features

To delete selected features:

1 Select one or more features with the Feature tool.
2 Choose *Edit → Features → Delete Features.*

Editing labels

We have previously described how to create, modify and delete labels (See *Labels* on page 6-1). As for landmarks, you may also **copy** / **paste** labels. You may select labels in a given gel image, then copy them to another gel. This is a simple means of creating a series of similar labels in a set of gels. Perform the following steps:

1 Select labels using the Label tool or one of the *Select → Labels* options.
2 Choose *Edit → Labels → Copy Labels.* This will copy the selected labels into memory.
3 Select one or more gels you want to copy the labels to.
4 Within these gels, select one or more features.
5 Choose *Edit → Labels → Paste Labels.* The copied labels will be pasted into each of the selected features.

Editing pairs

Manually creating and deleting pairs has already been explained in *Creating starting pairs* on page 8-2 .

Editing groups

A convenient way to add pairs between a set of gels and a reference gel is to create groups. You first have to set a master (See *Master gels* on page 13-2). Choose *DataBase → Set → Master* and choose your reference gel as the master. Select one feature in the reference gel and features in the other selected gels. Choosing *Edit → Groups → Group Features* adds pairs between the selected features and the selected feature in the reference gel.

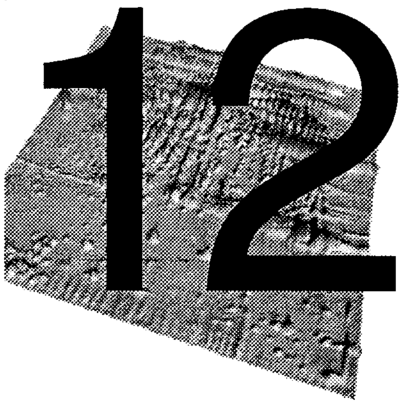
Select features that are all part of a given group. Choose *Edit → Groups → Ungroup Features*. This removes the selected features from the group.

To completely remove groups, select groups (see *Selecting groups* on page 9-2) and take *Edit → Groups → Delete Group*.

Editing / showing comments

You can associate a textual comment to individual gels. To add a new comment or modify an existing one, select one gel, then *Edit → Comments*. Write your comment and click *OK*.

To display the comments for a gel, select one gel, then *Show → Gels → Show Comments*.



12 Creating Images

The image menu

The menu is a special menu in the MelView program that processes existing 2-DE gel images on screen, but affects image files on disk. From this menu you may create new gel images from the selected ones. You can duplicate, crop and filter gels, create new synthetic or Gaussian gels, and flip or scale as well as erase gels. Except for the *Erase* option, all options in the *Image* menu create new gels that are automatically opened and displayed in the MelView window.

Duplicating gels

You can create copies of selected gels and include existing objects such as features, labels or landmarks.

Select the gels you want to duplicate. Choose *Image → Duplicate Gels.* A pop up window lets you select the objects you want to duplicate with the gels. Click *OK.* The Duplicate Gel window then lets you specify the new file names. You may also add an extension or replace the extensions of the original image files. Click *OK.* The gel image files will be copied to disk. They will then be opened and displayed in the MelView window.

Erasing gels

To delete existing gels, select them, then *Image → Erase Gels*. All selected gels will be removed from the screen as well as from the disk. Associated feature pairs are deleted as well.

Cropping gels

Define regions in selected gels using the Region tool. Choose *Image → Crop Gels*. The Crop Gels window asks you to choose what objects you want to keep with the cropped image, such as features, landmarks, or labels. You may also decide to keep the image alignment. Another pop up window then lets you specify the new file names. You may also add an extension or replace the extensions of the original image files. Click *OK*. New gels will be created containing only the part of the gels that lie within the specified regions, plus the specified objects.

Filtering gels

MelView has been optimized for analyzing 2-DE gel images without prior image processing or enhancement. If you analyze low quality images, MelView provides image processing filters that may help you to enhance the quality of your images.

Nevertheless, keep in mind that after filtering images, feature quantification values will change and will be hard to interpret.

Select the gels you want to filter, then pick a filter in the *Image → Filter Gels* options. This will create copies of the gels after they have been filtered, leaving the original images unaffected. The following filters are available:

1 **Smoothing** - Smooths the raw gel image. You have the choice between the following four smooth algorithms:
   - Gaussian smoothing
   - Diffusion smoothing
   - Polynomial smoothing
   - Adaptive smoothing

   You also have to choose how many times the smooth filter has to be applied.

2 **Histogram Equalization** - Performs histogram equalization on the selected gels. Various options are available and the following five equalization algorithms are provided:
   - Uniform

- Exponential
- Rayleigh
- Cube Root
- Logarithmic

3 **Remap Grey Levels** - Creates copies of the selected gels with modified grey levels mapping (see *Adjusting colors* on page 3-6). This is especially useful if you analyze images with a very low grey level range. For example, the Melanie II feature detection algorithm may not work properly on an image with only 25 grey levels. You can therefore remap it to an image with 4096 shades of grey so that features may correctly be detected.

4 **Contrast Enhancement** - Applies a contrast enhancing algorithm to the images in order to highlight small spots (Fig. 12-1). The following three contrast enhancement algorithms are available:

- Quadratic
- Sigmoid
- Wallis

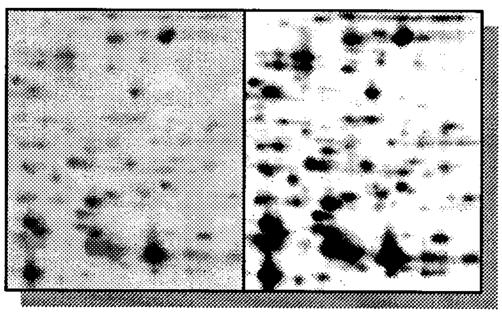


Figure 12-1. The original image (left) and the same image after contrast enhancement (right).

5 **Filter Background** - Applies one of the two following background filtering algorithms to the selected gels, in order to eliminate gel background (Fig. 12-2):

- Subtraction of global image minimum
- Background subtraction by polynomial transformation

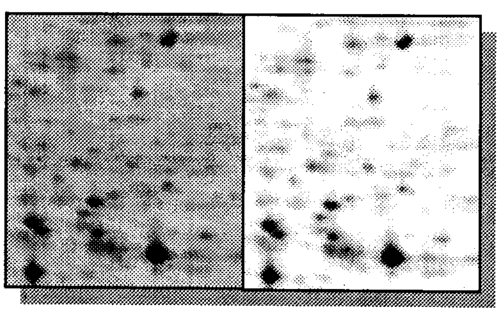


Figure 12-2. The original image (left) and the same image after background subtraction (right).

Creating synthetic gels

One of the best ways to study the variation in protein expression among a series of gels is to produce a **synthetic gel** by merging features from the studied gels, then to match all gels to the synthetic one, and finally to produce reports or histograms (see *Reports on groups* on page 9-2 and *Histograms on groups* on page 9-2).

To create a synthetic gel, perform the following steps:

1. Decide what gels will be used for creating the synthetic gel. Select them using the Select tool. The synthetic gel will contain features from these gels.
2. From these gels pick one out as reference gel. The synthetic gel will be based on it. This means that features will first be taken from the reference gel, then missing features will be extracted from the other gels. The resulting synthetic gel will most likely look similar to the reference gel. You should therefore choose a good quality gel image for the reference gel.
3. Pair wise match all gels chosen in step 1. For example, if you chose the three gels GelA, GelB and GelC, and pick GelA as reference gel, you will have to match GelA to GelB, GelA to GelC, and GelB to GelC.
4. Choose *Image → Create Gels → Create Synthetic Gels.*
5. When requested to do so, pick the reference gel and click *OK.*
6. The Create Synthetic Gel window then lets you specify the new file name. Click *OK.*

The resulting synthetic gel will be created and displayed (Fig. 12-3).

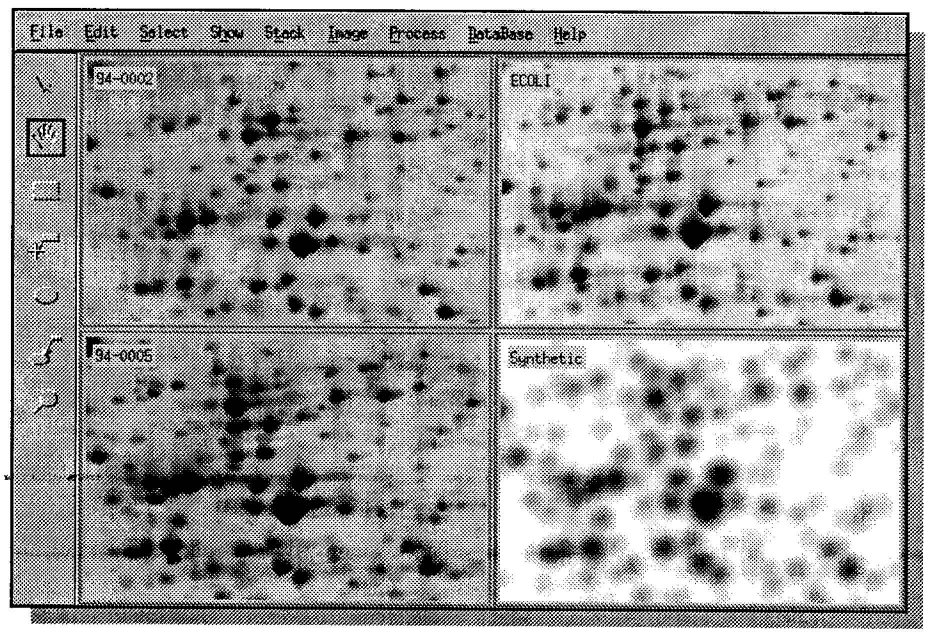


Figure 12-3. A synthetic gel and its three contributing gels. The reference gel is ECOLI.

You may now match your gel images to the synthetic one and perform a quantitative study on their features, using feature groups (see *Groups* on page 9-1).

Creating Gaussian gels

MelView's feature detection program lets you model features as Gaussian functions (see *Detecting features* on page 5-1). You may also create copies of gel images, where spots have been replaced by the Gaussian models of the detected features.

Select the gels you want to create Gaussian images from. Features need to be already detected. Choose *Image* → *Create Gels* → *Create Gaussian Gels.* The Create Gaussian Gels window lets you specify the new file names. You may also add an extension or replace the extensions of the original image files. Click *OK.* Gaussian images will be created and displayed (Fig. 12-4).

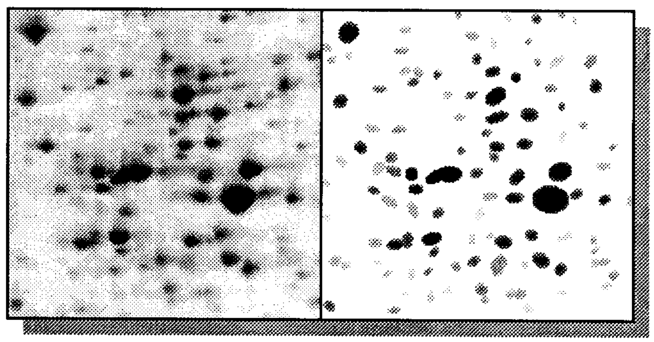


Figure 12-4. Part of a gel and its Gaussian transform.

Flipping gels

In the case where you scanned images in the wrong direction, you may correct this by applying the *Flip Gels* options. Select gels, then *Image* → *Flip Gels* and choose one of the flipping options (*Horizontally, Vertically* or *Both Directions*). As for other options in the *Image* menu, you will have to specify file names.

Scaling gels

The last option in the *Image* menu lets you create larger or smaller copies of selected gels. Select gels, then *Image* → *Scale Gels* and pick one of the possible scaling factors (1/4, 1/2, 2 or 4). You may also choose *Image* → *Scale Gels* → *Factor* and specify any custom scaling factor. As for other options in the *Image* menu, you will have to specify file names.

13 DATABASES

2-DE databases

A key aspect of the MelView program is its ability to link features on gel images to protein data in **2-DE databases**. 2-DE databases contain data on proteins identified on 2-DE images, such as *pI* and *Mw*, bibliographical references to protein related literature, information on protein functions, etc. Several specialized 2-DE databases exist that Melanie II can work in conjunction with. MelView is especially well suited to work with data from the SWISS-2DPAGE global 2-DE database that is accessible over the Internet (see *SWISS-2DPAGE master gels* in this chapter).

Features on 2-DE gels analyzed with Melanie II may be linked to three different types of 2-DE databases:

1. **Private 2-DE databases** - This is a database you build yourself by defining your own master gel on which you add protein related data. You may then match gels to your master protein map.
2. **The SWISS-2DPAGE database** - Twelve master 2-DE maps of Human samples, as well as of *Yeast* and *Escherichia Coli* have been provided on the Melanie II distribution tape. You may match your gels to any of these master images and identify proteins on them. SWISS-2DPAGE master gels may be updated by retrieving them from the Expasy FTP server.
3. **Remote 2-DE databases** - If your computer is connected to Internet, you may link your gels to remote databases such as SWISS-2DPAGE, and remotely retrieve protein data related to features on your gels.

The feature label

The **label** is a small tag associated to a feature and identifying the corresponding protein (see *Labels* on page 6-1). The label holds the protein's accession number (AC) and name taken from the 2-DE database, or it may just contain any textual data. The label is the link between the gel and the 2-DE database.

Note that two labels are the same if they have the same AC.

Master gels

A **master gel** contains a label for each identified protein. It holds the protein's accession number (AC) and its name (Fig. 13-1). You may define your own

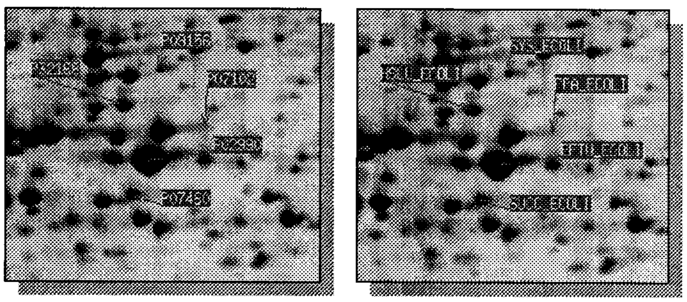


Figure 13-1. Labels on a master gel - with ACs (left) - with names (right).

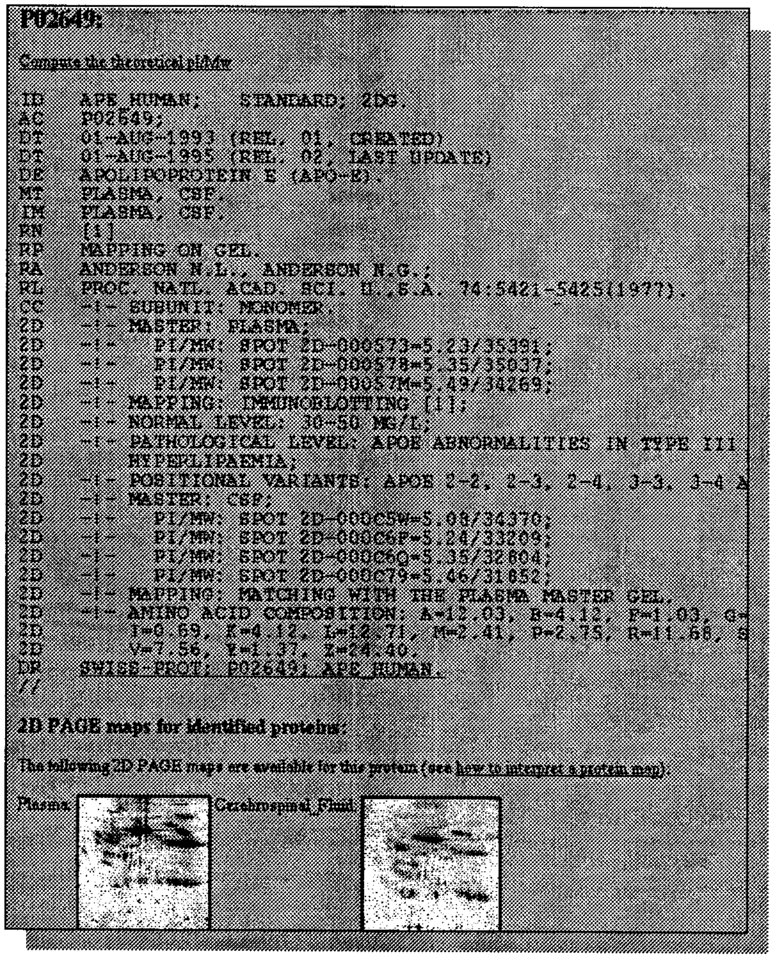
master gels by creating labels for the identified proteins and adding their names and AC (accession or unique identification number). The AC and names may be taken from existing databases, such as the SWISS-2DPAGE 2-DE database, the SWISS-PROT protein sequence database, or even gene databases such as EMBL or Genbank. Alternatively, you may match your images to existing master gels, such as the SWISS-2DPAGE master maps provided on the Melanie II distribution tape.

To work with master gels in MelView you have to designate the master among the open gels. Choose *Database → Set → Master* and click on the master image, then *OK*. The master's name is now printed with a red background.

SWISS-2DPAGE master gels

SWISS-2DPAGE master gels contain labels for all proteins corresponding to entries in the SWISS-2DPAGE 2-DE database. Each label holds the accession number (AC) of the database entry, as well as the entry's short name. For example, the Human Plasma protein map possesses a label whose AC is P02768 and name is ALBU_HUMAN, for each feature that corresponds to *albu-*

min. SWISS-2DPAGE entries contain detailed data on the related proteins, such as the protein's full name, bibliographical references, annotations (protein function, pathological variations, etc.) and the *pI* and *Mw* of the related spot on the 2-DE maps. They also hold cross-references to other databases such as the SWISS-PROT protein sequence database (that in turn has cross-references to numerous other databases such as gene databases, disease databases, etc.) or the YEPD *Yeast* 2-DE database and the ECO2DBASE *E. Coli* 2-DE database (Fig. 13-2).



P02649:

Compute the theoretical pI/Mw

```
ID   APE_HUMAN;    STANDARD; 2DG.
AC   P02649;
DT   01-AUG-1993 (REL. 01, CREATED)
DT   01-AUG-1995 (REL. 02, LAST UPDATE)
DE   APOLIPOPROTEIN E (APO-E).
MT   PLASMA, CSF.
IM   PLASMA, CSF.
RN   [1]
RP   MAPPING ON GEL.
RA   ANDERSON N.L., ANDERSON N.G.;
RL   PROC. NATL. ACAD. SCI. U.S.A. 74:5421-5425(1977).
CC   -!- SUBUNIT: MONOMER.
2D   -!- MASTER: PLASMA;
2D   -!-   PI/MW: SPOT 2D-000573=5.23/35391;
2D   -!-   PI/MW: SPOT 2D-000578=5.35/35037;
2D   -!-   PI/MW: SPOT 2D-00057M=5.49/34269;
2D   -!- MAPPING: IMMUNOBLOTTING [1];
2D   -!- NORMAL LEVEL: 30-50 MG/L;
2D   -!- PATHOLOGICAL LEVEL: APOE ABNORMALITIES IN TYPE III
2D       HYPERLIPAEMIA;
2D   -!- POSITIONAL VARIANTS: APOE 2-2, 2-3, 2-4, 3-3, 3-4 A
2D   -!- MASTER: CSF;
2D   -!-   PI/MW: SPOT 2D-000C5W=5.08/34370;
2D   -!-   PI/MW: SPOT 2D-000C6F=5.24/33209;
2D   -!-   PI/MW: SPOT 2D-000C6Q=5.35/32804;
2D   -!-   PI/MW: SPOT 2D-000C79=5.46/31852;
2D   -!- MAPPING: MATCHING WITH THE PLASMA MASTER GEL.
2D   -!- AMINO ACID COMPOSITION: A=12.03, B=4.12, F=1.03, G=
2D       I=0.69, K=4.12, L=12.71, M=2.41, P=2.75, R=11.68, S
2D       V=7.56, Y=1.37, Z=24.40.
DR   SWISS-PROT; P02649; APE_HUMAN.
//
```

2D PAGE maps for identified proteins:

The following 2D PAGE maps are available for this protein (see how to interpret a protein map):

Plasma: Cerebrospinal_Fluid:

Figure 13-2. The SWISS-2DPAGE entry for human *Apo E*.

In addition to labels containing data from the SWISS-2DPAGE database, features in SWISS-2DPAGE master gels also present the particularity of possessing **serial numbers**, an identification number that uniquely links a feature on the gel and protein data in the SWISS-2DPAGE database, to one spot on a specific SWISS-2DPAGE master gel. The serial number has the form *2D-xxxxxx*, where x is a digit or an upper case letter. For example, *2D-000C5W* and *2D-000C6F* identify two spots on the Human Plasma protein map that have been identified as *Apo E* (Fig. 13-2).

The serial numbers are embedded in the SWISS-2DPAGE master gels and may not be changed. You may at any time retrieve updated master gels from the Expasy FTP server (expasy.hcuge.ch) in the directory /pub/Masters. The serial numbers are shown in reports on features (see *Reports on features* on page 5-7).

Selecting CH2D features

You can select the features for all identified proteins in a SWISS-2DPAGE master by choosing *Database → Features → Select CH2D Features.*

Adding labels from a master

After having matched gels to a master, you may copy some or all labels from the master gel to any of the matched gels. Select one or more features in any gels, then take *Database → Labels → Add From Master.* MelView will look for features in the master that are paired to the selected ones. It will then copy their associated labels to the selected features. The selected features will thereafter have the same labels as their paired features in the master. This operation will in this way identify proteins on your gels. You may then display the AC or names of these proteins, or use the database query options described in *Querying a remote database through Internet* on page 13-5.

Updating labels from a master

This option lets you update the content of labels by copying data from related labels in the master. Select the features, the labels of which you want to update. Then take *Database → Labels → Update From Master.* The labels associated to the selected features will be updated. This operation affects the AC as well as the name of labels.

Deleting labels from a master

Like the two above described options, this option works also on selected features. It deletes labels associated to selected features that are identical to the corresponding labels in the master (identical = same AC). Select features, then choose *Database → Labels → Delete From Master.*

Querying a remote database through Internet

In MelView you may work with your own master images, or you can link your gels to any SWISS-2DPAGE master gels. An interesting possibility to link your images to 2-DE databases is offered if your computer is connected to the Internet. You may retrieve data from remote 2-DE databases for any label containing a valid accession number (AC). Also, you may automatically update the label names for these proteins. You have to follow the steps described in the remaining part of this chapter.

Setting the master

The master gel serves as the link to the remote database. You have to designate the master gel prior to querying the database. A gel may be master if it contains at least one label with a valid AC. Take *Database → Set → Master*, then choose the master from the open gels. Click *OK*.

Setting the browser

To retrieve data from remote databases you must have the Netscape Navigator® World-Wide Web browser installed. You also have to set the full path name for Netscape. Choose *Database → Set → Browser* and enter the program name and path. The program name alone may be sufficient if the directory in which Netscape resides is part of your path.

This option may also be made permanent from the *File → Preferences* menu option.

Setting the server

You can choose the World-Wide Web server on which the remote database resides. The server has to host a **federated 2-DE database** [see reference 8], that is a 2-DE database that may be remotely queried from within 2-DE analysis software. An example of such a WWW server is the Expasy WWW server. Choose *Database → Set → Server* and enter the server's host name. For Expasy, enter:

expasy.hcuge.ch

Setting the database

You also have to set the name of the remote database. Choose *Database → Set → Database* and enter the database name. On Expasy, you have the choice of the SWISS-2DPAGE 2-DE database (type: SWISS-2DPAGE) and the SWISS-PROT protein sequence database (type: SWISS-PROT).

The list of federated 2-DE databases and servers is maintained on the WORLD-2DPAGE WWW page at:

http://expasy.hcuge.ch/ch2d/2d-index.html

Querying the database

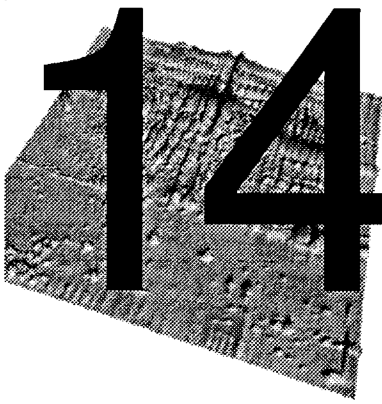
To query the remote database, select one feature that has got a label with a valid accession number (AC). Choose *Database → Query Server*. This will control the Netscape program on your computer that is automatically going to request the corresponding entry from the remote database. For example, if you have set the database **SWISS-2DPAGE** on the **expasy.hcuge.ch** server, and you select a feature whose label has the accession number **P02649**, then if you do *Database → Query Server*, you will obtain in Netscape the entry for Human *Apo E* (see Fig. 13-2).

Updating protein names

On most 2-DE databases, the accession number is the unique entry identifier, while the protein name may change. MelView provides a way for automatically updating the protein names in your gels. Select a gel containing labels with valid ACs. Set it as the master gel. Select all labels you want to update. Choose *Database → Labels → Update Short Name from Database*. For each selected label, MelView will request its short name from the remote database (using the AC as identifier) and then update the label accordingly.

PART IV

THE MELBATCH PROGRAM



14 THE MELBATCH PROGRAM

Batch processing

In the MelView program, actions may be carried out on open gels only. Certain of these actions (such as matching gel images, for example) may take time when performed on large sets of large images, but do not always require to be run interactively. **MelBatch** is a companion program to MelView that allows you to perform certain actions on gels without the need of displaying them. You may carry out whole series of operations on a set of gels without having first to open them.

For example, you can detect and quantify features on several gels, then match them to a reference gel, and finally print the images, all in batch (i.e. non interactive) mode. This saves memory and lets you use your computer for other purposes in the mean time.

Running MelBatch

To run the MelBatch program, type the following command:

```
> MelBatch &
```

The `&` sign indicates that the MelBatch program has to run in background. You have to make sure that the directory in which the MelBatch program resides is part of your path. Type

```
> echo $path
```

to verify this. If it is not in the path, please refer to the Unix manual to include it.

Upon loading the program the Melanie II logo appears. Click on the logo to get rid of it.

With MelBatch you may execute the following actions (see Fig 14-1):

- Convert images from the old Melanie 1 and Melanie II release 1 formats.
- Reduce images that are too large and eliminate parasite noise.
- Detect features.
- Quantify features
- Match gels.
- Print images.

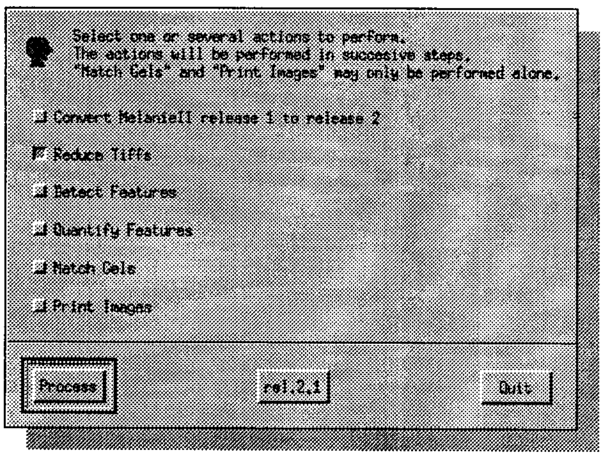


Figure 14-1. The MelBatch window

Converting images

MelView and MelBatch work with images stored in the Melanie II release 2 file format (see *The Melanie II 2-D PAGE file format* on page 1-6). MelBatch lets you convert gel images stored in the old Melanie 1 or Melanie II release 1 formats into the correct format.

Tick the button next to *Convert Melanie II release 1 to release 2*, then click *Process*. The Select window opens and you may choose images in the old format (see *Opening gels* on page 3-2). The Convert Gels window then pops up and you have to give names for the converted image files. You can either enter the names of the new files manually, or use the *Replace Extension* or the *Add Extension* options. *Add Extension* lets you specify an extension to add to each new file name, while *Replace Extension* lets you replace the file name extensions of the selected gels with any other extension.

The selected images will then be converted to the Melanie II release 2 format.

Reducing images

This option reduces the size of the original images. You may specify the reduction factor that divides the number of pixels in each direction. For example, reducing a 2000 by 2000 pixel image by a factor of 2 will produce a 1000 by 1000 pixel image, thus reducing the image size by 4 (2000*2000 = 4 million, while 1000*1000 = 1 million).

The reduction is performed by averaging neighboring pixels, thus eliminating parasite noise from the image. It is important to eliminate noise produced by the scanning device, in order to avoid the detection of artifacts in the feature detection process.

Tick the button next to *Reduce Tiffs*, then click *Process*. The Select window opens and you may choose gel images. You then have to give names for the reduced image files. You can either enter the names of the new files manually, or use the *Replace Extension* or the *Add Extension* options. *Add Extension* lets you specify an extension to add to each new file name, while *Replace Extension* lets you replace the file name extensions of the selected gels with any other extension. You will then be requested to set the reduction factor. Move the slider to specify it and click *OK*.

Detecting and quantifying features

With MelBatch you can detect and quantify features the same way you do in MelView, but without opening the gels. This is useful if you already know what detection parameter values to use.

Tick the button next to *Detect Features* and/or *Quantify Features*, then click *Process*. The Select window opens and you may choose gel images. For detecting features you will also be requested to specify the detection parameters (see *Detecting features* on page 5-1).

Matching gels

A set of gels may be matched at once to a reference gel (see *Matching gels* on page 8-1) Tick the button next to *Match Gels*, then click *Process*. The Select window opens and you may choose gel images. The next pop up window asks you to select the reference gel. Click on it, then *OK*. The reference gel will be matched to all other chosen gels.

Printing images

To print gels images, tick the button next to *Print Images*, then click *Process*. The Select window opens and you may choose gel images. Click *OK*. Each selected image will be sent to the printer.

MelBatch uses the Printer Settings you have set in your *Preferences* file (see *Printing gels* on page 3-12). To change them, you have to run MelView.

Combining operations

Actions in MelBatch may be combined. You may perform several actions sequentially without having to interfere. For example, you may select five gel images stored in the old file format, then ask MelBatch to convert them to the new format, to reduce their size, and finally to detect and quantify features. Just click the buttons corresponding to the operations you want to perform and click *OK*.

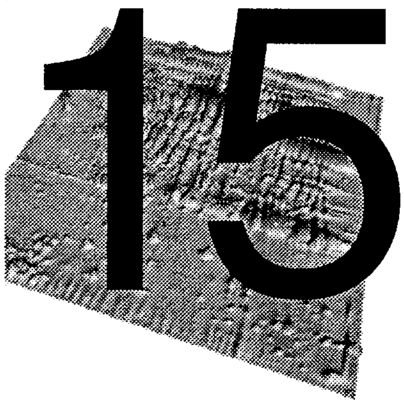
Note that the *Match Gels* and *Print Images* operations may not be combined with any other actions and have to be performed independently.

Exiting MelBatch

Click on *Quit* and confirm with *OK*.

PART V

TUTORIAL



TUTORIAL 1: LOOKING AT GELS

Introduction

This section describes a simple session with the **Melanie II** software using the gels provided on the distribution tape, in directory `Gels`. You may read this section and at the same time perform the described steps. The > sign indicates the command prompt on the Sun workstation.

This tutorial explains how to open gels, then how to manipulate them in order to improve their visualization. *Tutorial 2: analyzing gels* on page 16-1 presents more advanced features of the Melanie II software.

Manipulating two gels.

- Go to directory `Gels`, in which the sample gels are located. (for example: `/usr/local/MelanieII/Gels`):

```
> cd /usr/local/MelanieII/Gels
```

- Type the following command to run the **MelView** program:

```
> MelView &
```

The `&` sign indicates that the MelView program has to run in background.

Figure 15-1 shows the screen that appears when running MelView.

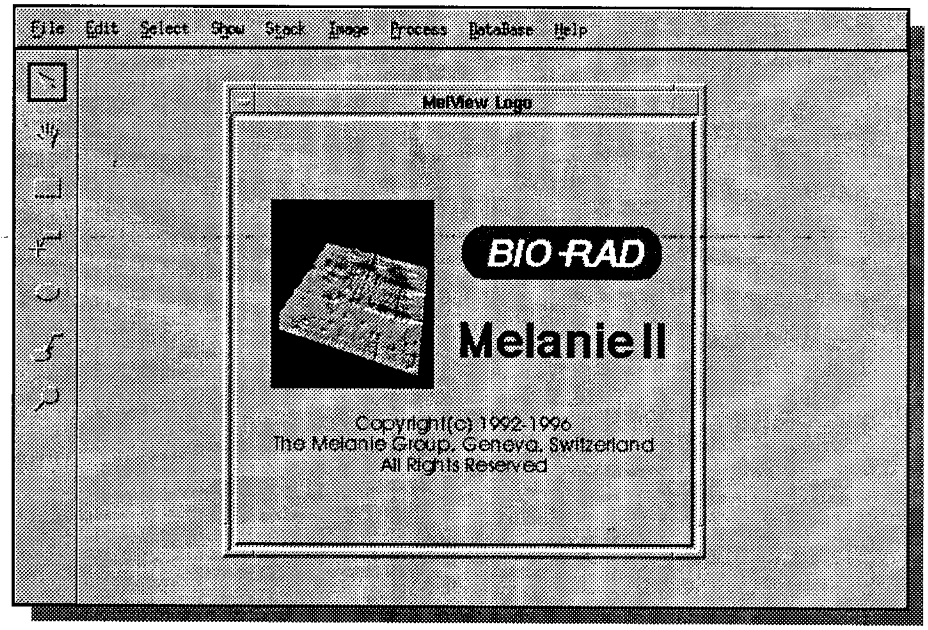


Figure 15-1. The MelView main screen

- Click on the Melanie II logo to get rid of it.

Opening two 2-DE images

- Click in the *File* menu and select the *Open* option. The *Open* window will be displayed (Fig. 15-2). Hold down the Control key and click on gels 94-0002 and

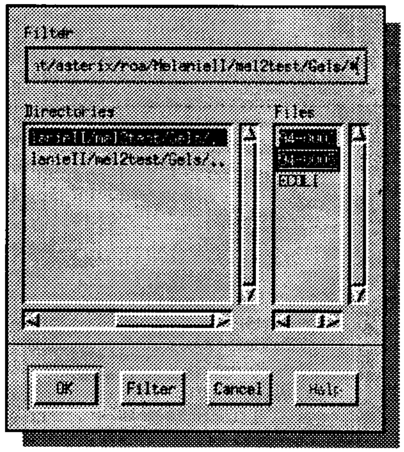


Figure 15-2. The Open window allows the selection of the gels that will be opened and loaded into MelView

94-0005 in the *Files* column. This selects the two gels. Click on the *OK* button. The gels will be opened.

- In the *Select* menu, choose the *Gels* → *Visible Gels* option. This will select all gels and draw a green line around each of them showing that they have been selected (Fig. 15-3). In MelView the green color always indicates selected objects.

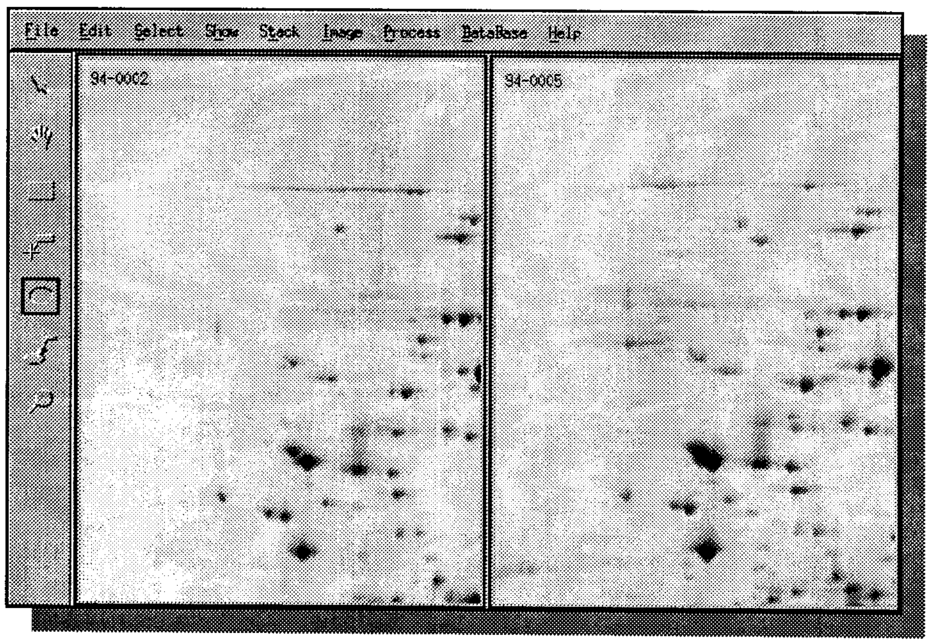


Figure 15-3. The two opened gels.

Adjusting colors

- Select the Hand tool (the second tool from the top in the left hand side tool bar). Put the cursor in one of the gels. Hold down the right mouse button and move the gels until you see a region with many spots.
- Now select the Region tool (the third tool from the top in the left hand side tool bar) and draw a small rectangle in one of the gels, using the mouse and the left button. In the *Select* menu, choose *Select Mode → Adjust Colors.* Select a logarithmic color function and adjust its parameter to the value 4. The change will directly be reflected in the selected region (Fig. 15-4).

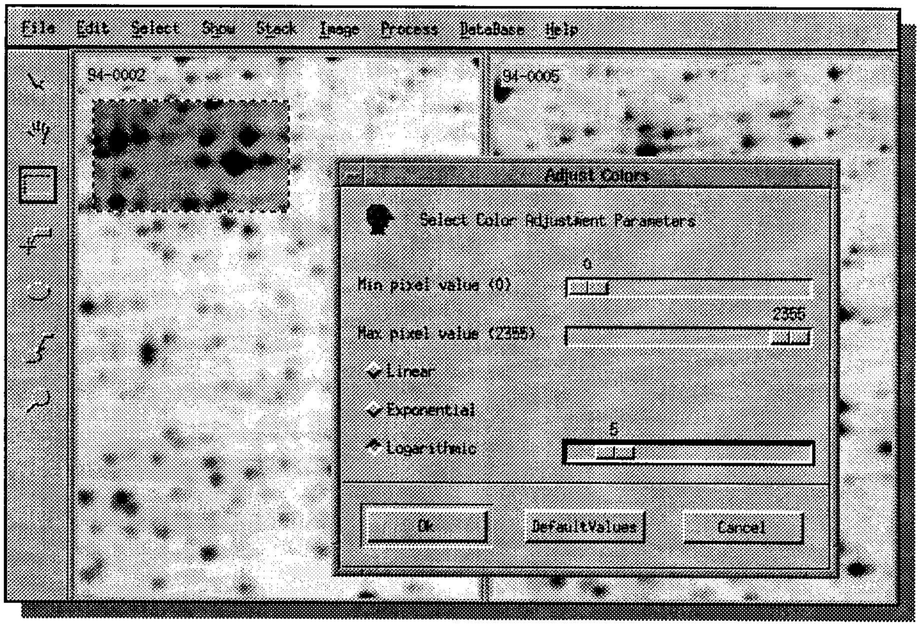


Figure 15-4. The *Adjust Colors* option. The modifications are directly reflected in the selected region.

- Click on the *OK* button. The colors will be adjusted in the two selected gels.

Zooming

- With the Select tool active (first tool in the tool bar), select only one gel. Then, in the *Show* menu, choose *Gels → Selected Gels*. Only the selected gel stays in the MelView window. The other gel is hidden. In menu *Select*, choose *Select Zoom → 1/2*. The gel is zoomed out by a factor of 2 and a larger part of the gel can be seen (Fig. 15-5).

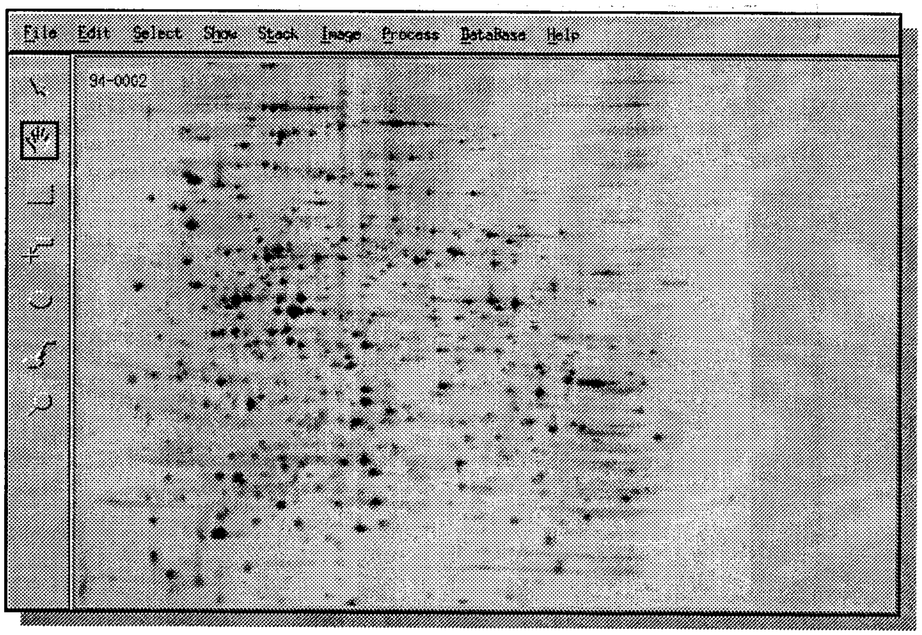


Figure 15-5. A gel after it has been zoomed out.

Stacking gels

- Choose *Show Gels → All Gels*, and *Select Gels → Visible Gels*, then *Stack → Stack Selected Gels*. The two gels are now displayed on top of each other. To switch from one gel to the other, choose *Stack → Front To Back*, or just Control-F. By pressing Control-F quickly several times, the differences between the two gels can be observed visually.

Aligning gels

We are now going to align the two gels in order to correct their relative distortion. The first step consists of setting a few landmarks in the two gels. Each landmark must correspond to the same feature in both gels. The landmarks must be well distributed over the gel. Follow the next steps:

- Put gel 94-0002 in front. Select the Landmark tool (fourth tool in the tool bar). Press Control-F to toggle.
- To add a landmark, double click, in the center of the gel, on a spot that is also in the other gel. The Create Landmark window pops up. Enter a landmark name (for example L1). Click *OK*. The landmark now appears on the gel image (Fig. 15-6).

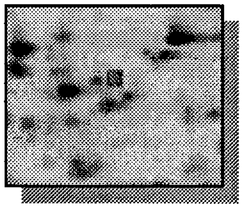


Figure 15-6. The L3 Landmark

- Press Control-F to toggle to gel 94-0005. To move the top gel so that a spot is exactly above the corresponding spot in the other gel, follow these steps: with the Landmark tool selected, put the cursor in the center of a feature (spot). Hold down the middle mouse button and at the same time press Control-F once and move the cursor to the center of the corresponding spot in the other gel. Release the mouse button.
- Double click on the same feature (spot) in gel 94-0005. The same landmark name should already be shown in the Create Landmark window. Click *OK*. The L1 landmark will be created on the second gel. Repeat this step four times, in order to add four landmarks around each of the four corners of the gels.
- Press Control-R to unstack the gels. Press Control-S to save the gels.
- Now that both gels contain related landmarks we are going to perform the alignment. Choose *Process —> Align Gels —> Align Gels*. In the *Align Gels* window, select gel 94-0002 as the reference gel. Click *OK*. Gel 94-0005 will then be aligned according to 94-0002.
- Choose *Stack —> Stack Selected Gels* to stack the gels again. To visually compare both gels, switch from one gel to the other by pressing Control-F quickly several times.

Magnifying glass

- Press Control-R to unstack the gels. Select the Magnify tool (last tool in the tool bar). Click the left mouse button and move the cursor in one of the gels. A small area under the cursor will be magnified. Now hold down the right mouse button, and move the cursor. The same region in both gels will be magnified at the same time (Fig. 15-7).

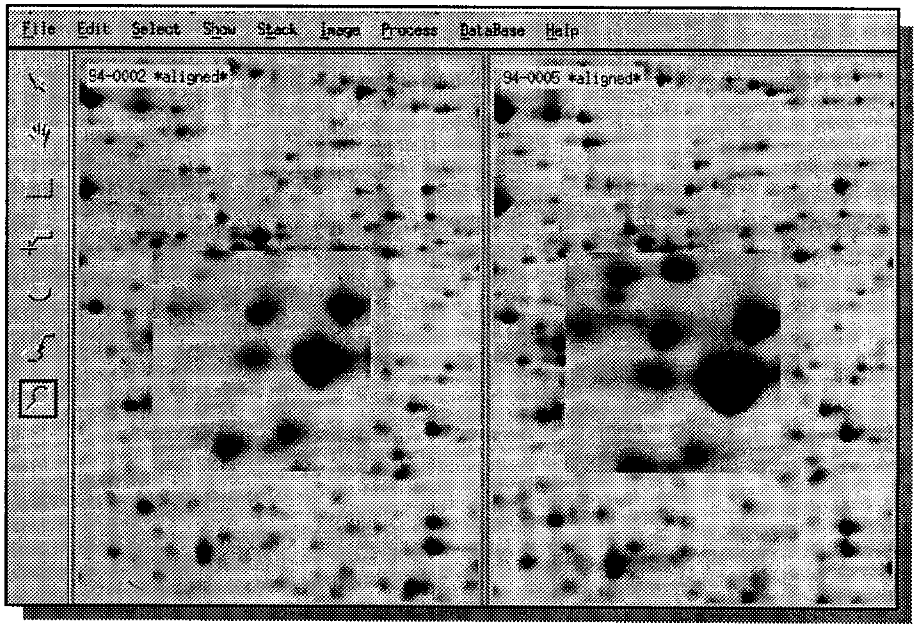
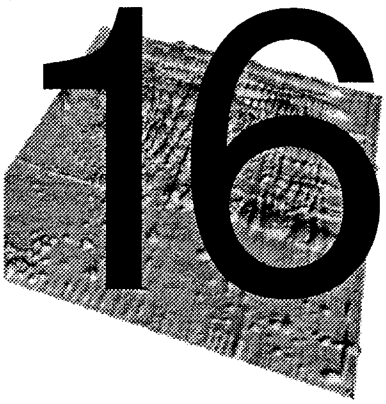


Figure 15-7. The Magnify tool applied to aligned gels.

Exiting MelView

To exit MelView, do *File* → *Quit*. To try out more advanced features, such as detecting spots and matching gels, you may proceed to *Tutorial 2: analyzing gels* on page 16-1.



16 TUTORIAL 2: ANALYZING GELS

Introduction

This section describes a typical session with the **Melanie II** software using the gels provided on the distribution tape, in directory `Gels`. You may read this section and at the same time perform the described steps. The directory `Gels-analyzed` contains the same gels after all the analysis described in this tutorial has been performed. You may therefore check what the result of this analysis should be. To do the complete analysis, perform the following steps. The > sign indicates the command prompt on the Sun workstation.

Analyzing a set of three gels.

- Go to directory `Gels`, in which the sample gels are located. (for example: `/usr/local/MelanieII/Gels`):

```
> cd /usr/local/MelanieII/Gels
```

- Type the following command to run the **MelView** program:

```
> MelView &
```

The `&` sign indicates that the MelView program has to run in background.

Figure 16-1 shows the screen that appears when running MelView.

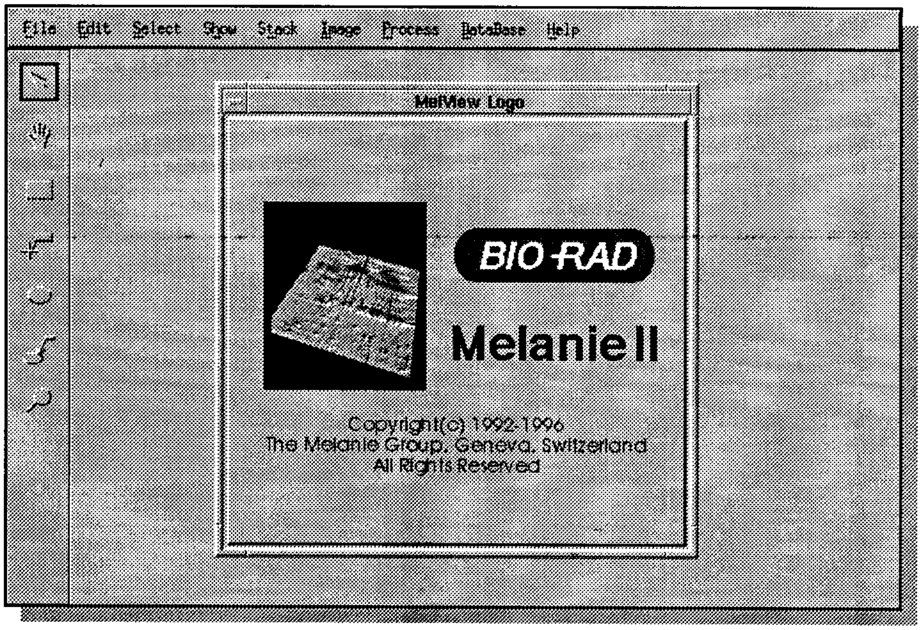


Figure 16-1. The MelView main screen

- Click on the Melanie II logo to get rid of it.

Setting preferences

- This tutorial is also available on-line. To consult it on-line, choose *Preferences* in the *File* menu. Set the *Melanie Home* directory to the directory in which the *man* directory resides, usually the same as the Melanie II binaries. Then, in the *Help* menu, choose *Tutorial.* On-line user manual and index are available too. To access the on-line tutorial and user manual, you must have the Netscape Navigator® World-Wide Web browser installed.

Opening 2-DE images

- Click in the *File* menu and select the *Open* option. The *Open* window will be displayed (Fig. 16-2). Hold down the Control key and click on each of the three

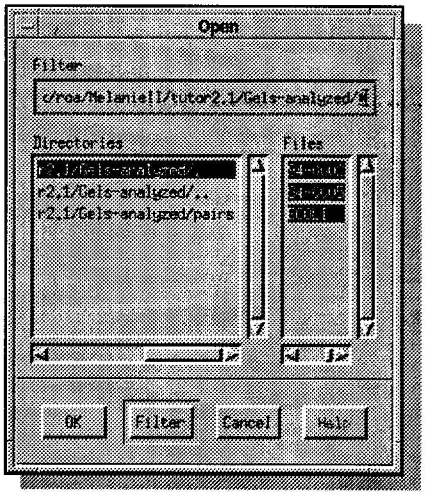


Figure 16-2. The Open window allows the selection of the gels that will be opened and loaded into MelView

file names in the *Files* column. This selects the three gels. Click on the *OK* button. The gels will be opened.

- In the *Select* menu, choose the *Gels* → *Visible Gels* option. This will select all gels and draw a green line around each of them showing that they have been selected (Fig. 16-3). In MelView the green color always indicates selected objects.

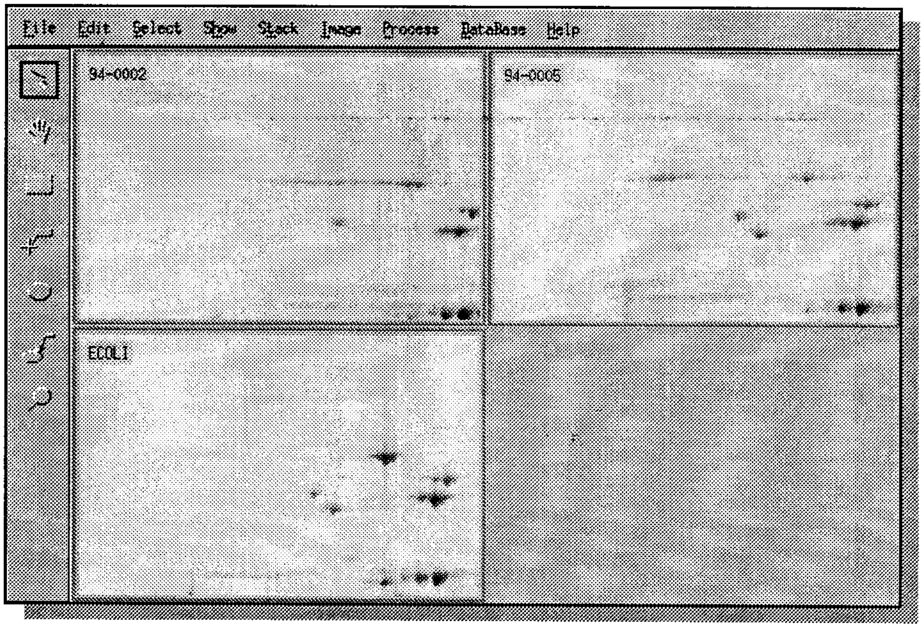


Figure 16-3. The three opened gels.

Adjusting colors

- The 94-0002 and 94-0005 gels are raw, not yet analyzed images. ECOLI is an already analyzed master 2-DE map of *Escherichia Coli* from the **SWISS-2DPAGE** database. In the *Show* menu, choose *Hide All* to hide the already detected and annotated features in the ECOLI image.
- Select the Hand tool (the second tool from the top in the left hand side tool bar). Put the cursor in one of the gels. Hold down the right mouse button and move the gels until you see a region with many spots.

- Now select the Region tool (the third tool from the top in the left hand side tool bar) and draw a small rectangle in one of the gels, using the mouse and the left button. In the *Select* menu, choose *Select Mode* → *Adjust Colors*. Select a logarithmic color function and adjust its parameter to the value 4. The change will directly be reflected in the selected region (Fig. 16-4).

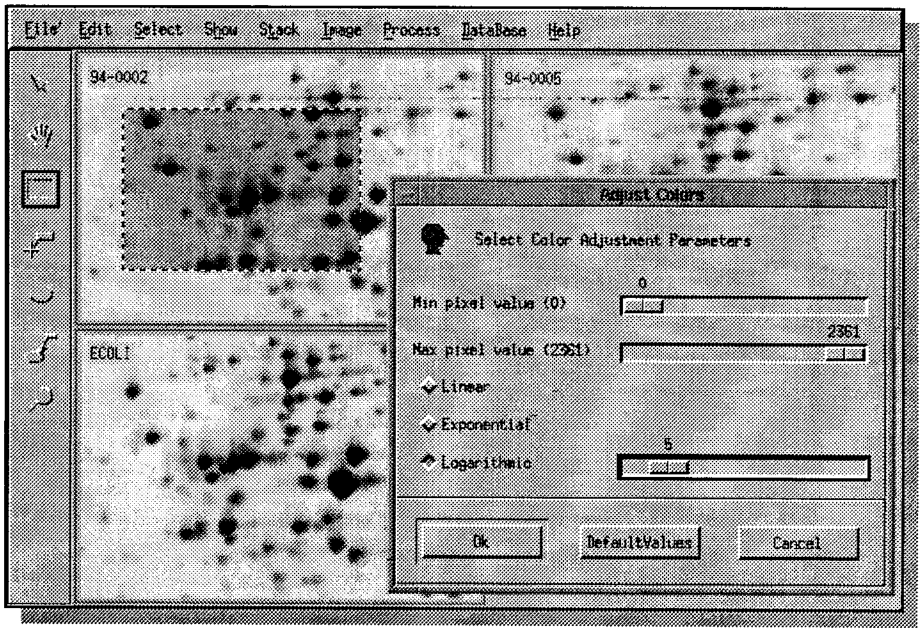


Figure 16-4. The *Adjust Colors* option. The modifications are directly reflected in the selected region.

- Click on the *OK* button. The colors will be adjusted in all three selected gels.

Reporting on gels

- Choose *Show —› Gels —› Report On Gels.* A window pops up giving basic information about the selected gels, such as image size in pixels, minimum and maximum grey levels, number of detected features (spots), etc. At this point, only the ECOLI gel has got detected features (Fig. 16-5).

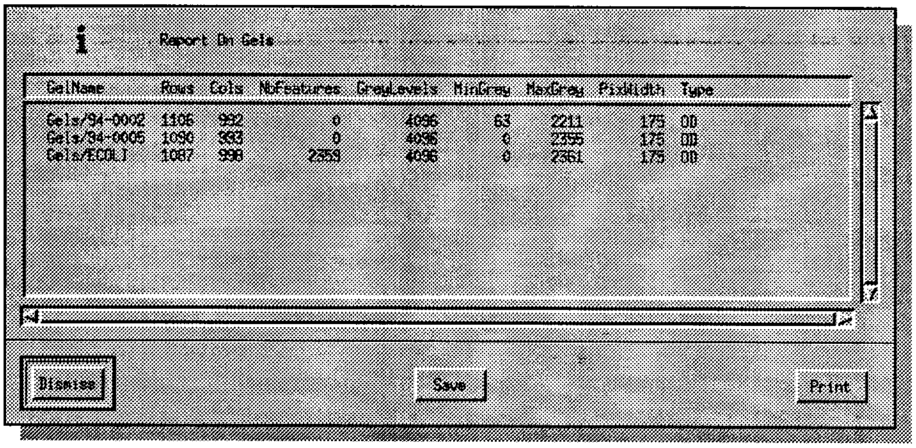


Figure 16-5. The *Report On Gels* window.

Printing gels

- You first have to set the printer. Select *File —› Preferences.* Make sure to define an existing printer (example: lp). Then select *File —› Print —› Print MelView Window.* The image of the whole MelView window is sent to the Postscript® printer.
- Using the Select tool, now select two gels. To achieve this, click on one gel, then hold down the shift key and click on the other gel. Choose *File —› Print —› Print Selected Gels.* The selected gels will be printed in full size.

Setting pseudo colors

- Choose *Select → Select Mode → Pseudo Colors*. Choose one of the color lookup table, for example FiveRamps. Click *OK*. The lookup table will be changed accordingly (Fig. 16-6). Choose *Select → Select Mode → Grey Levels* to go back to normal grey levels display.

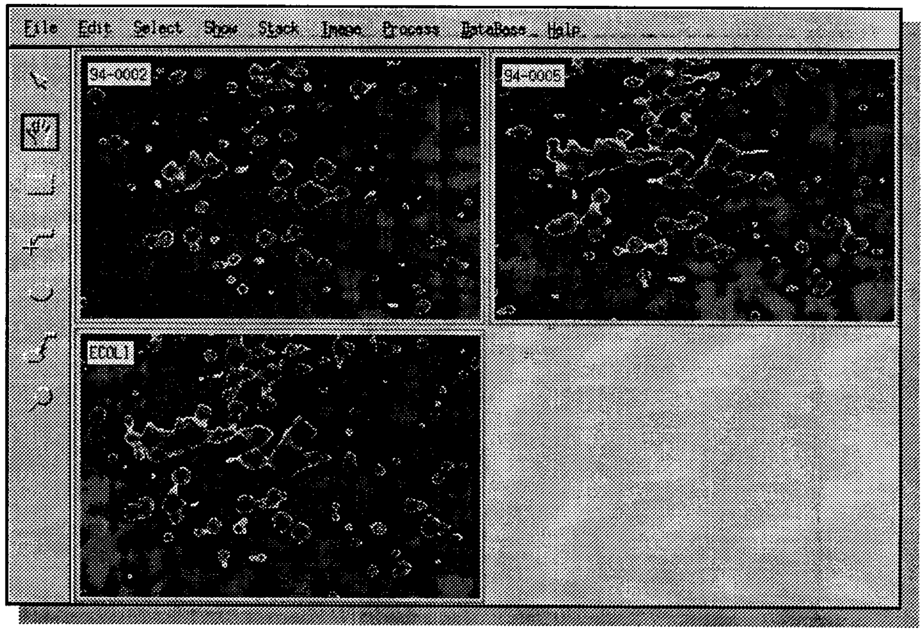


Figure 16-6. The gels displayed using the *FiveRamps* color lookup table.

Detecting and quantifying features

- Select the gels that have not yet been analyzed (94-0002 and 94-0005) using the Select tool and holding down the shift key. Choose *Process → Detect Features*. Click *OK*. Features (spots) will be detected using the default parameters.

Now choose *Process → Quantify Features* to compute the optical density, area and volume of the detected features (Fig. 16-7). Select *File → Save → Save Changes* to save the features and their associated quantification values.

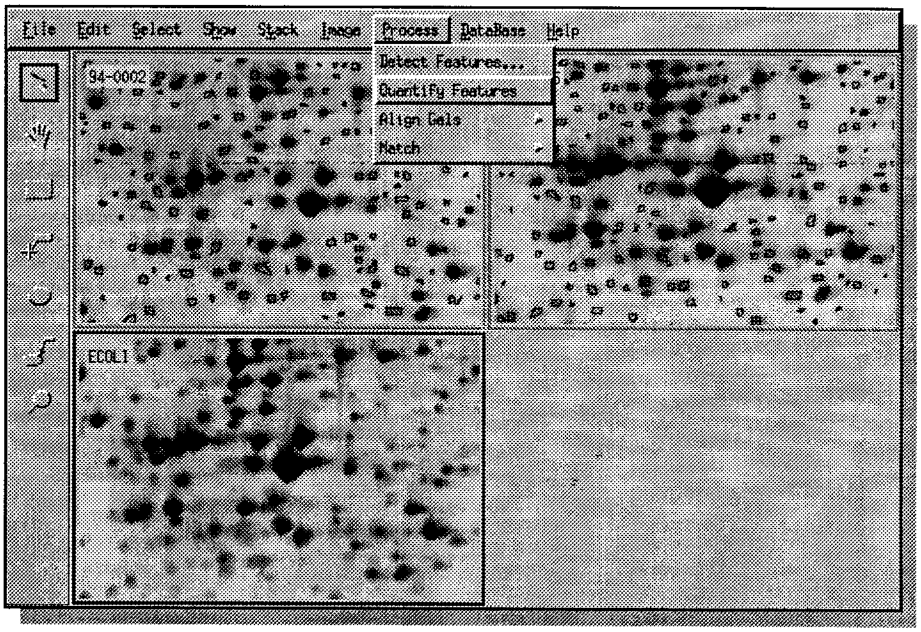
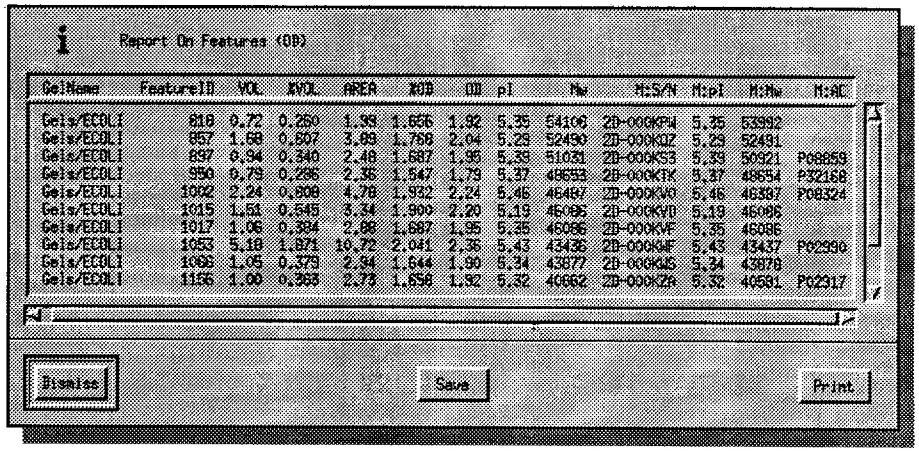


Figure 16-7. Features have been detected. The *Quantify Features* option has been selected.

Reporting on features

- With the Feature tool (fifth on the tool bar) select a few spots: click on one spot. It will be displayed in green, showing that it has been selected. Then hold down the shift key and click on a few other spots. Choose *Show → Features → Report On Features.* This brings up a report window showing specific information on the selected features (Fig. 16-8).



GelName	FeatureID	VOL	%VOL	AREA	%OD	OD	pI	Mw	M:S/N	M:pI	M:Mw	M:AC
Gels/ECOLI	818	0.72	0.260	1.39	1.656	1.92	5.35	54106	2D-000KPW	5.35	53992	
Gels/ECOLI	857	1.68	0.607	3.89	1.768	2.04	5.29	52490	2D-000KQZ	5.29	52491	
Gels/ECOLI	897	0.94	0.340	2.48	1.687	1.95	5.39	51031	2D-000KS3	5.39	50921	P08859
Gels/ECOLI	950	0.79	0.286	2.36	1.547	1.79	5.37	48653	2D-000KTK	5.37	48654	P32168
Gels/ECOLI	1002	2.24	0.808	4.78	1.932	2.24	5.46	46487	2D-000KV0	5.46	46387	P08324
Gels/ECOLI	1015	1.51	0.545	3.34	1.900	2.20	5.19	46085	2D-000KVD	5.19	46086	
Gels/ECOLI	1017	1.06	0.384	2.88	1.687	1.95	5.35	46086	2D-000KVF	5.35	46086	
Gels/ECOLI	1053	5.18	1.871	10.72	2.041	2.36	5.43	43436	2D-000KWF	5.43	43437	P02990
Gels/ECOLI	1096	1.05	0.379	2.94	1.644	1.90	5.34	43877	2D-000KW5	5.34	43878	
Gels/ECOLI	1156	1.00	0.363	2.73	1.658	1.92	5.32	40662	2D-000KZA	5.32	40581	P02317

Figure 16-8. Report on selected features.

Stacking gels

- Choose *Select → Gels → Visible Gels*, then *Show → Hide All* to hide the detected features. This also has the side effect of displaying the images faster.
- Select the ECOLI and 94-0002 gels. Choose *Show Gels → Selected Gels*, then *Stack → Stack Selected Gels*. The two gels are now displayed on top of each other. To switch from one gel to the other, choose *Stack → Front To Back*, or just Control-F. By pressing Control-F quickly several times, the differences between the two gels can be observed visually.

Aligning gels

This step is optional, but it is very useful when analyzing gels that are not completely superimposable. Gel alignment corrects gels pixel wise according to a reference gel, in order to make all gels superimposable.

The first step consists of setting a few landmarks in each of the three gels. Each landmark must correspond to the same feature in all gels. Start with the two currently stacked gels. The landmarks must be well distributed over the gel. Follow the next steps:

- Put the ECOLI gel in front. Select the Landmark tool (fourth tool in the tool bar). Press Control-F to toggle.
- To add a landmark, double click, in the center of the ECOLI gel, on a spot that is also in the other two gels. The Create Landmark window pops up. Enter a landmark name (for example L1). Click *OK*. The landmark now appears on the gel image (Fig. 16-9).

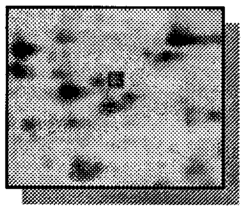


Figure 16-9. The L3 Landmark

- Press Control-F to toggle to gel 94-0002. To move the top gel so that a spot is exactly above the corresponding spot in the other gel, follow these steps: with the Landmark tool selected, put the cursor in the center of a feature (spot). Hold down the middle mouse button and at the same time press Control-F once and move the cursor to the center of the corresponding spot in the other gel. Release the mouse button.
- Double click on the same feature (spot) in gel 94-0002. The same landmark name should already be shown in the Create Landmark window. Click *OK*. The L1 landmark will be created on the second gel. Repeat this step four times, in order to add four landmarks around each of the four corners of the gels.
- Press Control-R to unstack the gels. Choose *Show* → *Gels* → *All Gels*. Then, with the Select tool, select the ECOLI and 94-0005 gels. Press Control-Z and Control-T to display the two selected gels in stack mode. Add the same five landmarks to gel 94-0005.
- Choose *Show* → *Gels* → *All Gels*, then *Select* → *Gels* → *Visible Gels*. Press Control-S to save the gels.
- Now that all three gels contain related landmarks we are going to perform the alignment. Choose *Process* → *Align Gels* → *Align Gels*. In the *Align Gels* window, select the ECOLI gel as the reference gel. Click *OK*. Gels will then be aligned according to the ECOLI gel.
- Once gels have been aligned, many tools and actions that work on one gel also work on all gels at the same time. Instead of pressing the left mouse button to use a tool on one gel, press the right mouse button. The tool will be applied to all aligned gels.

Magnifying glass

- Select the Region tool (third tool in the tool bar). In the ECOLI gel, hold down the left mouse button, and drag the mouse for about one inch. Release the button. A small rectangle will be drawn, defining a region on the ECOLI gel. Now hold down the right mouse button, drag the mouse. Release the mouse button. The same region in all gels will be defined.
- Select the Magnify tool (last tool in the tool bar). Click the left mouse button and move the cursor in any of the gels. A small area under the cursor will be magnified. Now hold down the right mouse button, and move the cursor. The same region in all aligned gels will be magnified at the same time (Fig. 16-10).

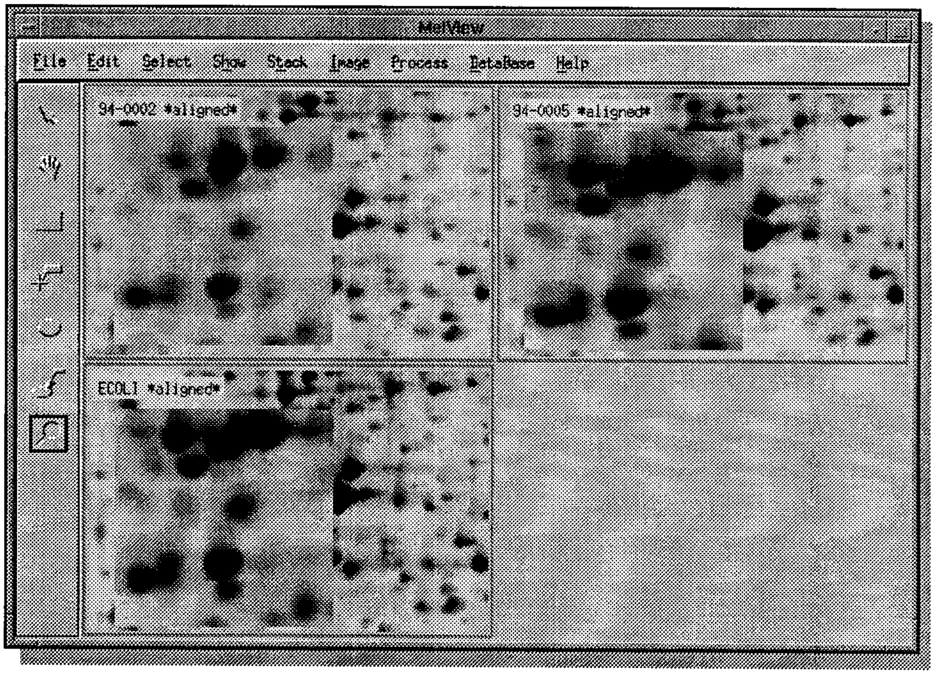


Figure 16-10. The Magnify tool applied to aligned gels.

Matching gels

We are now going to compare the features (spots) in gels 94-0002 and 94-0005 with those in the ECOLI gel. This process is called **gel matching**. It is going to pair each feature in each of the two gels with the corresponding features in ECOLI. The matching algorithm in Melanie II is able to automatically match features without any manual intervention. Nevertheless, when gels are quite dissimilar, or just to speed up the matching process, you may manually define starting feature pairs. As we already have defined landmarks in all three gels, we are going to use these as starting pairs.

- Select all three gels, using the Select tool (or Control-B).

- Choose *Process → Match → Match Landmarks.* This will pair wise match the features containing landmarks (in our case landmarks L1 to L5).
- Choose *Process → Match → Match Gels.* The Match Gels window pops up, asking for the **reference** gel, that is the gel the other selected gels are going to be matched to. Choose ECOLI and click *OK.* This will now compare all features in gels 94-0002 and 94-0005 to the features in gel ECOLI. Depending on the type of computer you are using, this might take a few minutes.
- After matching has been completed (and if it was successful), press Control-S to save the pairs.
- There are various ways of looking at the result. First choose *Show → Gels → Report On Matches.* A window shows how many features have been paired across the gels.
- The second way to observe the matching result is to select a few features in the reference gel (ECOLI) using the Feature tool, then to choose *Select → Pairs → For Features.* This will select (and display) the paired features in the other two gels (Fig. 16-11).

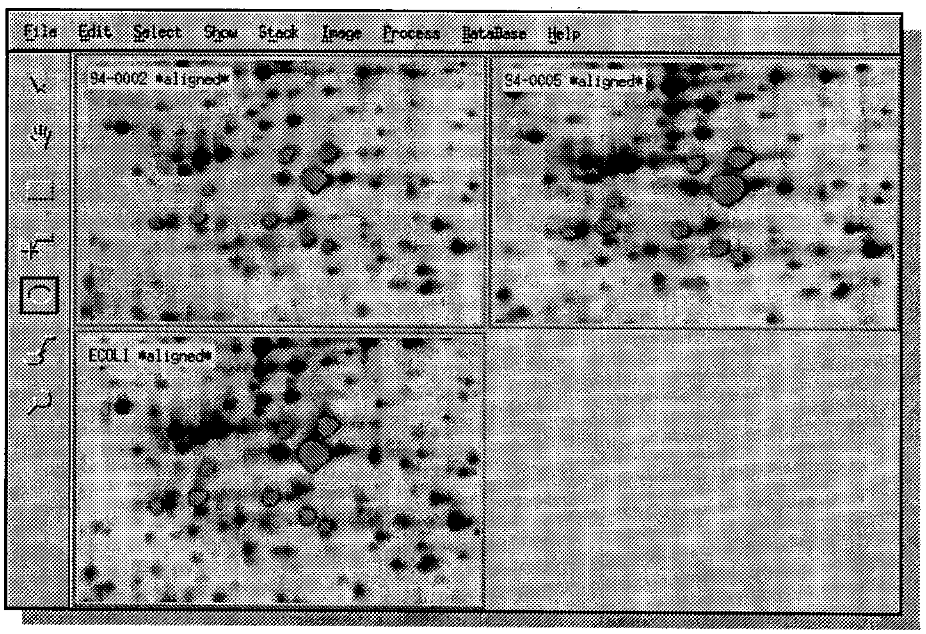


Figure 16-11. Selected paired features across the three gels.

- Third, you may select two gels, then select a few features on them, then *Show → Pairs → Report On Pairs* to get the features IDs of the pairs.

- Finally, select gel 94-0005 and ECOLI, then Control-Z and Control-T. The two selected gels are now stacked. Choose *Show → Pairs → Show Pairs.* All the paired features will be shown by small blue vectors (Fig. 16-12).

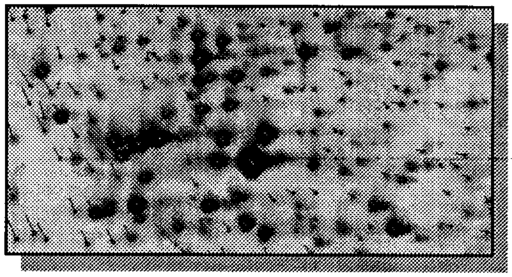


Figure 16-12. Matched gels: blue vectors indicate paired features.

Creating a synthetic gel

Let us now create a synthetic gel by merging the three gels. The gels have first to be pair wise matched. Therefore, you still have to match 94-0002 to 94-0005.

- Choose *Show → Gels → All Gels,* then select gels 94-0002 and 94-0005. Choose *Process → Match → Match Landmarks,* then *Process → Match → Match Gels.* Wait for the matching to finish.
- Press Control-S to save the pairs.
- Choose *Image → Create Gels → Create Synthetic Gels.* Then choose ECOLI as the reference gel. This means that the new synthetic gel will be based on ECOLI, adding spots from the other gels.
- In the next pop-up window, give a name for the synthetic gel, or accept the proposed one (keep the path). We will just call it *Synthetic.* Once created, the synthetic gel will be displayed. You might display it in the full screen to have a better look at it (select the gel, then Control-Z). See Figure 16-13.

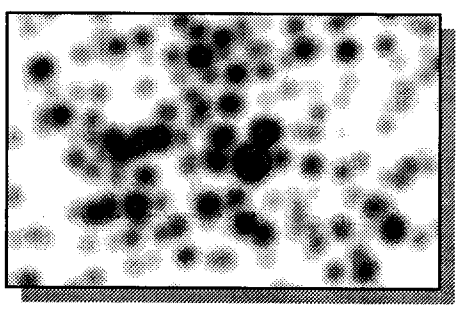


Figure 16-13. A part of the synthetic gel, displayed with GEColor.

Matching gels to the synthetic gel

We are now going to use the synthetic gel as a reference gel, because it contains most of the important features from the three other gels.

- Add the L1 to L5 landmarks to the synthetic gel (See *Aligning gels* on page 16-9).
- Match the three gels to Synthetic: Control-B, then *Process —> Match —> Match Landmarks*. Choose Synthetic as the reference gel. Then *Process —> Match —> Match Gels*, and choose Synthetic as reference gel again. The three gels now should be matched to Synthetic (Fig. 16-14).

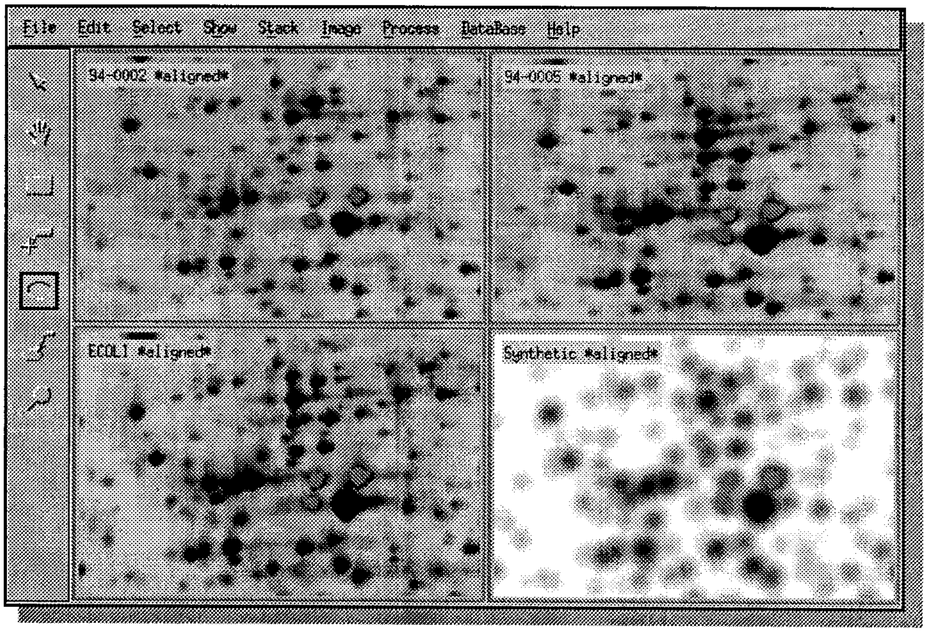


Figure 16-14. Three selected pairs.

- Once the gels have been matched, you may unalign the gels, in order to work with the original images. All pairs will be preserved: *Process —> Align Gels —> Unalign Gels.*

Grouping features

When several gels have been matched to one reference gel, the paired features form so-called **groups**. A group is a set of features, all paired together. Figure 16-14 shows three such groups. Groups may be visualized in reports and in histograms. They may also be exported to files, in order to be used in external programs such as Excel.

- To get a report on groups, choose *Show → Groups → Report On Groups.* Select a gel as reference gel, for example ECOLI. If a master gel has already been set, the latter will be taken as reference gel (See *Using a master gel* on page 16-15). Select the value type, for example the optical density (OD) or the feature area.
- To get histograms on groups, choose *Show → Groups → Histograms On Groups.* Figure 16-15 shows histograms for two of the selected features. The

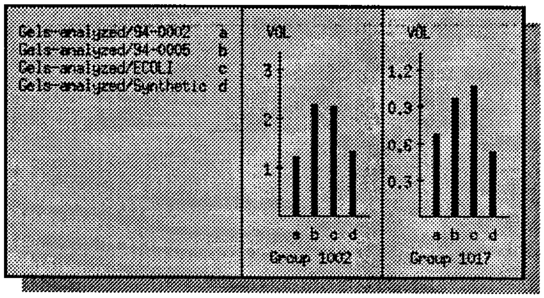


Figure 16-15. Histograms for two feature groups.

IDs of the selected groups are shown on the reference gel.

- To export the groups, choose *File → Export.* Select the export format. You may export to *Outgroups* format (this is the Melanie 1 table format), Excel tables and Excel lists.

Using a master gel

The ECOLI gel is a SWISS-2DPAGE master gel. That is several proteins have been identified on the 2-DE map. Corresponding entries exist in the SWISS-2DPAGE database on the Expasy World-Wide Web server. Each identified protein has a label associated to the corresponding feature.

- Choose *Database → Set → Master.* Select ECOLI. The ECOLI is now defined as the master gel. It's name is printed with a red background.
- Select ECOLI with the Select tool, then Control-Z, to have it alone on the screen.
- Choose *Show → Labels → Show Labels.* All existing labels will appear, showing the SWISS-2DPAGE accession number of each identified protein (Fig. 16-16).

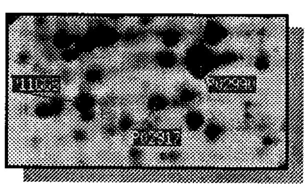


Figure 16-16. Labels showing the proteins' accession numbers.

- Choose *Show → Labels → Show Name*. The SWISS-2DPAGE short protein names will be shown instead of the accession numbers (Fig. 16-17).

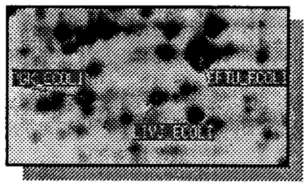


Figure 16-17. Labels showing the proteins' short names (ID).

- Choose *Show → Gels → All Gels* to display all gels again, then Control-B, then *Select → Features → All Features*. Finally, choose *Database → Labels → Add From Master*. Labels will be propagated to paired features. If necessary, choose *Show → Labels → Show Labels* (Fig. 16-18).

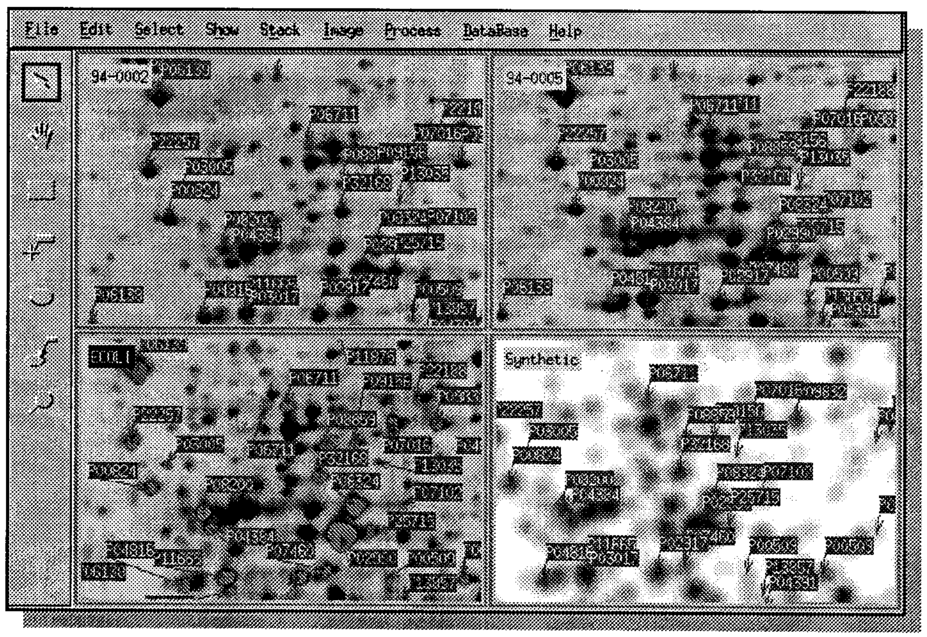


Figure 16-18. Labels propagated to other gels.

Using Internet

If your computer is connected to the Internet, you may directly access the SWISS-2DPAGE or SWISS-PROT databases on the Expasy server, as well as other databases on various servers.

- Choose *Database → Set Master*. Choose ECOLI. Then choose *Database → Set Browser* and verify that the right name and path are given for Netscape. Choose *Database → Set Server*. Enter **expasy.hcuge.ch** and finally, choose *Database → Set Database* and give **SWISS-2DPAGE**.

- Using the Feature tool (fifth tool on the tool bar), select one feature corresponding to an identified protein. Then choose *Database* → *Query Server*. MelView will remotely control Netscape Navigator® and request the corresponding entry from the SWISS-2DPAGE database on Expasy.

Exiting MelView

To exit MelView, do *File* → *Quit*.

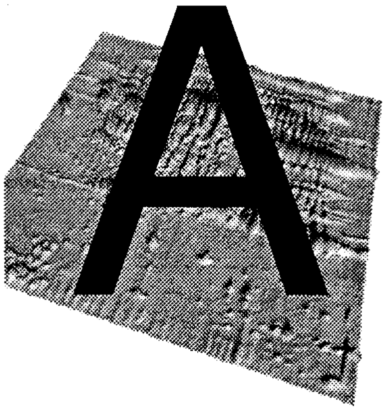
Other Melanie II features

Many other options and features are available in the MelView program, including:

- Import/export images from and to TIFF, GIF, PPM, etc.;
- Filters for smoothing, contrast enhancement, background subtraction, etc.;
- Managing gel images (flipping, scaling, duplicating, etc.);
- Landmark, feature and label selection by various criteria;
- Landmark, feature, label, pair and group editing.

PART VI

APPENDIXES



ALGORITHMS

This chapter presents the outline of algorithms implemented in the Melanie II software. For a detailed description, see [reference 4].

Detecting and quantifying features

The technique used in Melanie II for feature detection and quantification combines spot detection by a non-parametric method (NP), with spot quantification by a parametric method (P). This combined technique achieves detection and quantification in two successive operations as described in the following sections.

Spot detection

Spots are detected using a NP method based on the Laplacian and second derivatives:

Given a 2-DE image $I(x, y)$, a saturation threshold T, a spot S_i, and a point $\vec{p} = (x, y)$. To decide whether point $\vec{p}$ is part of a spot, two decision rules have been defined, depending on the intensity $I(\vec{p})$ at point $\vec{p}$:

1 $I(\vec{p}) \leq T$; this corresponds to a non-saturated value:

$$\vec{p} \in S_i \Leftrightarrow \begin{cases} -\Delta I(\vec{p}) - L < 0 \\ MIN\left(\frac{\partial^2}{\partial x} I(\vec{p}) - R, \frac{\partial^2}{\partial y} I(\vec{p}) - C\right) > 0 \quad \text{when} \quad -\Delta I(\vec{p}) - L \geq 0 \end{cases}$$

(Eq. A.1)

2 $I(p) > T$; this is a saturated value:

$$\vec{p} \in S_i \Leftrightarrow MIN\left(\frac{\partial^2}{\partial x} I(\vec{p}), \frac{\partial^2}{\partial y} I(\vec{p})\right) > 0 \,, \qquad \text{(Eq. A.2)}$$

where L, R, C are three small positive constants. L represents the threshold for the Laplacian of image I, R is the threshold for the second derivative along axis x, and C is the threshold for the second derivative along axis y. 2-DE images may be saturated due to the staining technique and/or image acquisition. Figure A-1 illustrates a 2-DE spot profile, where the ideal virtual shape is truncated due to the saturation effect.

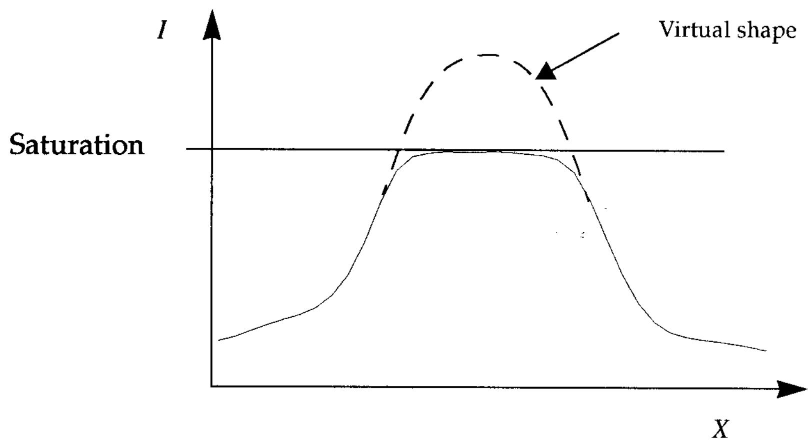


Figure A-1. Saturation of a 2-DE spot profile due to the staining technique or image acquisition device.

Threshold T is the value above which a pixel is considered to be saturated. The threshold T and laplacian ΔI in (Eq. A.1) and (Eq. A.2) are defined as follows:

$$\Delta I(p) = -\left(\frac{\partial^2}{\partial x} I(p) + \frac{\partial^2}{\partial xy} I(p)\right) \qquad \text{(Eq. A.3)}$$

$$T = max(I) - \frac{saturation}{100} \times (max(I) - min(I)) \qquad \text{(Eq. A.4)}$$

where *saturation* is a positive value between 0 and 100. *saturation* = 100 means that no pixel is saturated in the whole image.

Spot quantification

The second step in the combined technique may be achieved using one of two alternative methods: direct quantification (described in this section) and quantification by Gaussian modeling (described in *Gaussian fitting on page A-13*). In the first method, the area, volume, percent volume, optical density (OD) and percent OD of each spot are computed by considering the pixels inside their detected shape. These values are calculated as follows:

1 Spot area (AREA):

$$\mathrm{AREA} = \text{number of pixel} \times \text{pixel area}$$

2 Spot optical density (OD):

$$\mathrm{OD} = \max_{x, y \in spot} (I(x, y))$$

3 Spot percent optical density (%OD)

$$\%\mathrm{OD} = \frac{\mathrm{OD}}{\sum_{s=1}^{n} \mathrm{OD}_s} \times 100,$$

where OD_s is the optical density of spot s in a gel containing n spots.

4 Spot volume (VOL):

$$\mathrm{VOL} = \sum_{x, y \in spot} I(x, y)$$

5 Spot percent volume (%VOL):

$$\%\mathrm{VOL} = \frac{\mathrm{VOL}}{\sum_{s=1}^{n} \mathrm{VOL}_s} \times 100,$$

where VOL_s is the volume of spot s in a gel containing n spots.

Aligning

Image alignment is performed by polynomial image warping. The warping is performed in four stages. First, control points or landmarks are selected on one image (the reference image). Second, corresponding control points are given in the second image (the data image which is to be aligned to the reference image). Third, the warping transformation is determined by solving equation (Eq. A.6). Given a set of control points, we solve this equation using Least Squares minimization. In the fourth stage, warping is performed on the data image using the following transformation:

$$(x, y) \rightarrow (u(x, y), v(x, y)), \qquad \text{(Eq. A.5)}$$

where (x, y) are the pixel coordinates in the original data image, (u, v) are the pixel coordinates in the resulting warped data image aligned to the reference image. $u(x, y)$ and $v(x, y)$ are two polynomial functions (one for each axis) determined by the Least Squares method in order to minimize the distance between the transformed control points in the data image and those in the reference image.

To perform this, the following two systems of linear equations (u and v are linear with respect to a_i and b_i) have to be solved:

$$u = \sum_{i=0}^{n}\sum_{j=0}^{i} a_{ij}x^i y^{j-i} \qquad v = \sum_{i=0}^{n}\sum_{j=0}^{i} b_{ij}x^i y^{j-i} , \qquad \text{(Eq. A.6)}$$

where (x, y) are the landmark positions in the reference image, (u, v) are the positions of the corresponding landmarks in the data image, and a_{ij} and b_{ij} are the polynomial coefficients to be determined. The polynomial order n is specific to the problem. For M control points and polynomial functions of order n, the coefficients a_{ij} and b_{ij} are determined by two systems of M linear equations with T variables, where $T = (n+2)(n+1)/2$. The Least Squares method may be used to solve these equations only if $M \geq T$.

The image warping is computed by mapping the original image using the functions of (Eq. A.6). The inverse of this transformation is frequently required when performing comparisons. In the general form, the inverse of the function is difficult to compute. In the Melanie II system, only the first-order, second-order, third-order polynomials and one variable polynomials have been used. This means that warping along horizontal axis depends only on the landmarks' horizontal positions $u(x)$, while $v(y)$ depends only on the vertical ones. This simplifies the mapping of images, in which horizontal and vertical lines are mapped into horizontal and vertical lines respectively. Using this transformation, lines are mapped into curbs. With these simplifications, (Eq. A.6) becomes:

- First-order:

$$u = a_0 + a_1 x \qquad v = b_0 + b_1 y , \qquad \text{(Eq. A.7)}$$

- Second-order:

$$u = a_0 + a_1 x + a_2 x^2 \qquad v = b_0 + b_1 y + b_2 y^2 , \qquad \text{(Eq. A.8)}$$

- Third-order:

$$u = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \qquad v = b_0 + b_1 y + b_2 y^2 + b_3 y^3 . \qquad \text{(Eq. A.9)}$$

where (u, v) are the landmark positions in the image that is going to be aligned, (x, y) are the coordinates in the reference image, and a_i, b_i are the polynomial warping coefficients to be computed by minimizing the error between the positions (u, v) and (x, y). Given M landmarks in each gel, this can be expressed as follows:

$$error_x = \sum_{i=0}^{M} (u_i - (a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i^3))^2 \rightarrow minimum$$

$$error_y = \sum_{i=0}^{M} (v_i - (b_0 + b_1 y_i + b_2 y_i^2 + b_3 y_i^3))^2 \rightarrow minimum \quad , \qquad \text{(Eq. A.10)}$$

where (u_i, v_i) are the landmark positions in the first gel, and (x_i, y_i) are the corresponding landmark locations in the second gel.

The inverse of the above three expressions can easily be computed by algebraic methods. Figure A-2 illustrates the approximation of the horizontal position of landmarks in two different gels, using: (a) first-order, (b) second-order, (c) third-order polynomials; (d) the inverse third-order polynomial warping which in this case cannot align the two gels in a better way.

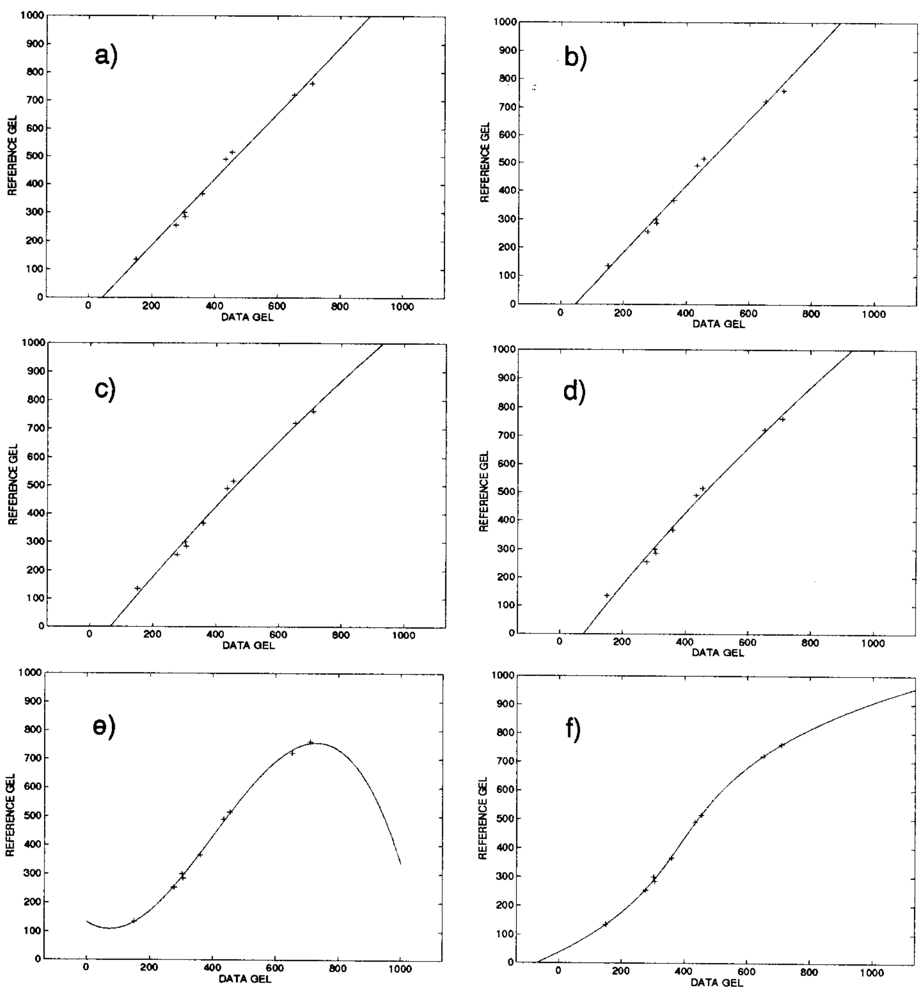


Figure A-2. One variable polynomial warping functions of: (a) first-order, (b) inverse first-order, (c) second-order, (d) inverse second-order, (e) third-order, (f) inverse third-order polynomial. Inverse polynomial mapping in this example aligns the gels best, and is monotone.

The minimum number of landmarks to compute a_i, b_i depends on the polynomial order. A polynomial order of n requires at least $n+1$ landmarks. For example, a first-order transformation requires at least two landmarks, while a third-order polynomial requires at least four landmarks. Figure A-3 illustrates one variable polynomial transformation of a grid representing a 2-DE image. This was performed using four landmarks and two polynomial functions of one variable.

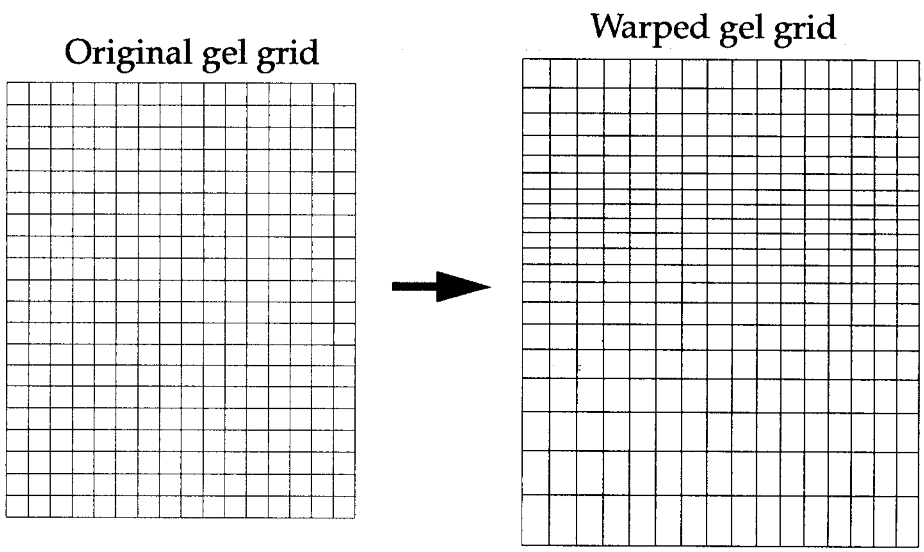


Figure A-3. : Transformation of a gel grid using two polynomial warping functions of one variable. A second order polynomial warping along X and third order polynomial warping along Y were used, as well as four landmarks.

A set of 2-DE images are aligned in the following steps:

1 A set of landmarks are selected on each image. In Figure A-4, landmarks are designate by labels $p1$ to $p9$.

2 The polynomial order for the horizontal and vertical axes are chosen, and one gel is selected as reference gel from the set of images. Then, the rest of the gels is aligned to the reference gel. In Figure A-4 the top two images illustrate

two gels that are not superimposable, while at the bottom, both gels are shown after alignment with a third and second order polynomial warping transformation along the horizontal and vertical axis, respectively.

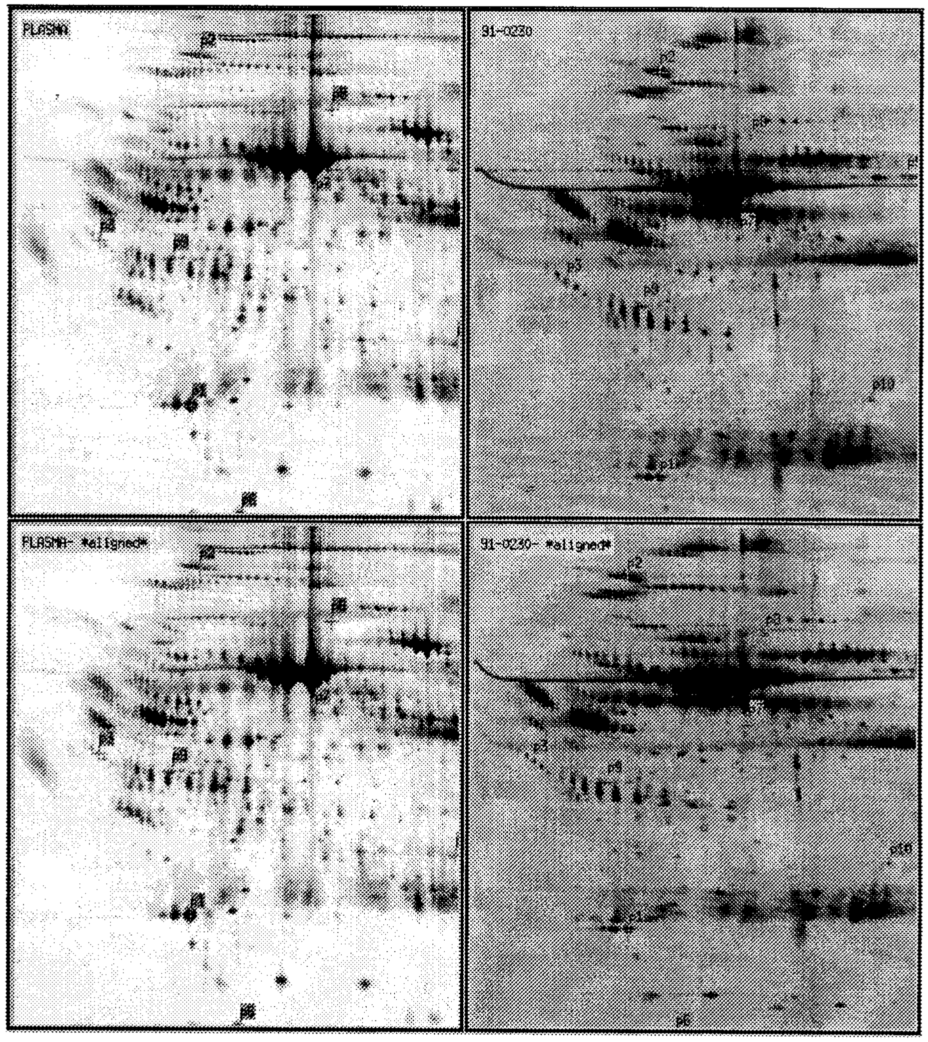


Figure A-4. : (top left) 2-DE reference image, (top right) image to warp, (bottom) both images after second order polynomial warping along Y, and third order warping along X. Nine landmarks and bilinear interpolation have been used. Visual correspondences are evident in the aligned images.

Matching

Matching algorithm

A powerful matching algorithm has been implemented in Melanie II. It is an improved version of the algorithm described in [reference 5]. Matching between two images is accomplished according to the following basic steps:

1 For each spot in a gel, a list of neighboring spots (cluster) is selected and associated to that spot (Fig. A-5). The central spot of a cluster is called the primary spot, and its surrounding spots are the secondary spots of the cluster. A given spot is considered to be part of a cluster if its centroid is inside a circle with a fixed radius. This radius is estimated by considering the image dimensions, the number of spots in the image, and a given minimum number of spots for a cluster. Clusters are represented in polar coordinates, and are built before matching, in order to be quickly compared.

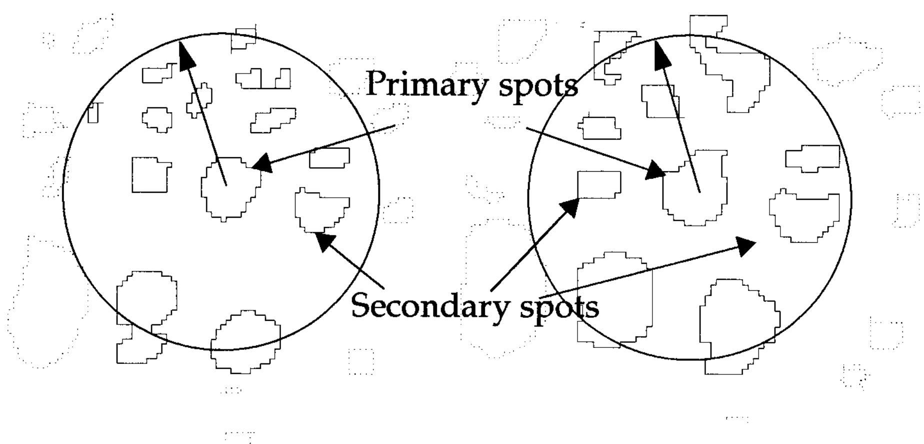


Figure A-5. Clusters are formed by considering the neighboring spots for each spot in both images. Secondary spots in each cluster are represented in polar coordinates relative to the primary spot.

2 The method starts to match clusters corresponding to the input pairs, if some exist. Otherwise, clusters whose primary spot has the highest intensity value are matched first.

3 Clusters are compared using a similarity measure based on a probabilistic method. This method is similar to calculating the probability K_m that if N darts land randomly on a board with an area A, at least m darts will land on the target whose area is T, with $m \le N$. This can be calculated by the expression:

$$K_m = \sum_{h=m}^{N} \binom{N}{h} p^h (1-p)^{N-h}, \text{ with } \binom{N}{h} = \frac{N!}{h!(N-h)!}, \qquad \text{(Eq. A.11)}$$

where $p = T/A$, which represents the probability that a dart will land on the target in one single try. In a similar way, the probability for the next random hit within a cluster where $m - 1$ spots have already been matched is given by the following equation:

$$p_m = \frac{A_s - A_{m-1}}{A_c - A_{m-1}}, \qquad \text{(Eq. A.12)}$$

where $m \geq 1$, A_s is the sum of the areas of the secondary spots in the cluster (highlighted spots in Figure A-5) A_c is the total area bounded by the cluster (circular area) and A_{m-1} is the total area of the matched spots in the cluster. Then, the probability that at least m spots were matched in the two clusters by a random process in N trials, where N is the number of spots in one of the clusters, can be estimated by the following expression:

$$K_m \approx \sum_{h=m}^{N} \binom{N}{h} p_G{}^h (1 - p_G)^{N-h}, \qquad \text{(Eq. A.13)}$$

where

$$p_G = \left|\prod_{i=1}^{m} p_i\right|^{\frac{1}{m}} \text{ is the geometric mean.} \qquad \text{(Eq. A.14)}$$

(Eq. A.13) is a good approximation of the real probability of randomly finding m or more matches in N tries:

$$K_m = \sum_{h=m}^{N} \binom{N}{h} \prod_{i=1}^{h} p_i \prod_{i=h+1}^{N} (1 - p_i), \qquad \text{(Eq. A.15)}$$

where p_i is defined by (Eq. A.12).

4 In given situations, the comparison of primary clusters may not be sufficient and mismatching may occur. In this case, secondary clusters are compared. Thus, more spots are considered, based on the idea that if matches in the primary cluster are correct, then the clusters surrounding them can also be matched.

5 A consistency check is performed in order to detect possible mismatching. Most of them occur among the set of most probable matches. Assuming that the great majority of matches are correct, the parameters of the following rigid transformation are calculated by considering the location of the matched spots and the Least Squares minimization:

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} t_y \\ t_y \end{bmatrix} + \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \qquad \text{(Eq. A.16)}$$

For a simple rotation and translation, this equation becomes:

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \end{bmatrix} + \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \qquad \text{(Eq. A.17)}$$

The test consists of checking whether or not

$$AxD - BxC \approx 1.0, \qquad \text{(Eq. A.18)}$$

which corresponds to a rotation of (Eq. A.17). For each primary cluster, parameters A,B,C and D are computed by considering each set of three matched spots in the cluster. If the evaluation of (Eq. A.18) is 1.0 ± 0.25 and $\theta = \pm 10$ degrees, then the three spots are considered to be *good*.

6 A transformation tries to match the remaining spots. Using the *good* matched spots in two clusters and the Least Squares method, parameters for (Eq. A.16) are computed. The location of unmatched spots in one gel are mapped using the resulting transformation. If the transformed location of a spot in the first gel lies within the boundaries of a spot in the second gel, given a proximity criteria, then both spots are matched. If unmatched spots cannot be matched, the criteria is relaxed. The unmatched spots are successively examined until no new matches are found. Multiple matches can also be accepted according to the user's wish.

Good / bad pairs

While the above described matching algorithm has been optimized for 2-DE gel images, mismatches may sometimes occur. Good matches may be discriminated from bad ones by the use of a quality measure. This measure is computed for each pair according to the following steps:

1 a neighboring region is considered around each pair,

2 corresponding regions are superimposed in order to minimize the vector lengths in the region, using Least Squares,

3 each pair of spots in the selected region is given a score representing the number of times it has been larger than a given threshold, otherwise it is marked *bad*, which reduces the score.

4 a pair is considered to be *good* if its final score is larger than a second threshold, otherwise it is considered *bad*. The threshold represents the pair quality.

Good and bad pairs may be selected for further operations. The quality pair evaluation keeps track of possible image alignment.

Creating synthetic gels

Synthetic gels may be created by merging a set of gel images that contain at least three pair wise matched gels. A synthetic gel is created as follows:

1 A reference gel is selected from the set of gels. This gel plays a special role in the creation of the synthetic gel, since the spot positions in the reference gel serve as representative positions for the synthetic gel.

2 Each spot s_R in the reference gel is tested to verify that it is *well matched*, that is, whether two other spots $s_1 \in G_1$ and $s_2 \in G_2$ exist, where G_1 and G_2 are two other gels from the given set of gels, such as $s \leftrightarrow s_1$ and $s \leftrightarrow s_2$ ($a \leftrightarrow b$ means that spot a is matched with spot b). This operation forms triangles of matched spots that constitute the starting groups (Fig. A-6).

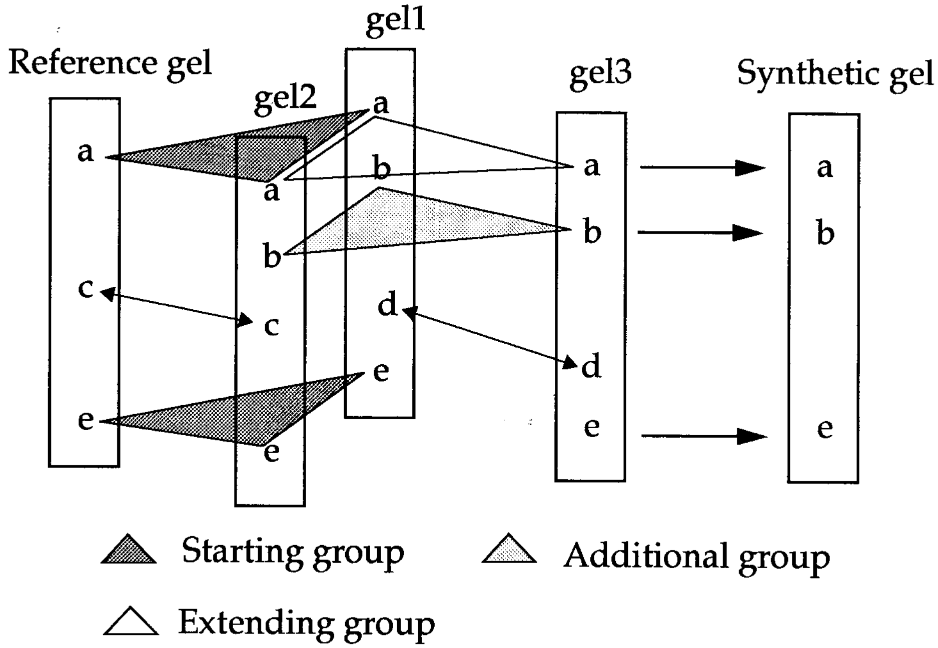


Figure A-6. Groups of spots are formed by first looking for triangle pairs. Then, after extending them, the synthetic gel is created. Each spot in the synthetic gel represents a formed group.

3 Starting groups are extended by adding supplementary spots, according to a connectivity test. This test verifies whether a spot is matched with at least one other spot in the initial group.

4 After examining all spots in the reference gel, additional groups are formed, starting with spots in the second gel that are not yet part of a group. Then, the same procedure repeats to form additional groups with the remaining gels.

5 A synthetic gel image is created that contains the same number of spots than the number of determined groups. Each spot in the synthetic gel represents a group of *well matched* spots. These representative spots are selected by considering positions, optical densities, and shapes of spots in a given group, according to the following procedure:

- Position: if the group has one spot in the reference gel, the position of this spot is taken. Otherwise, a set of neighbors of the first spot that are included in the group, and that have pairs on the reference gel are considered. Then, the spot closest to the first included spot is selected, and the translation vector between both spots is computed. Finally, the position of the spot representing the group on the reference gel is computed by adding this translation to the corresponding closest spot on the reference gel.
- Intensity: this value is equal to the average optical density value among all spots in the group.

- Shape: the spot shape is copied from the spot in the group that has got the area closest to the average area in the group.

6 Finally, the dimensions of the synthetic gel are recomputed, since all the spot coordinates have to be positive.

Gaussian fitting

This section describes the elements of 2D-Gaussians. In the next section, the Gaussian fitting algorithm for modeling spots is presented.

Gaussian model

The general form of the two-dimensional Gaussian equation is:

$$g(x, y) = e^{ax^2 + bxy + cy^2 + dx + ey + f} + h, \text{ with } b^2 - 4ac < 0, \qquad \text{(Eq. A.19)}$$

or in its simple form:

$$g(X, Y) = Ae^{\left(\frac{X - X_c}{\sigma_X}\right)^2 + \left(\frac{Y - Y_c}{\sigma_Y}\right)^2} + B, \qquad \text{(Eq. A.20)}$$

where (X_c, Y_c) is the Gaussian's center, A its amplitude, B its asymptotic horizontal plane, (σ_X, σ_Y) the diffusion coefficients along the principal axis.

Figure A-7 shows a graphic representation of (Eq. A.19). Each spot in a gel may be represented by one Gaussian function at position (x_c, y_c), with amplitude A, background B, the spread (σ_X, σ_Y) along the principal axes, and an orientation angle θ of its principal components.

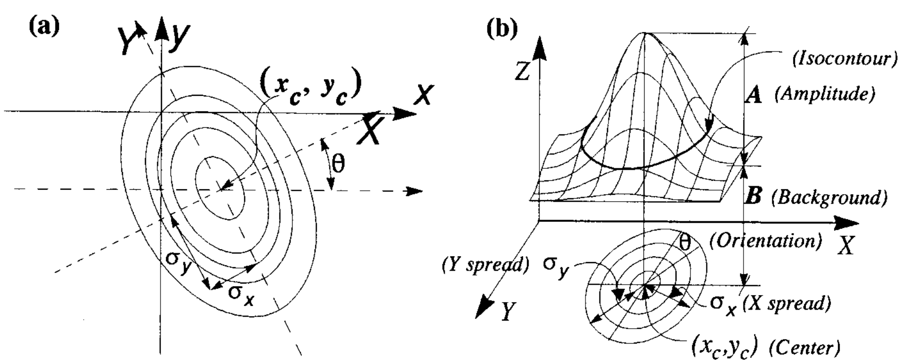


Figure A-7. (a) 2D-Gaussian contour levels with center (x_c, y_c) and rotation θ (X-Y axes). (b) 2D-Gaussian parameters representing a 2-DE spot gel.

Gaussian fitting

The general fitting problem may be described as follows:

Given a function F, a family of functions Φ, a discrete function I to be fitted, and a domain Ω, defined as:

$$F(\vec{a}) : \quad R^m \rightarrow \Phi$$
$$\phi(\vec{p}) : \quad \Omega \rightarrow R$$
$$I(\vec{p}) : \quad \Omega \rightarrow R \quad \text{, for } \forall\phi \in \Phi \text{ and } \Omega \subset R^n, \qquad \text{(Eq. A.21)}$$

find $\vec{a}_\infty \in R^m$, such that:

if $\phi_\infty = F(\vec{a}_\infty)$,

then: $\|\phi_\infty - I\| = \mathrm{Min}_{\vec{a} \in R^m}\|F(\vec{a}) - I\| = \|F(\vec{a}_\infty) - I\|$

when using a distance norm,

$$\text{or: } \|\phi_\infty - I\| = \mathrm{Min}_{\vec{a} \in R^m}\sqrt{\sum_{\vec{p} \in \Omega}(\phi_{\vec{a}}(\vec{p}) - I(\vec{p}))^2} \qquad \text{(Eq. A.22)}$$

when using the second order distance norm.

The function ϕ_∞ is the best approximation (best fit) of a discrete function I for a given distance norm.

Considering several vectors $\vec{a} \in R^7$ where $\vec{a} = (a, b, c, d, e, f, h)$ and the expression in (Eq. A.19), it is possible to define a set of Gaussian functions that form a 2D-Gaussian family of functions. The fitting problem using the second order norm and the 2D-Gaussian family of functions is defined as:

find $\vec{a}_\infty \in R^7$, such that:

$$\|G(\vec{a}_\infty) - I\| = \mathrm{Min}_{\vec{a} \in R^m}\sqrt{\sum_{\vec{p} \in \Omega}\left(G_{\vec{a}}(\vec{p}) - I(\vec{p})\right)^2} = \sqrt{\sum_{\vec{p} \in \Omega}\left(G_{\vec{a}_\infty}(\vec{p}) - I(\vec{p})\right)^2}, \qquad \text{(Eq. A.23)}$$

where $G(\vec{a}) = e^{a_1x^2 + a_2xy + a_3y^2 + a_4x + a_5y + a_6} + a_7$.

Figure A-8 shows the elements of the fitting process.

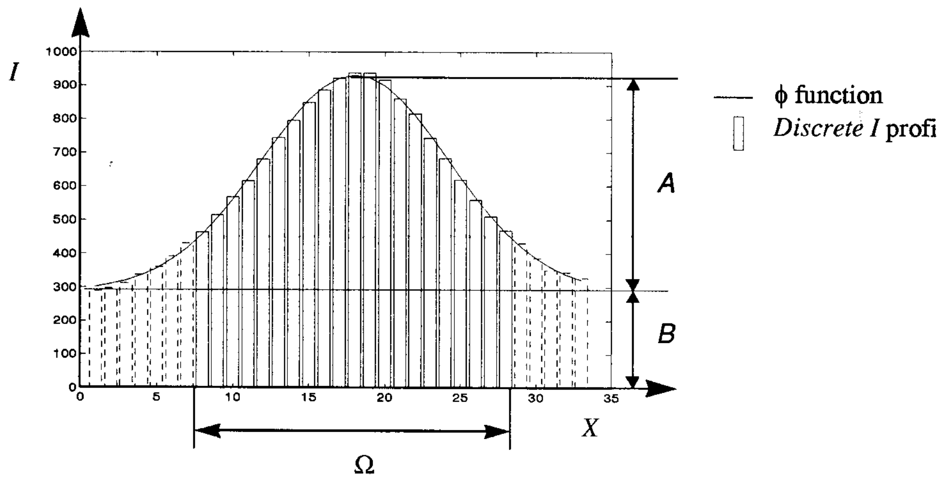


Figure A-8. Function ϕ, discrete function I, fitting domain Ω, background B, and distance between ϕ and I.

Several minimization methods may be used to find the minimum of (Eq. A.23). Most of them need a first approximation to start the iterative search of the real minimum. This means that the efficiency of these methods strongly depends on the first approximation. An efficient method to calculate a good starting point is to approximate (Eq. A.23):

Considering a_7 to be a constant, and $(I(x, y) - a_7) > 0$ for $\forall (x, y) \in \Omega$, then (Eq. A.23) may be approximated as follows:

$$\operatorname{Min}_{\vec{a} \in R^m} \sqrt{\sum_{(x, y) \in \Omega} (a_1 x^2 + a_2 xy + a_3 y^2 + a_4 x + a_5 y + a_6 - \log(I(x, y) - a_7))^2} .$$

which can be solved by the following system of linear equations:

$$Ax = b \quad \text{, or} \quad \begin{bmatrix} \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ x_i^2 & x_i y_j & y_j^2 & x_i & y_j & 1 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix} \times \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \end{bmatrix} = \begin{bmatrix} \cdots \\ \cdots \\ \log(I(x_i, y_j) - a_7) \\ \cdots \\ \cdots \end{bmatrix} . \qquad \text{(Eq. A.24)}$$

The dimension of matrix A is $[cardinal(\Omega) \times 6]$. (Eq. A.24) can be solved by the Least Squares method:

$$A^t A x = A^t b, \qquad \text{(Eq. A.25)}$$

where $A^t A$ is a $[6 \times 6]$ symmetric matrix.

The solution of (Eq. A.25) is illustrated by Figure A-9 for a 2-DE spot profile. This is an approximation of the exact solution of (Eq. A.23).

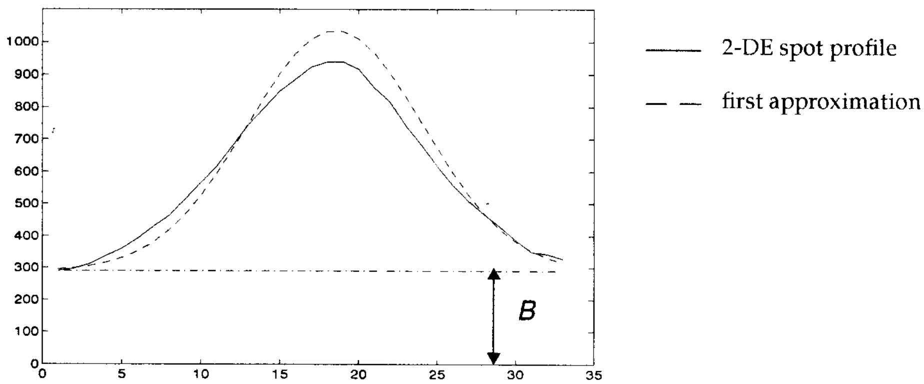


Figure A-9. First approximation of 2-DE spot profile by fitting the $\log(I(x) - B)$ where $B = min(I) + 1$ with a second degree polynomial.

Starting from this first approximation, (Eq. A.23) can be solved using a non-linear multidimensional minimization algorithm. Non-linear methods may be classified in two groups: methods using the first derivative of the distance function, and methods not using it. Methods in the first group are called the conjugate gradient methods and the quasi-Newton or variable metric methods. The second ones include the downhill simplex method and the Powell method. The Polak-Ribiere's variant of the conjugate gradient method has been chosen as a minimization method that can be described as follows:

Given a function F and an initial point $\vec{p}_0$ (a starting point for the first iteration), move from point $\vec{p}_i$ to point $\vec{p}_{i+1}$, as often as needed, by minimizing F along the line from $\vec{p}_i$ in the direction of the gradient $-\nabla F(\vec{p}_i)$. Stop when $\|F(\vec{p}_{n+1}) - F(\vec{p}_n)\| < \varepsilon$. At this point in time, $\vec{p}_{i+1}$ is the point where F has a minimum value with tolerance ε. The convergence and the result of this algorithm depend on the choice of the initial point $\vec{p}_0$.

The fitting algorithm is made of two steps. First, assuming h is equal to a constant equal to $min(I)$, the initial point $\vec{p}_0$ is computed by using the linear approximation method described above. Second, the real minimum is calculated using the conjugate gradient algorithm, with:

$$F(\vec{p}) = \sum_{(x,y) \in \Omega} \left(e^{ax^2 + bxy + cy^2 + dx + ey + f} + h - I(x,y)\right)^2,$$

where $\vec{p} = (a, b, c, d, e, f, h)$,

and the gradient of $F(\vec{p})$:

$$\nabla F(\vec{p}) = \left(\frac{\partial}{\partial a}F(\vec{p}), \frac{\partial}{\partial b}F(\vec{p}), \ldots, \frac{\partial}{\partial h}F(\vec{p})\right) = 2\begin{bmatrix} \sum_{x,y \in \Omega} x^2 \cdot G(x,y)(G(x,y) + h - I(x,y)) \\ \sum_{x,y \in \Omega} xy \cdot G(x,y)(G(x,y) + h - I(x,y)) \\ \sum_{x,y \in \Omega} y^2 \cdot G(x,y)(G(x,y) + h - I(x,y)) \\ \sum_{x,y \in \Omega} x \cdot G(x,y)(G(x,y) + h - I(x,y)) \\ \sum_{x,y \in \Omega} y \cdot G(x,y)(G(x,y) + h - I(x,y)) \\ \sum_{x,y \in \Omega} G(x,y)(G(x,y) + h - I(x,y)) \\ \sum_{x,y \in \Omega} (G(x,y) + h - I(x,y)) \end{bmatrix}, \quad \text{(Eq. A.26)}$$

where:

$$G(x, y) = e^{ax^2 + bxy + cy^2 + dx + ey + f} + h.$$

Figure A-10 illustrates the final solution of the conjugate gradient minimization algorithm, using the first linear approximation shown in Figure A-9. The combination of both methods gives an optimal fit.

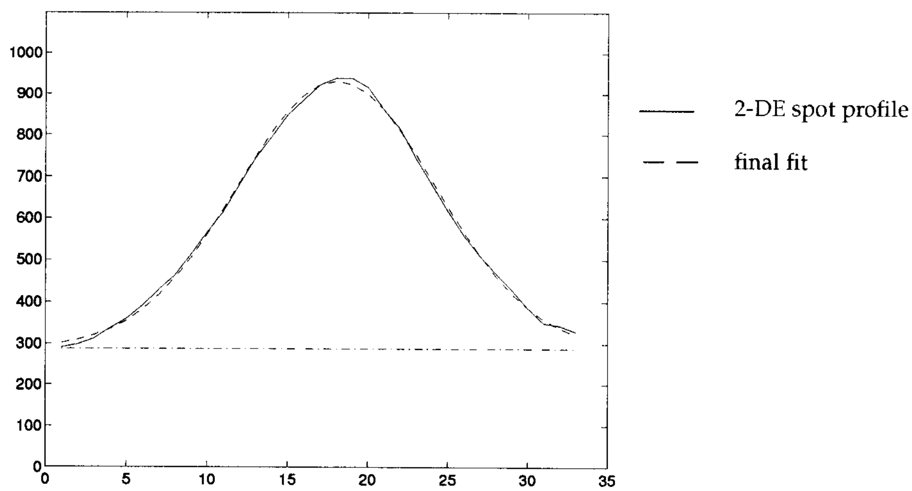


Figure A-10. Final solution of the conjugate gradient minimization algorithm that best fits a 2-DE spot profile.

Filtering gel images

Smoothing

Melanie II implements four smoothing filters to reduce high frequency noise in images: Gaussian, diffusion, polynomial and adaptive smoothing. The last two methods present the advantage of not affecting image contrast.

Gaussian smoothing

Gaussian smooth of a 2-DE image is performed by convolving it with the following 3x3 mask operator:

$$mask = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \quad \text{(Eq. A.27)}$$

Diffusion smoothing

Isotropic diffusion is a recursive low-pass filter that performs smoothing in two successive passes, using two different recursive operators. For one-dimensional signals, the filter has the following form:

$$I^{(t+1)}(x) = \frac{1}{2}(I^{(t+1)}(x-1) + I^{(t)}(x+1)) \quad \text{(Eq. A.28)}$$

as a left to right operator, then

$$I^{(t+2)}(x) = \frac{1}{2}(I^{(t+1)}(x-1) + I^{(t+2)}(x+1)) \quad \text{(Eq. A.29)}$$

as a right to left operator, where $I^{(t)}(x)$ represents the intensity value of a one-dimensional signal at position x after t iterations.

Both operators establish an approximation to the solution of the isotropic thermic diffusion equation:

$$\frac{\partial u}{\partial t} = C\frac{\partial^2 u}{\partial x^2}, \quad \text{(Eq. A.30)}$$

or in its discrete form:

$$u_{t+\Delta t}(x) = 2u_t(x) - u_t(x-1) - u_t(x+1), \quad \text{(Eq. A.31)}$$

where $u_t(x)$ represents the temperature in x at time t.

The thermic diffusion equation for a two-dimensional signal is:

$$\frac{\partial u}{\partial t} = C\Delta u\text{, where }\Delta u = -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} \quad \text{(Eq. A.32)}$$

and its discrete form is:

$$u_{t+\Delta t}(x, y) = 4u_t(x, y) - u_t(x-1, y) - u_t(x+1, y) - u_t(x, y-1) - u_t(x, y+1) \quad \text{(Eq. A.33)}$$

For a two-dimensional signal, this filter is composed of four operators which have the following forms:

1 Left to right:

$$I^{(t+1)}(x, y) = \frac{1}{2}(I^{(t+1)}(x-1, y) + I^{(t)}(x+1, y)), \quad \text{(Eq. A.34)}$$

2 Right to left:

$$I^{(t+2)}(x, y) = \frac{1}{2}(I^{(t+1)}(x-1, y) + I^{(t+2)}(x+1, y)), \quad \text{(Eq. A.35)}$$

3 Top to bottom:

$$I^{(t+3)}(x, y) = \frac{1}{2}(I^{(t+3)}(x, y-1) + I^{(t+2)}(x, y+1)), \quad \text{(Eq. A.36)}$$

4 Bottom to right:

$$I^{(t+2)}(x, y) = \frac{1}{2}(I^{(t+1)}(x, y-1) + I^{(t+2)}(x, y+1)). \quad \text{(Eq. A.37)}$$

The frequency response of the recursive filter is typical of a low pass filter. High frequencies are better filtered with the isotropic diffusion filter than with the Gaussian or mean filter (Fig. A-11).

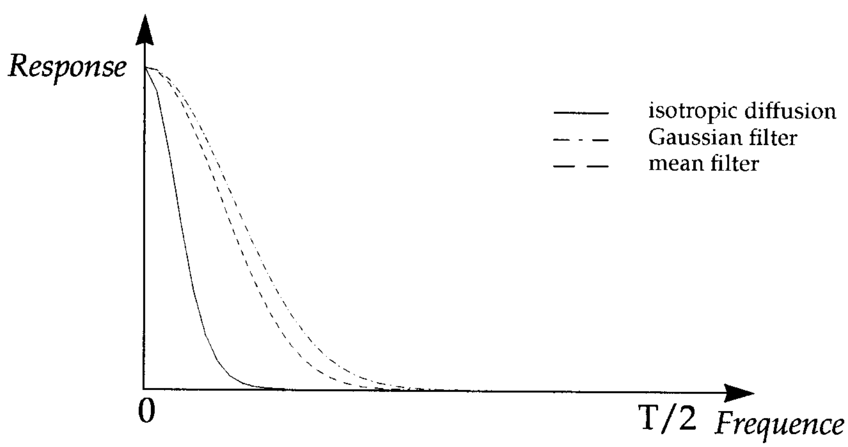


Figure A-11. Frequency response of the isotropic diffusion filter compared with the Gaussian filter [1/4, 2/4, 1/4] **and the mean filter** [1/3, 1/3, 1/3] **. Each filter has convolved ten times with Dirac.**

Polynomial smoothing

This smoothing technique is based on the approximation of pixel intensity values in a small area A (3x3, 5x5 or 7x7) by a second degree polynomial function. For a polynomial function of degree n, the general equation can be expressed as:

$$P(x, y) = \sum_{i,j}^{n} a_{ij} x^i y^j . \qquad \text{(Eq. A.38)}$$

The principle of the fitting process is to find coefficients a_{ij} of a polynomial function P that best fits the pixels in a small neighborhood A, where the central pixel is (x, y). Choosing $x = 0$ and $y = 0$ for equation (Eq. A.38), only coefficient a_{oo} remains, giving a linear equation that can easily be resolved. This value represents the new value of the central pixel of region A. Coefficient a_{oo} is calculated by minimizing the following expression with the Least Squares method, considering the intensity values $I(x, y)$ of a given image in a small area $A = \{(x, y)/-m \le x, y \le m\}$, centered in $(0, 0)$, as mentioned above:

$$\varepsilon = \sum_{x=-m}^{m} \sum_{y=-m}^{m} (P(x, y) - I(x, y))^2 \rightarrow minimum. \qquad \text{(Eq. A.39)}$$

For a quadratic polynomial function, the minimization of ε results in the following convolution operators, allowing coefficient a_{oo} to be calculated for areas of size 3x3, 5x5 and 7x7 respectively:

$$a_{oo} = \frac{1}{9}\begin{bmatrix} -1 & 2 & -1 \\ 2 & 5 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

$$a_{oo} = \frac{1}{175}\begin{bmatrix} -13 & 2 & 7 & 2 & -13 \\ 2 & 17 & 22 & 17 & 2 \\ 7 & 22 & 27 & 22 & 7 \\ 2 & 17 & 22 & 17 & 2 \\ -13 & 2 & 7 & 2 & -13 \end{bmatrix}$$

$$a_{oo} = \frac{1}{147}\begin{bmatrix} -7 & -2 & 1 & 2 & 1 & -2 & -7 \\ -2 & 3 & 6 & 7 & 6 & 3 & -2 \\ 1 & 6 & 9 & 10 & 9 & 6 & 1 \\ 2 & 7 & 10 & 11 & 10 & 7 & 2 \\ 1 & 6 & 9 & 10 & 9 & 6 & 1 \\ -2 & 3 & 6 & 7 & 6 & 3 & -2 \\ -7 & -2 & 1 & 2 & 1 & -2 & -7 \end{bmatrix}$$

(Eq. A.40)

Polynomial approximation functions of the second degree are appropriate, considering the curbed shape of 2-DE images.

Adaptive smooth

The adaptive smoothing technique is based on the anisotropic diffusion equation. This method smooths images, but it preserves pertinent discontinuities. The idea is to weight the smooth operator according to the magnitude of the discontinuity d. This means that, for each point in the image, a continuity value w is calculated using a decreasing function $f(d)$, such as $f(0) = 1$ and $f(d) \rightarrow 0$ as d increases, where d represents the amount of discontinuity at each point.

For an image $I(x, y)$, the two-dimensional adaptive filter can be expressed by the following equation:

$$I^{(t+1)}(x, y) = \frac{1}{N^{(t)}} \sum_{i=-1}^{1} \sum_{j=-1}^{1} I(x+i, y+j) w^{(t)}, \qquad \text{(Eq. A.41)}$$

where

$$w^{(t)}(x, y) = f(d^{(t)}(x, y)) = e^{-\left(\frac{|d^{(t)}(x, y)|^2}{2K^2}\right)}, \qquad \text{(Eq. A.42)}$$

with $d^{(t)}(x, y) = \sqrt{G_x^{\,2} + G_y^2}$, (Eq. A.43)

where G_x and G_y are the gradients along the x and y axes respectively, and

$$N^{(t)} = \sum_{i=-1}^{1} \sum_{j=-1}^{1} w(x+i, y+j) w^{(t)}. \qquad \text{(Eq. A.44)}$$

Figure A-12 illustrates the isotropic and anisotropic diffusion smooth operators on a 1D signal representing the gray-level profile of one row in a 2-DE gel image. (a1) and (b1) show the original profile. (a2) and (a3) show the profile after two, respectively five iterations of the isotropic smooth operator. (b2) and (b3) display the profile after four, respectively ten iterations of the anisotropic smooth operator, with parameter $k = 2$. As one can observe, the second method preserves significant discontinuities and signal shape better than the first one.

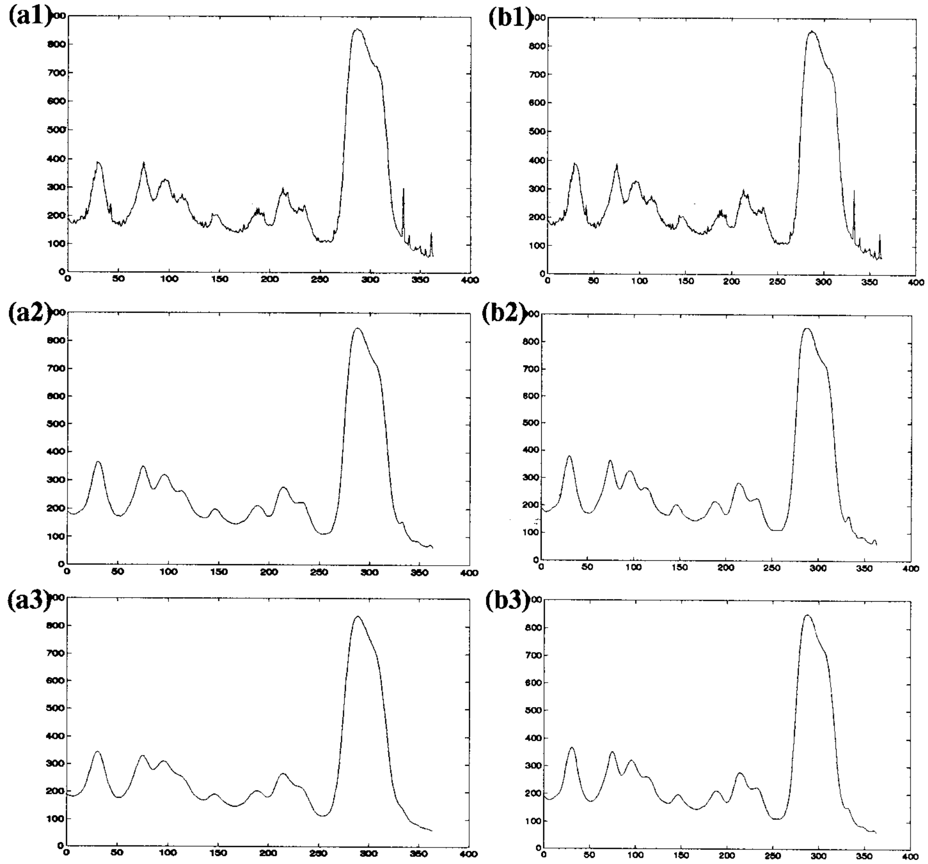


Figure A-12. (a1) and (b1) represent the original profile of a 2-DE image. (a2), (a3) show the result of the application of isotropic diffusion smooth operator to the original profile after 2, respectively 5 iterations. In (b2),(b3) the anisotropic adaptive smooth operator was applied to the original profile 4, respectively 10 times with k=2. Significant discontinuities and the signal shape are better preserved by the anisotropic smooth method.

Histogram equalization

Histogram equalization is used to improve the visualization aspect of images. This operation attempts to correct the grey level values of a given image by changing its normalized histogram. For a given grey level distribution $p_x(i)$, with $i = 0 \ldots n_x$, a new distribution $p_y(j)$ with $j = 0 \ldots n_y$ is computed by searching a value N for each M such that:

$$\sum_i^M p_x(i) \le \sum_j^N p_y(j) \qquad \text{(Eq. A.45)}$$

which defines a requantification law $y(x) = N(M)$, and

$$P_x(M) = \sum_i^M p_x(i) \qquad \text{(Eq. A.46)}$$

which is the cumulative law of $p_x(i)$.

Using this formalism, the following transformations may be defined:

- Uniform:

$$p_y(y) = \frac{1}{(y_{max} + y_{min})} \text{ with } y_{min} \le y \le y_{max}$$
$$\Rightarrow \qquad y(x) = (y_{max} + y_{min})P_x(x) + y_{min} \qquad \text{(Eq. A.47)}$$

- Exponential:

$$p_y(y) = \alpha \cdot \exp(-\alpha(y - y_{min})) \text{ with } y_{min} \le y$$
$$\Rightarrow \qquad y(x) = y_{min} + \frac{1}{\alpha}\log(1 - P_x(x)) \qquad \text{(Eq. A.48)}$$

- Logarithmic:

$$p_y(y) = (y(\log(y_{max}) - \log(y_{min})))^{-1}$$
$$\Rightarrow \qquad y(x) = y_{min}(y_{max}/y_{min})^{P_x(x)} \qquad \text{(Eq. A.49)}$$

- Cube-Root:

$$p_y(y) = \frac{1}{3}\frac{\sqrt[3]{y^{-2}}}{\sqrt[3]{y_{max}} - \sqrt[3]{y_{min}}}$$
$$\Rightarrow \qquad y(x) = \sqrt[3]{(\sqrt[3]{y_{max}} - \sqrt[3]{y_{min}})P_x(x) + \sqrt[3]{y_{min}}} \qquad \text{(Eq. A.50)}$$

where y_{max}, y_{min} are the maximum, respectively minimum possible value of the requantification law $y(x)$, and

$$P_x(x) = \sum_i^x p_x(i) \qquad \text{(Eq. A.51)}$$

is the cumulative law.

The Melanie II system equalizes an image by first splitting it into several regions. Then, for each region, the histogram is computed. Next, histograms corresponding to regions are equalized using one of the transformation functions described above. Finally, for each pixel, the new equalized value is computed by bilinear interpolation on the four possible transformations of the value using its neighbor equalized histograms.

Grey levels remapping

This filter corrects intensity values by mapping each pixel value of an image I with a dynamic range $[min, max]$ into an image I' with a dynamic range $[MIN, MAX]$ according to a mapping function F, using the following expression:

$$I'(x, y) = F\left(\frac{I(x, y) - min}{max - min}\right)(MAX - MIN) + MIN \qquad \text{(Eq. A.52)}$$

where F is a bijective monotone function into $[0, 1]$ that maps this interval into $[0, 1]$. The function F may be:

- Linear: $F(x) = x$,

$$I'(x, y) = \frac{I(x, y) - min}{max - min}(MAX - MIN) + MIN; \qquad \text{(Eq. A.53)}$$

- Logarithmic: $F(x) = \frac{\log(ax + 1)}{\log(n)}$ in base n,

$$I'(x, y) = \log\left(a \cdot \left(\frac{I(x, y) - min}{max - min}\right) + 1\right)\frac{(MAX - MIN)}{\log(n)} + MIN \qquad \text{(Eq. A.54)}$$

where $a = e^{\log(n)} - 1$;

- Exponential: $F(x) = \frac{\exp(ax) - 1}{\exp(n) - 1}$,

$$I'(x, y) = \left(\exp\left(a \cdot \left(\frac{I(x, y) - min}{max - min}\right)\right) - 1\right)\frac{(MAX - MIN)}{\exp(n) - 1} + MIN \qquad \text{(Eq. A.55)}$$

where $a = n$.

Contrast enhancement

The contrast enhancement filter normalizes pixel values of a given image $I(x, y)$. The interval $[min, max]$ is first mapped to the interval $[0, 1]$, then it is remapped to $[0, 1]$ using a transformation function F. Finally, the new pixel values of the contrasted image are computed using the following expression:

$$I'(x, y) = F\left(\frac{I(x, y) - min}{max - min}\right) \times (\text{Highest grey}), \qquad \text{(Eq. A.56)}$$

where max and min are the highest, respectively lowest intensity values of image $I(x, y)$, *Highest grey* is the highest intensity value supported by the current image format, and F is one of the following transformation functions:

- Quadratic:

  $F(z) = z^2$ with $z \in [0, 1]$ a real value.

- Sigmoid:

$$F(z) = \frac{erf\left(\alpha\left(x - \frac{1}{2}\right)\right)}{2 \cdot erf\left(\frac{\alpha}{2}\right)} + \frac{1}{2}, \qquad \text{(Eq. A.57)}$$

where $z \in [0, 1]$, $\alpha = 4$, and

$$erf(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}dt. \qquad \text{(Eq. A.58)}$$

- Wallis:

The contrast enhancement is performed using a statistical approach. The contrasted image is computed using the following expression:

$$I'(x, y) = \frac{I(x, y)}{S(x, y)}, \qquad \text{(Eq. A.59)}$$

where $S(x, y)$ is the standard deviation given by:

$$S(x, y) = \frac{1}{(2k+1)^2}\sum_{i=-k}^{k}\sum_{j=-k}^{k}(I(x+i, y+j) - M(x+i, y+j))^2 \qquad \text{(Eq. A.60)}$$

with

$$M(x, y) = \frac{1}{(2k+1)^2}\sum_{i=-k}^{k}\sum_{j=-k}^{k}I(x, y). \qquad \text{(Eq. A.61)}$$

Background filtering

The background of a 2-DE image may be eliminated by using one of two background subtraction algorithms: subtraction of global image minimum and subtraction using polynomial functions.

Subtraction of global image minimum

The minimum pixel value in the gel is subtracted from all pixel values in the image.

Background estimation using polynomial functions

Background estimation of a 2-DE image is achieved by fitting the pixel values located outside the spot shapes with a third order polynomial function of the following form:

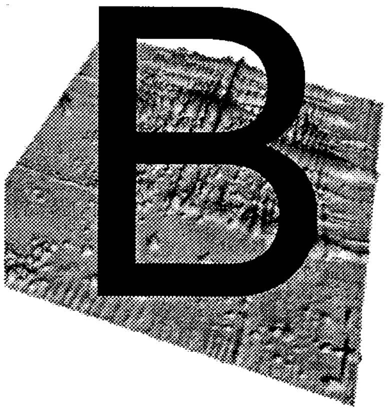
$$B(x, y) = a_1 + a_2x + a_3y + a_4x^2 + a_5xy + a_6y^2 + a_7x^3 + a_8x^2y + a_9xy^2 + a_{10}y^3, \qquad \text{(Eq. A.62)}$$

where $B(x, y)$ is the estimated background pixel value, x is the horizontal position, and y is the vertical position of the pixel in the image, and $a_1, \ldots, a_{10}$ are the polynomial parameters. These parameters are determined for a given gel in the following steps:

1 spot shapes are detected by the method described in *Spot detection on page A-1*,

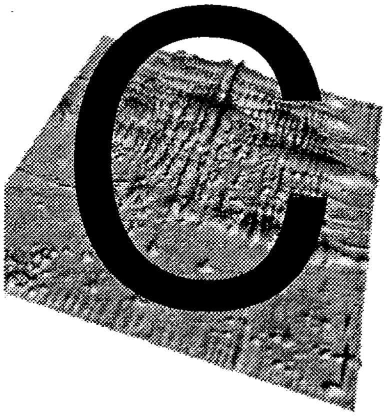
2 a grid of 3x3 pixels is chosen in order to select pixels in the image. Only the pixels that belong outside the spot shapes are selected,

3 selected pixels are fitted with (Eq. A.62) using the Least Squares method. This is achieved in a similar way than has been described *Gaussian fitting on page A-13*.



B KNOWN BUGS

- Netscape Navigator® loads its own color table according to the colors already in use at load time. If you have loaded Netscape Navigator® prior to using Melanie II, and you feel that the colors are badly displayed in Netscape, you should quit the Netscape Navigator® browser and run it again.



REFERENCES

The Melanie project

1 *The Melanie II 2-DE analysis software WWW page*:
http://expasy.hcuge.ch/melanie/melanie-top.html

2 Appel R.D., Hochstrasser D.F., Funk M., Vargas R.J., Pellegrini C., Muller A.F., Scherrer J-R. *The MELANIE project - From a Biopsy to Automatic Protein Map Interpretation by Computer.* Electrophoresis 12, 1991, 722-735.

3 Appel R.D., Bologna G., Hochstrasser D.F. *Classification tools for diagnostic rule formation from protein maps.* Proc. MIE'93, 11th International Congress of the European Federation for Medical Informatics, Jerusalem, April 18-22, 1993, 40-44.

4 Vargas R.J. *Two-dimensional gel electrophoresis computer analysis systems: from image acquisition to protein identification.* Ph.D. thesis, Faculty of science, Geneva University, 1996.

5 Miller M.J., Olson A.D., Thorgeirsson S.S. *Computer analysis of two-dimensional gels: Automatic matching.* Electrophoresis 5, 297-303, 1984.

Two-dimensional electrophoresis (2-DE)

6 *2-D PAGE protocols*:
http://expasy.hcuge.ch/ch2d/technical-info.html

2-DE databases

7 *The SWISS-2DPAGE 2-DE database*:
http://expasy.hcuge.ch/ch2d/ch2d-top.html

8 Appel R.D., Bairoch A., Sanchez J.C., Vargas J.R., Golaz O., Pasquali C., Hochstrasser D.H. *Federated 2-DE database: a simple means of publishing 2-DE data*. Electrophoresis 17, 1996, in press.

9 *WORLD-2DPAGE: index to federated 2-DE databases*:
http://expasy.hcuge.ch/ch2d/2d-index.html

10 Appel R.D., Sanchez J.C., Bairoch A., Golaz O., Miu M., Pasquali C., Vargas R.J., Hughes G., Hochstrasser D.F. *SWISS-2DPAGE: a database of two-dimensional gel electrophoresis images*. Electrophoresis 14, 1993, 1232-1238.

11 Appel R.D, Sanchez J-C., Bairoch A., Golaz O., Ravier F., Pasquali C., Hughes G.J., Hochstrasser D. *The SWISS-2DPAGE database of two-dimensional polyacrylamide gel electrophoresis*. Nucleic Acids Research, 1994, Vol. 22, No. 17, 3581-3582.

12 Sanchez J.-C., Appel R.D., Golaz O., Pasquali C., Ravier F., Bairoch A., Hochstrasser D.F. *Inside SWISS-2DPAGE database*. Electrophoresis 16, 1995, 1131-1151.

13 Hughes, G.J., Frutiger, S., Paquet, N., Pasquali, C., Sanchez, J-C., Tissot, J.D., Bairoch, A., Appel, R.D., Hochstrasser, D.F. *Human Liver Protein Map: The 1993 Update*. Electrophoresis 14, 1993, 1216-1222.

14 Golaz, O., Hughes, G.J., Frutiger, S., Paquet, N., Bairoch, A., Pasquali, C., Sanchez, J.-C., Tissot, J.-D., Appel, R.D., Walzer, C., Balant, L., Hochstrasser, D.F. *Plasma & Red Blood Cell Protein Maps: The 1993 Update*. Electrophoresis 14, 1993, 1223-1231.

15 Sanchez J.C., Golaz O., Frutiger S., Schaller D., Appel R.D., Bairoch A., Hughes G.J., Hochstrasser D.F. *The YEAST SWISS-2DPAGE database*. Electrophoresis 17, 1996, in press.

16 Pasquali C., VanBogelen R.A., Wilkins M., Frutiger S., Appel R.D., Vargas R., Sanchez J.C., Hochstrasser D.F. *The ESCHERICHIA COLI SWISS-2DPAGE database*. Electrophoresis 17, 1996, in press.

17 Gravel P., Sanchez J.C., Walzer C., Golaz O., Hochstrasser D.F., Balant L.P., Hughes G.J., Garcia-Sevilla J., Guimon J. *Human blood platelet protein map established by two-dimensional polyacrylamide gel electrophoresis*. Electrophoresis 16, 1995, 1152-1159.

The ExPASy WWW server

18 *The ExPASy WWW server*:
http://expasy.hcuge.ch/

19 Appel R.D., Bairoch A., Hochstrasser D.F. *A new generation of information retrieval tools for biologists: the example of the expasy WWW server*. Trends in Biochemical Sciences TiBS June 1994 (222), Vol. 19, No. 6, pp. 258-260.

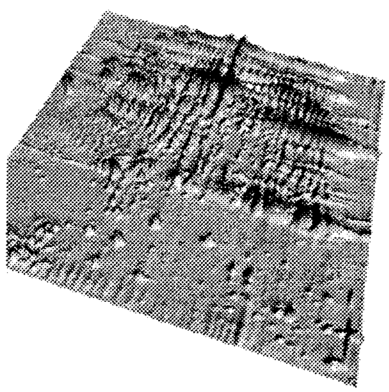
20 Hochstrasser D.F., Appel R.D., Golaz O., Pasquali C., Sanchez J.-C., Bairoch A. *Sharing of Worldwide Spread Knowledge Using Hypermedia Facilities & Fast Communications Protocols (Mosaic & World-Wide Web): The Example of ExPASy.* Meth. Inform. Med., 1995, 34: 75-78.

D

Technical Support

Bio-Rad provides technical support on the features of the Melanie II software. If you have any problems or any questions on the use of the features, contact your local Bio-Rad office, or in the U.S. call Bio-Rad Technical Service at 1-800-4BIORAD (1-800-424-6723).

Please report any reproducible software problems to Bio-Rad Laboratories.



INDEX

H

I

L

M